Because the only systems to be analysed are those that have some bearing on the new problem as a whole or on parts of it, we could call this way of collecting information the systematic exploitation of proven ideas, or of experience. It proves particularly helpful for finding a first solution concept as a starting point for further variations. It must, however, be said that this approach carries the danger of causing designers to stick with known solutions instead of pursuing new paths.

4. Analogies

In the search for solutions and in the analysis of system properties, it is often useful to substitute an analogous problem (or system) for the one under consideration, and to treat it as a model. In technical systems, analogies may be obtained, for instance, by changing the type of energy used [3.3, 3.64]. Analogies chosen from the nontechnical sphere may prove very useful as well.

Besides helping in the search for solutions, analogies are also helpful in the study of the behaviour of a system during the early stages of its development by means of simulation and model techniques, and in the subsequent identification of essential new subsolutions and the introduction of early optimisations.

If the model is to be applied to systems of markedly different dimensions and conditions, a supportive similarity (dimensional) analysis should be undertaken (see Section 9.1.1).

5. Measurements and Model Tests

Measurements on existing systems, model tests supported by similarity analyses and other experimental studies are among the most important sources of information. Rodenacker [3.59] in particular stresses the importance of experimental studies, arguing that design can be interpreted as the reversal of physical experiment.

In the precision engineering and mass production industries, including those where micromechanisms and electronic products are developed, experimental investigations are an important and established means of arriving at solutions. This approach has organisational repercussions since, in the creation of such products, experimental development is often incorporated within the design activity (see Figure 1.3).

In a similar way, the testing and subsequent modification of software solutions belong to this empirically based group of methods.

3.2.2 Intuitive Methods

Designers often seek and discover solutions for difficult problems by intuition—that is, solutions come to them in a flash after a period of search and reflection. These solutions suddenly appear as conscious thoughts and often their origins cannot be traced. As Galtung of the International Peace Research Institute in Oslo has put it: “The good idea is not discovered or undiscovered; it comes, it happens”.

It is then developed, modified and amended, until such time as it leads to the solution of the problem.

Good ideas are always scrutinised by the subconscious or preconscious in the light of expert knowledge, experience and the task in hand, and often the simple impetus resulting from the association of ideas suffices to force them into consciousness. That impetus can also come from apparently unconnected external events or discussions. Frequently, a sudden idea will hit the bull's eye, so that all that needs to be done is to make changes or adaptations that lead straight to a final solution. If that is indeed the case and a successful product is created, then this represents the optimum procedure. Very many good solutions are born in that way and developed successfully. A good design method, far from trying to eliminate this process, should serve to back it up.

An industrial concern should nevertheless beware of exclusive reliance on the intuition of its designers, nor should designers themselves leave everything to chance or rare inspiration. Purely intuitive methods have the following disadvantages:

- The right idea does not always come at the right time, since it cannot be forced.
- Current conventions and personal prejudices may inhibit original developments.
- Because of inadequate information, new technologies or procedures may fail to reach the consciousness of the designer.

These dangers increase with specialisation, the division of tasks and with time pressure.

There are several methods of encouraging intuition and opening new paths by the association of ideas. The simplest and most common of these involves critical discussions with colleagues. Provided that such discussions are not allowed to stray too far and are based on the general methods of persistent questions, negation, forward steps, etc. (see Section 2.2.5), they can be very helpful and effective.

Methods with an intuitive bias such as Brainstorming, Synectics, Gallery Method, Method 635 and many others involve group dynamics that are used to generate the widest possible range of ideas. One of the effects of group dynamics is the uninhibited exchange of associated ideas between the members.

Most of these techniques were originally devised for the solution of nontechnical problems. They are, however, applicable to any field that demands new, unconventional ideas.

1. Brainstorming

Brainstorming can be described as a method of generating a flood of new ideas. It was originally suggested by Osborn [3.51] and provides conditions in which a group of open-minded people from as many different spheres of life as possible bring up, without prejudice, any thoughts that occur to them and thus trigger off new ideas in the minds of the other participants [3.74]. Brainstorming relies strongly on stimulation of the memory and on the association of ideas that have never been considered in the current context or have never been allowed to reach consciousness.

For maximum effect, brainstorming sessions should be run along the following lines:

Composition of the Group

- The group should have a leader and consist of a minimum of five and a maximum of 15 people. Fewer than five constitute a spectrum of opinion and experience that is too small, and hence produce too few stimuli. With more than 15, close collaboration may decline because of individual passivity and withdrawal.
- The group must not be confined to experts. It is important that as many fields and activities as possible are represented, the involvement of nontechnical members adding a rich new dimension.
- The group should not be hierarchically structured but, if possible, made up of equals in order to prevent the censoring of such thoughts as might give offence to superiors or subordinates.

Leadership of the Group

- The leader of the group should only take the initiative when dealing with organisational problems (invitation, composition, duration and evaluation). Before the actual brainstorming session, the leader must outline the problem and, during the session, must see to it that the rules are observed and, in particular, that the atmosphere remains free and easy. To that end the leader should start the session by expressing a few absurd ideas, or mentioning an example from another brainstorming session, but should never lead in the expression of ideas. On the other hand, the flow of new ideas should be encouraged whenever the productivity of the group slackens. The leader must ensure that no one criticises the ideas of other participants, and should appoint one or two members to take minutes.

Procedure

- All participants must try to shed their intellectual inhibitions; that is, they should avoid rejecting as absurd, false, embarrassing, stupid, well-known or redundant any ideas expressed spontaneously by themselves or by other members of the group.
- No participant should criticise any ideas that are brought up, and everyone must refrain from using such killer phrases as “we’ve heard it all before”, “it can’t be done”, “it will never work” and “this has nothing to do with the problem”.
- New ideas will be taken up by the other participants, who may change and develop them at will. It is also useful to combine several ideas into new proposals.
- All ideas should be written down, sketched out, or recorded.
- All suggestions should be concrete enough to allow the emergence of specific solution ideas.
- The practicability of the suggestions should be ignored at first.

- A session should not generally last for more than 30 to 45 minutes. Experience has shown that longer sessions produce nothing new and lead to unnecessary repetitions. It is better to make a fresh start with new ideas or with other participants later.

Evaluation

- The results should be reviewed by experts to find potential solution elements. If possible, these should be classified and graded in order of feasibility and then developed further.
- The final result should be reviewed with the entire group to avoid possible misunderstandings or one-sided interpretations on the part of the experts. New and more advanced ideas may well be expressed or developed during such a review session.

Brainstorming is indicated [3.56] whenever:

- No practical solution principle has been discovered.
- The physical process underlying a possible solution has not yet been identified.
- There is a general feeling that deadlock has been reached.
- A radical departure from the conventional approach is required.

Brainstorming is even useful in the solution of subproblems arising in known or existing systems. Moreover, it has a beneficial side-effect: all of the participants are supplied with new data, or at least with fresh ideas on possible procedures, applications, materials, combinations, etc., because the group represents a broad spectrum of opinion and expertise (for instance, designers, production engineers, sales persons, materials experts and buyers). It is astonishing what a profusion and range of ideas such a group can generate. The designers will remember the ideas brought up during brainstorming sessions on many future occasions. Brainstorming triggers off new lines of thought, stimulates interest and represents a break in the normal routine.

It should, however, be stressed that no miracles must be expected from brainstorming sessions. Most of the ideas expressed will not be technically or economically feasible, and those that are will often be familiar to the experts. Brainstorming is meant first of all to trigger off new ideas, but it cannot be expected to produce ready-made solutions because problems are generally too complex and too difficult to be solved by spontaneous ideas alone. However, if a session should produce one or two useful new ideas, or even some hints in what direction to go looking for the solution, it will have achieved a great deal.

An example of a solution obtained by Brainstorming can be found in Section 6.6, which also shows how the resulting ideas were evaluated and how classifying criteria for the subsequent search for solutions were derived from them.

2. Method 635

Brainstorming has been developed into Method 635 by Rohrbach [3.60]. After familiarising themselves with the task, and after careful analysis, each of six par-

ticipants is asked to write down three rough solutions in the form of keywords. After some time, the solutions are handed to each participant's neighbour who, after reading the previous suggestions, enters three further solutions or developments. This process is continued until each original set of three solutions has been completed or developed through association by the five other participants, hence the name of the method.

Method 635 has the following advantages over Brainstorming:

- A good idea can be developed more systematically.
- It is possible to follow the development of an idea and to determine more or less reliably who originated the successful solution principle, which might prove advisable for legal reasons.
- The problem of group leadership rarely arises.

The method has the following disadvantage:

- Reduced creativity by the individual participants owing to isolation, and lack of stimulation in the absence of overt group activity.

3. Gallery Method

The Gallery Method developed by Hellfritz [3.27] combines individual work with group work, and is particularly suitable for any stage of the design process where solution proposals can be expressed in the form of sketches or drawings. The organisation and team building are similar to Brainstorming. The method consists of the following steps.

Introduction Step: The group leader presents the problem and explains the context.

Idea Generation Step 1: For 15 minutes the individual group members create solutions intuitively and without prejudice using sketches supported, where necessary, by text.

Association Step: The results from idea generation step 1 are hung on a wall as in an art gallery so that all group members can see and discuss them. The purpose of this 15-minute association step is to find new ideas or to identify complementary or improved proposals through negation and reappraisal.

Idea Generation Step 2: The ideas and insights from the association step are further developed individually by each of the group members.

Selection Step: All ideas generated are reviewed, classified and, if necessary, finalised. Promising solutions are then selected (see Section 3.3.1). It is also possible to identify potential solution characteristics that can be developed later using a discursive method (see Section 3.2.3).

The Gallery Method has the following advantages:

- Intuitive group working takes place without unduly lengthy discussions.
- An effective exchange of ideas using sketches is possible.
- Individual contributions can be identified.
- Documentary records are easily assessed and stored.

4. Delphi Method

In this method, experts in a particular field are asked for written opinions [3.7].

The requests take the following form:

First Round: What starting points for solving the given problem do you suggest? Please make spontaneous suggestions.

Second Round: Here is a list of various starting points for solving the given problem. Please go through this list and make what further suggestions occur to you.

Third Round: Here is the final evaluation of the first two rounds. Please go through the list and write down what suggestions you consider most practicable.

This elaborate procedure must be planned very carefully and is usually confined to general problems bearing on fundamental questions or on company policy. In the field of engineering design, the Delphi Method should be reserved for fundamental studies of long-term developments.

5. Synectics

Synectics is a word derived from Greek and it refers to the activity of combining various and apparently independent concepts. Synectics is comparable to Brainstorming, with the difference that its aim is to trigger off fruitful ideas with the help of analogies from nontechnical or semi-technical fields.

The method was first proposed by Gordon [3.25]. It is more systematic than Brainstorming, with its arbitrary flow of ideas. However, both methods call for complete frankness and lack of inhibition or criticism.

A synectics group should consist of no more than seven members, otherwise the ideas expressed will run away with themselves. The leader of the group has the additional task of helping the group to develop the proposed analogies by guiding them through the following steps:

- Presentation of the problem.
- Familiarisation with the problem (analysis).
- Grasping the problem.
- Rejection of familiar assumptions with the help of analogies drawn from other spheres.
- Analysis of one of the analogies.
- Comparison of the analogy with the existing problem.
- Development of a new idea from that comparison.
- Development of a possible solution.

If the result is unsatisfactory, the process may have to be repeated with a different analogy.

An example may help to illustrate this method. In a seminar set up for the purpose of discovering the best method of removing urinary calculi from the human body,

several mechanical devices for gripping, holding and extracting these stones were mentioned. The device would have to stretch and open up inside the urethra. The keywords "stretch" and "open up" suggested the idea of an umbrella to one of the participants (see Figure 3.12).

Question: How can the umbrella analogy—(a) in Figure 3.12—be applied?

Possible answer 1: By (b) drilling through the stone, pushing the umbrella through the hole and opening it up. Not very feasible.

Possible answer 2: By (c) pushing a tube through the hole and blowing it up (balloon) behind the stone. Drilling of hole not feasible.

Possible answer 3: By (d) pushing the tube past the stone. When the tube is withdrawn the resistance may seriously damage the urethra.

Possible answer 4: By (e) adding a second balloon as a guide and by (f) embedding the stone in a gel between the two balloons and then pulling it out? This was found to be the best solution.

This example shows the association with a semi-technical analogy (umbrella) from which a solution was developed that took into account the special constraints that existed in this case. The solution shown here is not the final solution resulting from the seminar but represents an example of how the method was used.

Characteristic of this approach is the unrestricted use of analogies which, in the case of technical problems, are selected from nontechnical or semi-technical spheres. Such analogies will generally suggest themselves quite spontaneously at the first attempt but, during subsequent development and analysis, they will generally be derived more systematically.

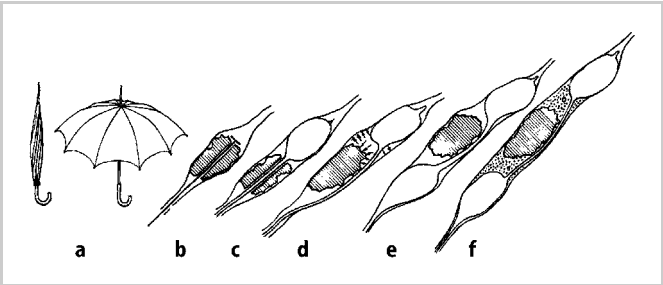


Figure 3.12. Step-by-step development of a solution principle for the removal of urinary calculi based on analogy and stepwise improvement

6. Combination of Methods

Any one of these methods taken by itself may not lead to the required goal. Experience has shown that:

- The group leader of, or another participant in, a brainstorming session may, when the flow of ideas dries up, introduce synectic procedures—deriving analogies, rejection of familiar assumptions, etc.—to release a new flood of ideas.

- A new idea or an analogy may radically change the approach and ideas of the group.
- A summary of what has been agreed so far may lead to new ideas.
- The explicit use of the methods of negation and reappraisal and of forward steps (see Section 2.2.5) can enrich and extend the variety of ideas.

In the seminar we mentioned, the presentation of the idea "destroy stone" produced a host of new suggestions, such as drilling, smashing, hammering, ultrasonic disintegration and so on. When the flow of ideas eventually dried up, the group leader asked, "How does nature destroy?", which immediately evoked a number of new suggestions, including weathering, heating and cooling, decay, putrefaction, bacterial action, ice expansion and chemical decomposition. A combination of the two principles "clasp stone" and "destroy stone" provoked the question, "What else?" This produced the answer "contact stone rather than clasp", which in turn threw up such new ideas as sucking, gluing, and applying various contact forces.

The different methods should be combined so as to best address particular cases. A pragmatic approach ensures the best results.

3.2.3 Discursive Methods

Methods with a discursive bias provide solutions in a deliberate step-by-step approach that can be influenced and communicated. Discursive methods do not exclude intuition, which can make its influence felt during individual steps and in the solution of individual problems, but not in the direct implementation of the overall task.

1. Systematic Study of Physical Processes

If the solution of a problem involves a known physical (chemical, biological) effect represented by an equation, and especially when several physical variables are involved, various solutions can be derived from the analysis of their interrelationships, that is, of the *relationship* between a dependent and an independent variable, all other quantities being kept constant. Thus, if we have an equation in the form $y = f(u, v, w)$, then, according to this method, we investigate solution variants for the relationships $y_1 = f(u, \underline{v}, \underline{w})$, $y_2 = f(\underline{u}, v, \underline{w})$ and $y_3 = f(\underline{u}, \underline{v}, w)$, the underlined quantities being kept constant.

Rodenacker has given several examples of this procedure, one of which concerns the development of a capillary viscometer [3.59]. Four solution variants can be derived from the well-known law of capillary action $\eta \sim \Delta p \cdot r^4/(\dot{V} \cdot l)$. They are shown schematically in Figure 3.13.

1. A solution in which the differential pressure Δp serves as a measure of the viscosity: $\Delta p \sim \eta$ ($\dot{V}$, r and l = constant).
2. A solution based on changes in radius of the capillary tube: $\Delta r \sim \eta$ ($\dot{V}$, Δp and l = constant).

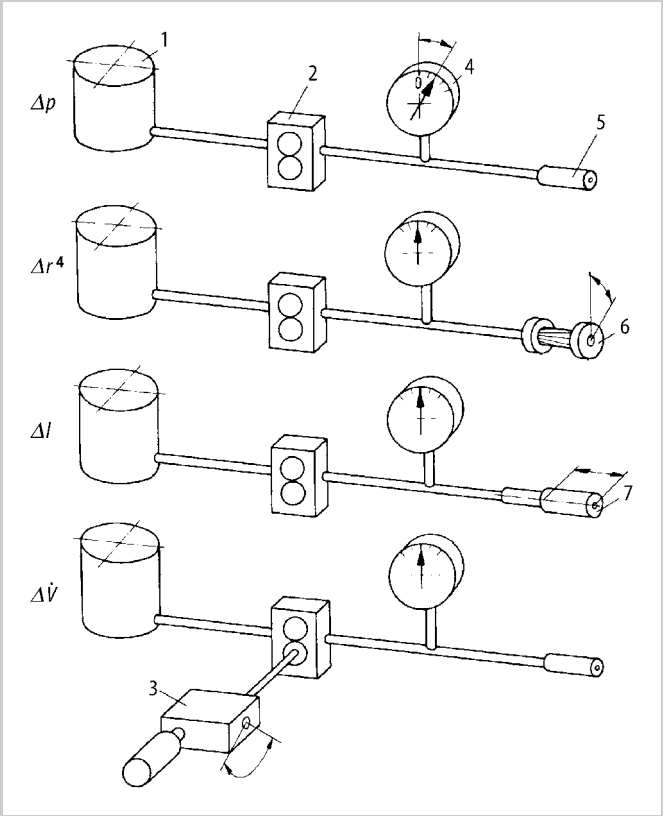


Figure 3.13. Schematic representation of four viscometers, after [3.59]. *1*, container; *2*, gear pump; *3*, variable drive; *4*, pressure gauge; *5*, fixed capillary tube; *6*, capillary tube with variable diameter; *7*, capillary tube with variable length

3. A solution based on changes in the length of the capillary tube: $\Delta l \sim \eta$ (Δp, $\dot{V}$ and r = constant).
4. A solution based on changes in the volume flow rate: $\Delta \dot{V} \sim \eta$ (Δp, r and l = constant).

Another way of obtaining new or improved solutions by the analysis of physical equations is the resolution of known physical effects into their individual components. Rodenacker [3.59], in particular, has used this approach in the design of novel devices and the development of new applications for existing ones.

By way of example, let us look at the development of a frictional thread locking device, based on the analysis of the equation governing the torque needed to release a threaded fastener:

$$T = P[(d/2)\tan(\phi_v - \beta) + (D/2)\mu_f] \tag{3.1}$$

The torque given by Equation (3.1) is made up of the following components:

Frictional torque in the thread:

$$T_t \sim P(d/2)\tan\phi_v = P(d/2)\mu_v \tag{3.2}$$

where

$$\tan\phi_v = \mu_t/\cos(\alpha/2) = \mu_v$$

Frictional torque on the bolt head or nut face:

$$T_f = P(D/2)\tan\phi_f = P(D/2)\mu_f \tag{3.3}$$

Release torque of the thread due to pre-load and thread pitch:

$$T_r \sim P(d/2)\tan(-\beta) = -P \cdot \frac{P}{2\pi} \tag{3.4}$$

(where p = thread pitch, β = helix angle, d = mean thread (t) diameter, P = pre-load, D = mean face (f) diameter, μ_v = virtual (v) coefficient of friction in the thread, μ_t = actual coefficient of friction in the thread, μ_f = coefficient of friction on the head or nut face, α = flank angle, ϕ = angle of friction).

To discover solution principles for the improvement of the locking properties of a threaded fastener, we must analyse the physical relationships further so as to identify the physical effects involved. The individual effects involved in Equations (3.2) and (3.3) are:

- the friction effect (Coulomb friction)

$$F_t = \mu_v P \quad \text{and} \quad F_f = \mu_f P$$

- the lever effect

$$T_t = F_t\, d/2 \quad \text{and} \quad T_f = F_f\, D/2$$

- the wedge effect

$$\mu_v = \mu_t/\cos(\alpha/2)$$

The individual effects in Equation (3.4) are:

- the wedge effect

$$F_r \sim P\tan(-\beta)$$

- the lever effect

$$T_r = F_r\, d/2$$

An examination of the individual physical effects will yield the following solution principles for the improvement of the locking properties of the fastener:

- Use of the wedge effect to reduce the tendency to loosen by decreasing the helix angle β.

- Use of the lever effect to increase the frictional moment on the head or nut face by increasing the mean face diameter D.
- Use of the friction effect to increase the frictional force by increasing the coefficient of friction μ.
- Use of the wedge effect to increase the frictional force on the face by means of conical surfaces ($P\mu_f/\sin\gamma$ with cone angle = 2γ). This method is used with automobile wheel attachment nuts.
- Increase of the flank angle α to increase the virtual coefficient of friction in the thread.

2. Systematic Search with the Help of Classification Schemes

In Section 2.2.5 we showed that the systematic presentation of information and data is helpful in two respects. On the one hand it stimulates the search for further solutions in various directions; on the other hand it facilitates the identification and combination of essential solution characteristics. Because of these advantages, a number of classification schemes have been drawn up, all with a similar basic structure. Dreibholz [3.10] has published a comprehensive survey of the possible applications of such classification schemes.

Classifying criterion for labelling the columns / Classifying criterion for labelling the rows		Column parameters				
		C 1	C 2	C 3	C 4	
Row parameters	R 1					
	R 2					
	R 3					
	R 4					

a

Consecutive numbers / Classifying criterion for labelling the rows		1	2	3	4	
Row parameters	R 1					
	R 2					
	R 3					
	R 4					

b

Figure 3.14. General structure of classification schemes. After [3.10]

The usual two-dimensional scheme consists of rows and columns of parameters used as classifying criteria. Figure 3.14 illustrates the general structure of classification schemes: (a) when parameters are provided for both the rows and the columns; and (b) when parameters are provided for the rows only, because the columns cannot be arranged in any apparent order. If necessary, the classifying criteria can be extended by a further breakdown of the parameters or characteristics (see Figure 3.15); this, however, often tends to confuse the general picture. By allocating the column parameters to the rows it is possible to trans-

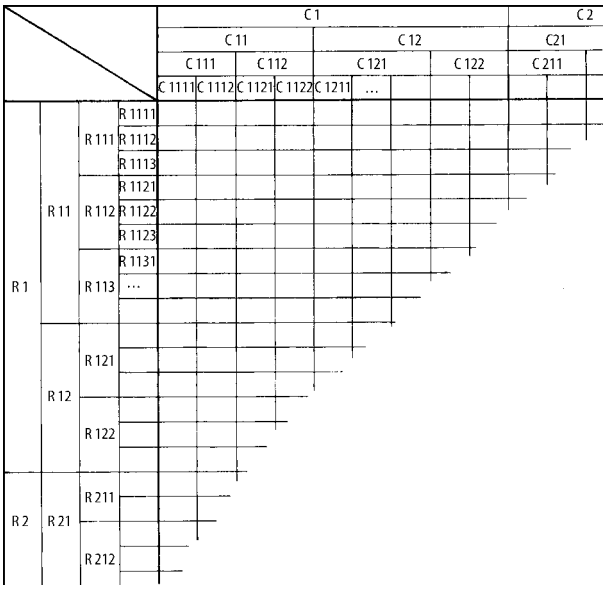


Figure 3.15. Classification scheme with further subdivision of parameters. After [3.10]

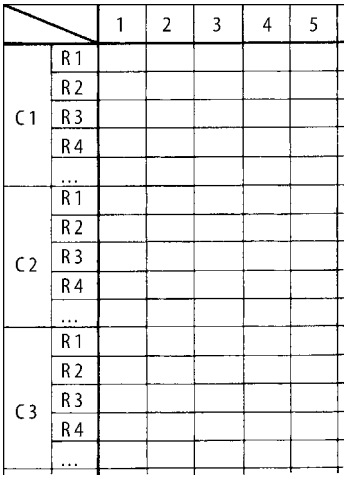


Figure 3.16. Modified classification scheme. After [3.10]

form every classification scheme based on row and column into a scheme in which only the row parameters are retained, and the columns are merely numbered (see Figure 3.16).

Such classification schemes help the design process in a great many ways. In particular, they can serve as design catalogues during all phases of the search for a solution, and they can also help in the combination of subsolutions into overall solutions (see Section 3.2.4). Zwicky [3.77] has referred to them as "morphological matrices".

The choice of classifying criteria or their parameters is of crucial importance. In establishing a classification scheme it is best to use the following step-by-step procedure:

Step 1: Solution proposals are entered in the rows in random order.

Step 2: These proposals are analysed in the light of the main headings (characteristics), such as type of energy, working geometry, working motion, etc.

Step 3: They are classified in accordance with these headings.

The criteria and their parameters can also be obtained from an earlier use of intuitive methods to analyse known solutions or solution ideas.

This procedure not only helps with the identification of compatible combinations but, more importantly, encourages the opening up of the widest possible solution

Classifiing criteria:
Types of energy, physical effects and manifestations

Headings	*Examples*
Mechanical	Gravitation, inertia, centripetal force
Hydraulic	Hydrostatic, hydrodynamic
Pneumatic	Aerostatic, aerodynamic
Electrical	Electrostatic, electrodynamic, inductive, capacitative, piezo-electric, transformation, rectification
Magnetic	Ferromagnetic, electromagnetic
Optical	Reflection, refraction, diffraction, interference, polarisation, infra-red, visible, ultra-violet
Thermal	Expansion, bimetal effect, heat storage, heat transfer, heat conduction, heat insulation
Chemical	Combustion, oxidation, reduction, dissolution, combination, transformation, electrolysis, exothermic and endothermic reactions
Nuclear	Radiation, isotopes, source of energy
Biological	Fermentation, putrefaction, decomposition

Figure 3.17. Classifying criteria and headings (characteristics) for variation in the physical search area

fields. The classifying criteria and characteristics listed in Figures 3.17 and 3.18 can be useful when searching systematically for solutions and the variation of solution ideas for technical systems. They refer to types of energy, physical effects, manifestations, as well as the characteristics of the working geometry, working motions, and the basic material properties (see Section 2.1.4).

Figure 3.19 provides a simple example of searching for a solution to satisfy a subfunction. Here the answer was obtained by varying the type of energy against a number of working principles.

Classifying criteria

Working geometry, working motions and basic material properties

Working geometry

Headings	*Examples*
Type	Point, line, surface, body
Shape	Curve, circle, ellipse, hyperbola, parabola Triangle, square, rectangle, pentagon, hexagon, octagon Cylinder, cone, rhomb, cube, sphere Symmetrical, asymmetrical
Position	Axial, radial, tangential, vertical, horizontal Parallel, sequential
Size	Small, large, narrow, broad, tall, low
Number	Undivided, divided Simple, double, multiple

Working motions

Headings	*Examples*
Type	Stationary, translational, rotational
Nature	Uniform, non-uniform, oscillating Plane or three-dimensional
Direction	In x,y,z direction and/or about x,y,z axis
Magnitude	Velocity
Number	One, several, composite movements

Basic material properties

Headings	*Examples*
State	Solid, liquid, gaseous
Behaviour	Rigid, elastic, plastic, viscous
Form	Solid body, grains, powder, dust

Figure 3.18. Classifying criteria and headings (characteristics) for variation in the form design search area

Type of energy / Working principle	mechanical	hydraulic	electrical	thermal
1	Pot. energy	Liquid reservoir (pot. energy)	Battery	Mass
2	Moving mass (transl.)	Flowing liquid	Capacitor (electr. field)	Heated liquid
3	Flywheel (rot.)			Superheated steam
4	Wheel on inclined plane (rot. + transl. + pot.)			
5	Metal spring	Other springs (compression of fluid + gas)		
6		Hydraulic reservoir a. Bladder b. Piston c. Membrane (Pressure energy)		

Figure 3.19. Different working principles that satisfy the function "store energy" obtained by varying the type of energy

Figure 3.20 is an example of variation based on working motions.

Figure 3.21 shows the variation in the working geometry in the design of shaft–hub connections. Thanks to such arrangements, the multiplicity of solutions obtained, for instance by the method of forward steps (see Section 2.2.5 and Figure 2.21), can be put into order and completed.

To sum up, the following recommendations are given:

- Classification schemes should be built up step-by-step and as comprehensively as possible. Incompatibilities should be discarded, and only the most promising solution proposals pursued. In so doing, designers should try to determine which classifying criteria contribute to the discovery of a solution, and to examine further variations by modifying the parameters.
- The most promising solutions should be chosen and labelled using a special selection procedure (see Section 3.3.1).
- If possible, the most comprehensive classification schemes should be drawn up (those schemes intended for repeated use), but systems should never be built for the sake of systematics alone.

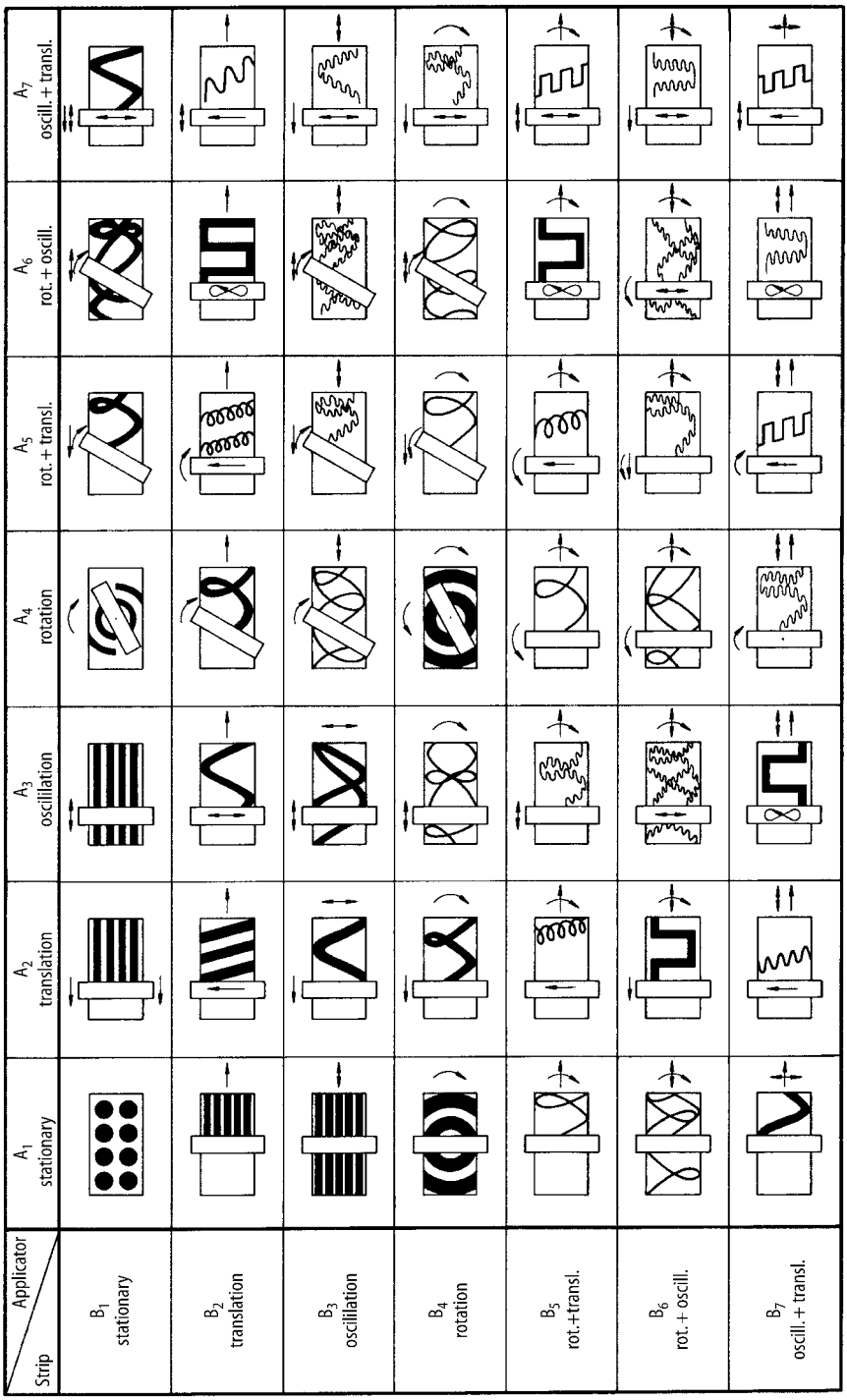


Figure 3.20. Means of coating the backs of carpets by combining the motions of the carpet (strip) and those of the applicator

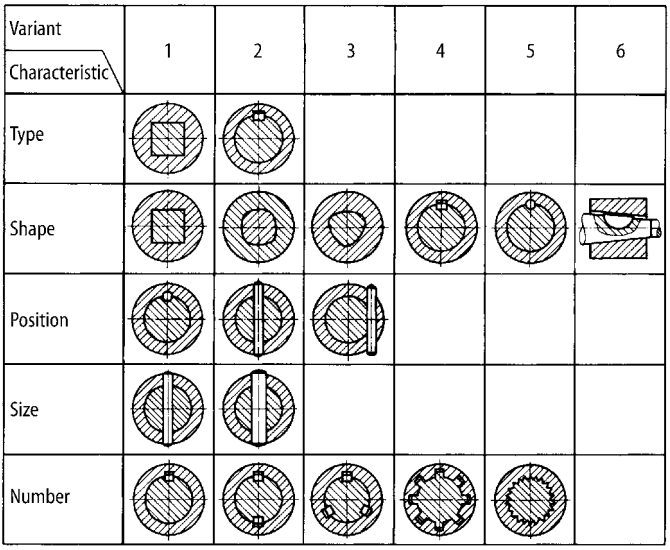


Figure 3.21. Variations in the working geometry for shaft–hub connections

3. Use of Design Catalogues

Design catalogues are collections of known and proven solutions to design problems. They contain data of various types and solutions at distinct levels of embodiment. Thus they may cover physical effects, working principles, principle solutions, machine elements, standard parts, materials, bought-out components, etc. In the past, such data were usually found in textbooks and handbooks, company catalogues, brochures and standards. Some of these contained, apart from purely objective data and suggested solutions, examples of calculations, solution methods and other design procedures. It is also possible to imagine catalogue-like collections for methods and procedures.

Design catalogues should provide:

- quicker, more problem-oriented access to the accumulated solutions or data
- the most comprehensive range of solutions possible, or, at the very least, the most essential ones, which can be extended later
- the greatest possible range of interdisciplinary applications
- data for conventional design procedures as well as for computer-aided methods.

The construction of design catalogues has been studied, above all, by Roth and collaborators [3.62]. Roth suggests that a design catalogue of the type shown in Figure 3.22 is most likely to satisfy all of the demands listed above.

The classifying criteria determine the structure of the catalogue. They influence the ease with which catalogues can be handled and reflect the level of complexity of particular solutions, as well as their degree of embodiment. In the conceptual design phase, for instance, it is advisable to select as classifying criteria the

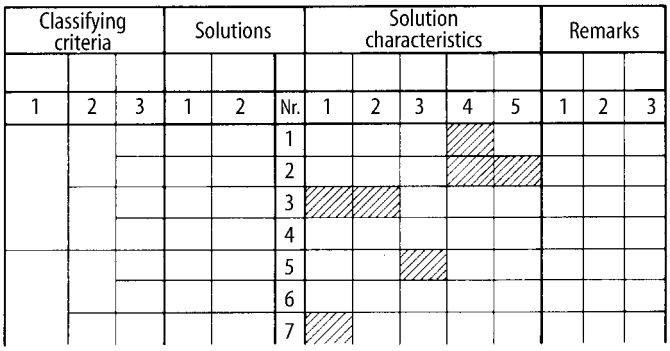


Figure 3.22. Basic structure of a design catalogue. After [3.62]

functions to be fulfilled by the solutions. This is because the conceptual design is based on the underlying subfunctions. When classifying characteristics it is best to choose generally valid functions (see Section 2.1.3), which help to elicit the most product-independent solutions.

Further classifying criteria might include the types and characteristics of energy (mechanical, electrical, optical, etc.), of material or signals, of working geometries, of working motions and of basic material properties. In the case of design catalogues intended for the embodiment design phase, useful classifying criteria include the properties of materials and the characteristics of particular machine elements, such as types of coupling.

The solution column is the main part of the catalogue and contains the solutions. Depending on the level of abstraction, the solutions are represented as sketches, with or without physical equations, or as more or less complete drawings or illustrations. The type and completeness of the information given once again depends on the intended application. It is important that all data is of the same level of abstraction and omits side issues.

The column covering the solution characteristics is important for the choice of solutions.

The remarks column can be used for information about the origin of the data and for additional comments.

The characteristics used for selection may involve a great variety of properties—for instance typical dimensions, reliability, response, number of elements, etc. They help designers in the preliminary selection and evaluation of solutions and, in the case of computer-based catalogues, they can also be used in the final selection and evaluation.

Another important requirement of design catalogues is that they should have uniform and clear definitions and symbols.

The more concrete and detailed the stored information, the more direct but also the more limited the application of a catalogue. With increasing degree of embodiment, data for a given solution become more comprehensive. However, the chances of arriving at a complete solution field decreases because the number of details, for example embodiment variants, increases rapidly. Thus, it may be

Table 3.2. Available design catalogues

Application	Object	Author and reference
General	Construction of catalogues	Roth [3.62]
	List of available catalogues and solutions	Roth [3.62]
Principle solutions	Physical effects	Roth [3.62]
	Solutions to functions	Koller [3.39]
Connections	Types of connections	Roth [3.62]
	Connections	Ewald [3.14]
	Fixed connections	Roth [3.62]
	Wedded joints for steel profiles	Wölse and Kastner [3.75]
	Riveted joints	Roth [3.62], Kopowski [3.41], Grandt [3.26]
	Adhesive joints	Fuhrmann and Hinterwalder [3.18]
	Clamping elements	Ersoy [3.13]
	Principles of threaded joints	Kopowski [3.41]
	Threaded fasteners	Kopowski [3.41]
	Elimination of backlash in threaded joints	Ewald [3.14]
	Elastic joints	Gießner [3.24]
	Shaft–hub connections	Roth [3.62], Diekhöner and Lohkamp [3.9], Kollmann [3.40]
Guides and bearings	Linear guides	Roth [3.62]
	Rotational guides	Roth [3.62]
	Plain and roller bearings	Diekhöner [3.8]
	Bearings and guides	Ewald [3.14]
Power generation, power transmission	Electric motors (small)	Jung and Schneider [3.32]
	Drives (general)	Schneider [3.65]
	Power generators (mechanical)	Ewald [3.14]
	Effects to generate power	Roth [3.62]
	Single-stage power multiplication	Roth [3.62], VDI 2222 [3.70]
	Lifting mechanisms	Raab and Schneider [3.57]
	Screw drives	Kopowski [3.41]
	Friction systems	Roth [3.62]
Kinematics, mechanisms	Solving motion problems using mechanisms	VDI 2727, part 2 [3.72]
	Chain drives and mechanisms	Roth [3.62]
	4-bar mechanisms	VDI 2222, part 2 [3.70]
	Logical inverse mechanisms	Roth [3.62]
	Logical conjunctive and disjunctive mechanisms	Roth [3.62]
	Mechanical flip-flops	Roth [3.62]
	Mechanical non-return safety devices	Roth [3.62], VDI 2222, part 2 [3.70]
	Lifting mechanisms	Raab and Schneider [3.57]
	Uniform-motion transmissions	Roth [3.62]
	Handling devices	VDI 2740 [3.73]
Gearboxes	Spur gears	VDI 2222, part 2 [3.70], Ewald [3.14]
	Mechanical single-stage gearboxes with constant gear ratio	Diekhoner and Lohkamp [3.9]
	Elimination of backlash in spur gears	Ewald [3.14]
Safety technology	Danger situations	Neudorfer [3.52]
	Protective barriers	Neudorfer [3.53]
Ergonomics	Indicators, controls	Neudorfer [3.51]
Production processes	Casting	Ersoy [3.13]
	Drop forging	Roth [3.62]
	Press forging	Roth [3.62]

Function	Input	Output	Physical effects						
E_{mech} → E_{mech}	Force, pressure, torque	Length, angle	Hooke (Tension/ compression/ bending)	Shear, torsion	Upthrust Poisson´s effect	Boyle-Mariotte	Coulomb I and II	...	...
		Speed	Energy Law	Conservation of momentum	Conservation of angular momentum	...	...	...	...
		Accele-ration	Newton´s Law	...	...	...	...	...	...
	Length, angle	Force, pressure, torque	Hooke	Shear, torsion	Gravity	Upthrust	Boyle-Mariotte	Capillary	
			Coulomb I and II	...	...	...	...	...	...
	Speed		Coriolis force	Conservation of momentum	Magnus-effect	Energy law	Centrifugal force	Eddy current	
	Accelera-tion		Newton´s Law	...	...	...	...	...	...
E_{mech} → E_{hyd}	Force, length, speed, pressure	Speed, pressure	Bernoulli	Viscosity (Newton)	Torricelli	Gravitatio-nal pressure	Boyle-Mariotte	Conservation of momentum	...
E_{hyd} → E_{mech}	Speed	Force, length	Profile lift	Turbulence	Magnus-effect	Flow resistance	Back pressure	Reaction principle	...
E_{mech} → E_{therm}	Force, speed	Temperature, quantity of heat	Friction (Coulomb)	1st law	Thomson-Joule	Hysteresis (damping)	Plastic deformation	...	...
E_{therm} → E_{mech}	Temperature, heat	Force, pressure, lenght	Thermal expansion	Steam pressure	Gas Law	Osmotic pressure	...	...	...
E_{elektr} → E_{mech}	Voltage, current, magn. field	Force, speed, pressure	Biot-Savart-effect	Electro-kinetic effect	Coulomb I	Capacitance effect	Johnsen-Rhabeck-effect	Piezoeffect	...
E_{mech} → E_{elektr}	Force, length, speed, pressure	Voltage, current	Induction	Electro-kinetics	Electro-dynamic effect	Piezoeffect	Frictional electricity	Capacitance effect	...
E_{elektr} → E_{therm}	Voltage, current	Temperature, heat	Joule heating	Peltier-effect	Electric arc	Eddy current	...	...	...
E_{therm} → E_{elektr}	Temperature, heat	Voltage, current	Electr. conduction	Thermo-effect	Thermionic emission	Pyroelectri-city	Noise-effect	Semiconductor, Super-conductor	...
E_{mech} ↔ E_{mech}	Force, length, pressure, speed	Force, length, pressure, speed	Lever	Wedge	Poisson´s effect	Friction	Crank	Hydraulic effect	...
E_{hyd} ↔ E_{hyd}	Pressure, speed	Pressure, speed	Continuity	Bernoulli	...	...	...	...	...
E_{therm} ↔ E_{therm}	Temperature, heat	Temperature, heat	Heat conduction	Convection	Radiation	Condensa-tion	Evaporation	Freezing	...
E_{elektr} ↔ E_{elektr}	Voltage, current	Voltage, current	Transformer	Valve	Transistor	Transducer	Thermogal-vanometer	Ohm´s law	...
...	...	...	...	...	...	...	...	...	...

Figure 3.23. Design catalogue of physical effects based on [3.39, 3.48] for the generally applicable functions "change energy" and "vary energy component". Also applicable to flow of signals

Classifying criteria		Solutions				Solution characteristics															Remarks		
Type of interface	Type of force transmission	Equation	Name	Configuration		Transferable torque	Torque transmission depending on		Axial forces	Stress concentration	Applicable for	Behaviour at overload	Centering possible	Unbalanced force	Axial displacement of hub	Hub moveable	Joint adjustable	Shaft diameter (mm)	Material	Manufacturing effort	Assembly effort	Standard (DIN)	Application examples	Remarks
1	2	1	2	3	Nr.	1	2		3		4	5	6	7	8	9	10	11	12	13	14	15	16	17
Normal (form fit)	Direct	$T \le K \cdot \frac{d}{2} \cdot A_s \cdot \tau_{max}$ or $T \le K \cdot \frac{d}{2} \cdot A_p \sigma_{max}$	spline shaft		1	large	h i l	shaft diameter, (form) factor, centering factor, material	—	large	pulsating or alternating	fracture	yes	no	clearance fit	yes	no	10 – 150	shaft: 37 Cr 4 41 Cr 4 42 CrMo	high	small	5461/63 5471/72	toothed wheels	exterior, flank, interior centering possible
			involute spline shaft		2				—													5480, 5482		short hub possible
			serrated shaft		3				—									150 – 500				5481		
			3-sided polygon-shaft		4		e i l		—	medium	pulsating or alternating		self-centering	no	clearance fit without load		yes for taper	10 – 100		small, special machinery necessary		—		used for short and thin hubs, conical shaft end possible, broaching or grinding necessary
			4-sided polygon-shaft		5				—									10 – 100				—		
	Indirect		transverse pin	d_{st} d_m D	6	small	d_p D		are supported	large	—		yes	yes	—	no	yes for taper pin	0.5 – 50	pin: 4D, 5S 6S, 8G 9S, 20K St 50 K St 70 St 60	medium	medium	1,7 1470 – 77 1481, 6324, 7346	power stretcher, machine tools, vehicles	taper and grooved pin possible
			tangential pin		7						—				—									
			in line pin		8		d_p l		—	medium	—				clearance fit						small			
			key joint		9		h l b		—	large	—						no	5 – 500	spring St 60 shaft, hub: GG, GS ST			6885		
			Woodruff key		10				—		—											6888		

Figure 3.24. Extract of a catalogue for shaft–hub connections. After [3.62]

possible to provide a full list of physical effects fulfilling the function "channel", but it would hardly be possible to list all of the potential embodiments of bearings (channelling a force from a rotating to a stationary system).

Table 3.2 lists the currently available design catalogues that satisfy the requirements and structure described above. Therefore, in what follows we include just a few examples of, or extracts from, available design catalogues.

Figure 3.23 shows a catalogue of physical effects associated with the functions "change energy" and "vary energy component". It is based on Koller [3.39] and Krumhauer [3.48]. The catalogue makes it possible to derive these effects from the classifying criteria, that is, the "inputs and outputs" of the functions. The characteristics on which the selection is based must be derived from the technical literature.

Figure 3.24 shows an extract of a catalogue for shaft–hub connections based on [3.62]. In this, unlike the previous catalogue, the solutions are concrete enough, thanks to the specification of the form design features, for the embodiment design phase to start with a scale layout drawing.

Computer-based systems are used to facilitate searching through catalogues, company brochures, supplier information and other documents. Hypermedia software provides a way of structuring, storing and retrieving the contents of such documents. It allows the flexible manipulation of chunks of information, and the representation and linking of objects and procedures in a specific knowledge domain, using different representation principles. This is called navigating in a hypermedia system [3.58]. To use distributed sources of information, a global network is required, such as the internet (www). Using the internet, so-called "virtual markets" or "virtual supply chains" can be created with which designers can communicate from their work places [3.4].

3.2.4 Methods for Combining Solutions

As described in Sections 2.1.3 and 2.2.5, it is often useful to divide problems, tasks and functions into subproblems, subtasks and subfunctions and to solve these individually (factorisation method) (see also Section 6.3). Once the solutions for subproblems, subtasks or subfunctions are available, they have to be combined in order to arrive at an overall solution.

The methods we have been describing, particularly those with an intuitive bias, may have led to the discovery of suitable combinations. However, there are special methods for arriving at such syntheses more directly. In principle, they must permit a clear combination of solution principles with the help of the associated physical and other quantities and the appropriate geometrical and material characteristics. When analysing combinations that involve software elements, it is important to identify and use appropriate solution characteristics.

The main problem with such combinations is ensuring the physical and geometrical compatibility of the solution principles to be combined, which in turn ensures the smooth flow of energy, material and signals, and avoids geometrical interference in mechanical systems. For information systems, the main problem is the compatibility requirements of the information flow.

A further problem is the selection of technically and economically favourable combinations of principles from the large field of theoretically possible combinations. This aspect will be discussed at greater length in Section 3.3.1.

1. Systematic Combination

For the purpose of systematic combination, the classification scheme to which Zwicky [3.77] refers as the "morphological matrix" (see Figure 3.25) is particularly useful. Here, the subfunctions, usually limited to the main functions, and appropriate solutions (solution principles) are entered in the rows of the scheme.

If this scheme is to be used for the elaboration of overall solutions, then at least one solution principle must be chosen for every subfunction (that is, for every row). To provide the overall solution, these principles (subsolutions) must then be combined systematically into an overall solution. If there are m_1 solution principles for the subfunction F_1, m_2 for the subfunction F_2, and so on, then after complete combination we have $N = m_1 \cdot m_2 \cdot m_3 \cdots m_n$ theoretically possible overall solution variants.

The main problem with this method of combination is to decide which solution principles are compatible; that is, to narrow down the theoretically possible search field to the practically possible search field.

The identification of compatible subsolutions is facilitated if:

- the subfunctions are listed in the order in which they occur in the function structure, if necessary separated according to flow of energy, material and signals
- the solution principles are suitably arranged with the help of additional column parameters, for example the types of energy
- the solution principles are not merely expressed in words but also in rough sketches

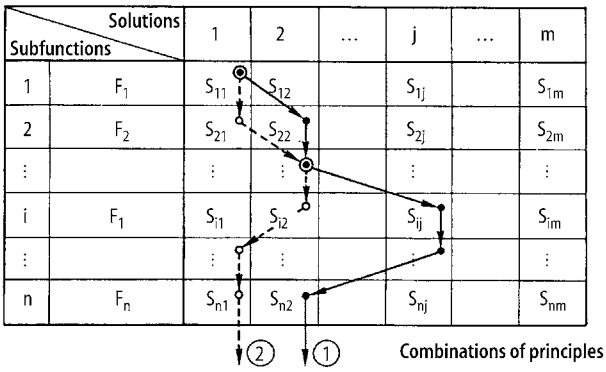


Figure 3.25. Combining solution principles into combinations of principles: Combination 1: $S_{11} + S_{22} + \ldots S_{n2}$; Combination 2: $S_{11} + S_{21} \ldots S_{n1}$

- the most important characteristics and properties of the solution principles are recorded as well.

The verification of compatibilities, too, is facilitated by classification schemes. If two subfunctions to be combined—for instance, "change energy" and "vary mechanical energy component"—are entered respectively in the column and row headings of a matrix with their characteristics in the appropriate cells, then the compatibility of the subsolutions can be verified more easily than it could be if such examinations were to be confined to the designer's head. Figure 3.26 illustrates this type of compatibility matrix. Further examples of this method of combination will be found in Section 6.4.2 (Figures 6.15 and 6.19).

To sum up:

- Combine only compatible subfunctions.
- Pursue only such solutions as meet the demands of the requirements list and fall within the available resources (see selection procedures in Section 3.3.1).
- Concentrate on promising combinations and establish why these should be preferred above the rest.

In conclusion, it must be emphasised that what we have been discussing is a generally valid method of combining subsolutions into overall solutions. The method can be used for the combination of working principles during the conceptual phase, and of subsolutions or even of components and assemblies during the embodiment phase. Because it is essentially a method of information processing, it

Vary mechan. energy component \ Change energy		Electric motor	Oscillating solenoid	Bimetal spiral in hot water	Oscillating hydraulic piston	...
		1	2	3	4	...
Four-bar linkage	A	if A capable of rotating	slow motion	yes	additional lever linkage but only for low piston speeds	...
Chain drive Spur gear drive	B	yes	slow rotation only through additional elements (free wheeling etc.), difficult to reverse direction	gear segments suffice, depending on angle of rotation	with a rack and swivel, but only for low piston speeds	...
Maltese drive	C	yes look out for shock loads	see B2	yes (when angle of rotation is small lever with sliding block)	lever with sliding block, but only for low piston speeds	...
Friction wheel drive	D	yes	see B2	large forces because of torque during slow movement, imprecise positioning	see D3	...
...	...	...	...	...	...	...

☒ very difficult to apply (do not pursue further)

⊠ can only be applied under certain circumstances (defer)

Figure 3.26. Compatibility matrix for combination possibilities of the subfunctions "change energy" and "vary mechanical energy component". After [3.10]

is not confined to technical problems but can also be used in the development of management systems and in other fields.

2. Combining With the Help of Mathematical Methods

Mathematical methods and computers should only be used for the combination of solution principles if real advantages can be expected from them. Thus, at the relatively abstract conceptual phase, when the nature of the solution is not yet fully understood, a quantitative elaboration—that is, a mathematical combination along with an optimisation—is quite out of place and can be misleading. The exceptions are combinations of known elements and assemblies, for instance in variant or circuit design. In the case of purely logical functions, combinations can be performed with the help of Boolean algebra [3.17, 3.59] in, say, the layout of safety systems or the optimisation of electronic or hydraulic circuits.

In principle, the combination of subsolutions into overall solutions with the help of mathematical methods calls for knowledge of the characteristics or properties of the subsolutions that are expected to correspond with the relevant properties of the neighbouring subsolutions. These properties must be unambiguous and quantifiable. In the formation of principle solutions (for example working structures), data about the physical relationships may be insufficient, since the geometrical relationships may have a limiting effect and hence may, in certain circumstances, lead to incompatibilities. In that case, physical equations and geometrical structure must first be matched mathematically, and this is not generally possible except for systems of low complexity. For systems of higher complexity, in contrast, such correlations often become ambiguous, so that designers must once again choose between variants. We may, accordingly, speak of dialogue systems in which the process of combination consists of mathematical and creative steps.

This makes it clear that, with increasing physical realisation or embodiment of a solution, it becomes simpler to establish quantitative combination rules. However, the number of properties increases and with them the number of constraints and optimisation criteria, so that the mathematical effort becomes very great and requires computer support.

3.3 Selection and Evaluation Methods

3.3.1 Selecting Solution Variants

For the systematic approach, the solution field should be as wide as possible. By considering all possible classifying criteria and characteristics, designers are often led to a larger number of possible solutions. This profusion constitutes the strength and also the weakness of the systematic approach. The very great theoretically admissible, but practically unattainable, number of solutions must be reduced at the earliest possible moment. On the other hand, care must be taken not to eliminate valuable working principles, because it is often only through their combination

with others that an advantageous working structure will emerge. While there is no absolutely safe procedure, the use of a systematic and verifiable selection procedure greatly facilitates the choice of promising solutions from a wealth of proposals [3.55].

This selection procedure involves two steps, namely *elimination* and *preference*.

First, all totally unsuitable proposals are eliminated. If too many possible solutions still remain, those that are patently better than the rest must be given preference. Only these solutions are evaluated at the end of the conceptual design phase.

If faced with a large number of solution proposals, the designer should compile a selection chart (see Figure 3.27). In principle, after every step—that is, even after establishing function structures—the only solution proposals pursued should:

- be compatible with the overall task and with one another (Criterion A)
- fulfil the demands of the requirements list (Criterion B)
- be realisable in respect of performance, layout, etc. (Criterion C)
- be expected to be within permissible costs (Criterion D).

Unsuitable solutions are eliminated in accordance with these four criteria applied in the above sequence. Criteria A and B are suitable for yes/no decisions and their application poses relatively few problems. Criteria C and D often need a more quantitative approach, which should only be used once Criteria A and B have been satisfied.

Since Criteria C and D involve quantitative considerations, they may lead not only to the elimination of proposed solutions that fail to meet the requirements, but also of those that exceed the requirements by an unnecessary margin.

A preference is justified if, among the very large number of possible solutions, there are some that:

- incorporate direct safety measures or introduce favourable ergonomic conditions (Criterion E)
- are preferred by the designer's company; that is, can be readily developed with the usual know-how, materials, procedures and under favourable patent conditions (Criterion F).

Additional selection criteria can be used if they help decisions to be made.

It must be stressed that selection based on preferential criteria is only advisable when there are so many variants that a full evaluation would involve too much time and effort.

If, in the suggested sequence, one criterion leads to the elimination of a proposal, then the other criteria need not be applied to it there and then. At first, only the solution variants that satisfy all of the criteria should be pursued. Sometimes, however, it is impossible to settle the issue because of lack of information. In the case of promising variants that satisfy Criteria A and B, the gap will have to be filled by a reevaluation of the proposal, which will ensure that no good solutions are passed over.

TH Darmstadt

SELECTION CHART

for *Fuel gauge*

Page: *1*

Solution variants (Sv) evaluated by

SELECTION CRITERIA

(+) Yes
(–) No
(?) Lack of information
(!) Check requirements list

DECISION

Mark solution variants (Sv)

(+) Pursue solution
(–) Eliminate solution
(?) Collect information (re-evaluate solution)
(!) Check requirements list for changes

Enter solution variant (Sv):

Sv		A Compatibility assured	B Fulfils demands of requirements list	C Realisable in principle	D Within permissible costs	E Incorporates direct safety measures	F Preferred by designer's company	G Adequate information	Remarks (Indications, Reasons)	DECISION
1	1	+	+	+	?				Number of measuring positions	?
2	2	+	–						Storing the mass	–
3	3	–							Radioactivity	–
4	4	+	+	+	+		(+)		(Further development of existing solutions)	+
5	5	+	+	+	+					+
6	6	–							Fluid not conducting	–
7	7	+	+	+	+					+
8	8	+	+	+	+				see Sv 7	+
	9									
	10									
	11									
	12									
	13									
	14									
	15									
	16									
	17									
	18									
	19									
	20									

Date: *9.85* Initials: *la*

Figure 3.27. Systematic selection chart: 1, 2, 3, etc. are solution variants of the proposals made in Table 3.3. The column reserved for remarks lists reasons for lack of information or elimination

Table 3.3. Extract from a list of solutions for a fuel gauge

No.	Solution principle	Signal
	1. Measuring the quantity of fluid	
	1.1. Mechanical, static	
1.	Fix container at three points. Measure vertical forces (weight). (Measuring at one support may be sufficient)	Force
2.	Mutual attraction. The force is proportional to the masses and therefore to the fluid mass	Force
	1.2. Atomic	
3.	Distribution of radioactive material in the fluid	Concentration of radiation intensity
	2. Measuring the fluid level	
	2.1. Mechanical, static	
4.	Float with or without lever effect. Lever output: linear or angular displacement Potentiometer resistance represents fluid level within the container	Displacement
	2.2. Electrical	
5.	Resistance wire: hot in air, cold in fluid. Level of fluid determines: overall resistance, volume (dependent on temperature and length of wire)	Ohmic resistance
6.	Fluid as ohmic resistance (level-dependent). Changing the level of the (conducting) fluid changes the resistance	Ohmic resistance
	2.3. Optical	
7.	Photocells in the container. Fluid covers a certain number of photocells. The number of light signals is a measure of the fluid level	Light signal (discrete)
8.	Light transmission or light reflection. Transmission in the presence of fluid. Total reflection in presence of air	Light signal (discrete)

The criteria are listed in the order shown above as a labour-saving device, and not in order of importance.

The selection procedure has been systematised for easier implementation and verification (see Figure 3.27). Here, the criteria are applied in sequence and the reasons for eliminating any solution proposal is recorded. Experience has shown that the selection procedure we have described can be applied very quickly, that it gives a good picture of the reasons for selection, and that it provides suitable documentation in the form of a selection chart.

If the number of solution proposals is small, elimination may be based on the same criteria, but less formally recorded.

The example we have chosen concerns solution proposals for a fuel gauge in accordance with the requirements in Figure 6.4. An extract from the list of proposals is given in Table 3.3.

Further examples of selection charts can be found in Section 6.4.3 (see Figure 6.17) and Section 6.6.2 (see Figure 6.48).

3.3.2 Evaluating Solution Variants

The promising solutions that result from the selection procedure usually have to be firmed up before a final evaluation is made using criteria that are more detailed and possibly quantified. This evaluation involves an assessment of technical,

safety, environmental and economic values. For this purpose, evaluation procedures have been developed that can be used to evaluate technical and nontechnical systems, and that can be applied in all phases of product development. Evaluation procedures are by their very nature more elaborate than selection procedures (see Section 3.3.1) and are therefore only applied at the end of the main working steps to determine the current value of a solution. This occurs, in general, when preparing for a fundamental decision concerning the direction of a solution path, or at the end of the conceptual and embodiment phases [3.61].

1. Basic Principles

An evaluation is meant to determine the "value", "usefulness" or "strength" of a solution with respect to a given objective. An objective is indispensable since the value of a solution is not absolute, but must be gauged in terms of certain requirements. An evaluation involves a comparison of concept variants or, in the case of a comparison with an imaginary ideal solution, a "rating" or degree of approximation to that ideal.

The evaluation should not be based on individual aspects such as production cost, safety, ergonomics or environment, but should, in accordance with the overall aim (see Section 2.1.7), consider all aspects in an appropriate balance.

Hence there is a need for methods that allow a more comprehensive evaluation, or in other words cover a broad spectrum of objectives (task-specific requirements and general constraints). These methods are intended to elaborate not only the quantitative but also the qualitative properties of the variants, thus making it possible to apply them during the conceptual phase, with its low level of embodiment and correspondingly low state of information. The results must be reliable, cost-effective, easily understood and reproducible. The most important methods to date are Cost–Benefit Analysis based on the systems approach [3.76], and the combined technical and economic evaluation technique specified in Guideline VDI 2225 [3.71], which essentially originates from Kesselring [3.36].

In what follows, we shall outline a basic evaluation procedure incorporating the concepts of Cost–Benefit Analysis and of Guideline VDI 2225. At the end the similarities and differences between both methods will be discussed.

Identifying Evaluation Criteria

The first step in any evaluation is the drawing up of a set of objectives from which evaluation criteria can be derived. In the technical field, such objectives are mainly derived from the requirements list and from general constraints (see guidelines in Section 2.1.7), which are identified while working on a particular solution.

A set of objectives usually comprises several elements that not only introduce a variety of technical, economic and safety factors, but that also differ greatly in importance.

A range of objectives should satisfy as far as possible the following conditions:

- The objectives must cover the decision-relevant requirements and general constraints as completely as possible, so that no essential criteria are ignored.

- The individual objectives on which the evaluation must be based should be as independent of one another as possible; that is, provisions to increase the value of one variant with respect to one objective must not influence its values with respect to the other objectives.
- The properties of the system to be evaluated must, if possible, be expressed in concrete quantitative or at least qualitative (verbal) terms.

The tabulation of such objectives depends very much on the purpose of the particular evaluation, that is, on the design phase and the relative novelty of the product.

Evaluation criteria can be derived directly from the objectives. Because of the subsequent assignment of values, all criteria must be given a positive formulation, i.e. such that a higher value indicates better, for example:

- "low noise" not "loudness level"
- "high efficiency" not "magnitude of losses"
- "low maintenance" not "maintenance requirements".

Cost–Benefit Analysis systematises this step by means of an objectives tree, in which the individual objectives are arranged in hierarchical order. The subobjectives are arranged vertically into levels of decreasing complexity, and horizontally into objective areas—for instance, technical, economic—or even into major and minor objectives (see Figure 3.28). Because of their required independence, subobjectives of a higher level may only be connected with an objective of the next lowest level. This hierarchical order helps the designer to determine whether or not all decision-relevant subobjectives have been covered. Moreover, it simplifies the assessment of the relative importance of the subobjectives. The evaluation criteria (called objective criteria in Cost–Benefit Analysis) can then be derived from the subobjectives of the stage with the lowest complexity.

Guideline VDI 2225, on the contrary, introduces no hierarchical order for the evaluation criteria, but derives a list of them from minimum demands and wishes and also from general technical properties.

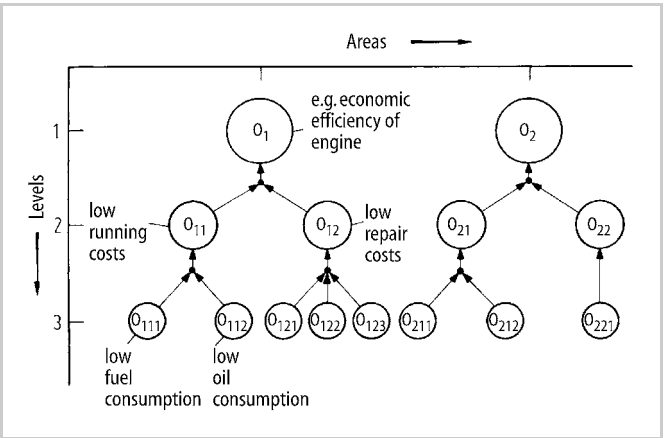


Figure 3.28. Structure of an objectives tree

Weighting Evaluation Criteria

To establish evaluation criteria, we must first assess their relative contribution (weighting) to the overall value of the solution, so that relatively unimportant criteria can be eliminated before the evaluation proper begins. The evaluation criteria retained are given "weighting factors" which must be taken into consideration during the subsequent evaluation step. A weighting factor is a real, positive number. It indicates the relative importance of a particular evaluation criterion (objective).

It has been suggested that such weightings should be assigned to the wishes when they are recorded in the requirements list [3.62, 3.63], but that is only possible if such wishes can be ranked in order of importance when the requirements list is first drawn up. That, however, rarely happens at this early stage—experience has shown that many evaluation criteria emerge during the development of the solution, and that their relative importance changes. It is nevertheless most helpful to include rough estimates of the importance of wishes when drawing up the requirements list, because, as a rule, all the persons concerned are available at that time (see Section 5.2.2).

In Cost–Benefit Analysis, weightings are based on factors ranging from 0 to 1 (or from 0 to 100). The sum of the factors of all evaluation criteria (subobjectives at the lowest level) must be equal to 1 (or 100) so that a percentage weighting can be attached to all of the subobjectives. The drawing up of an objectives tree greatly facilitates this process.

Figure 3.29 illustrates the procedure. Here the objectives have been set out on four levels of decreasing complexity and provided with weighting factors. The evaluation proceeds step-by-step from a level of higher complexity to the next lowest level. Thus the three subobjectives O_{11}, O_{12} and O_{13} of the second level are first weighted with respect to the objective O_1. In this particular case the

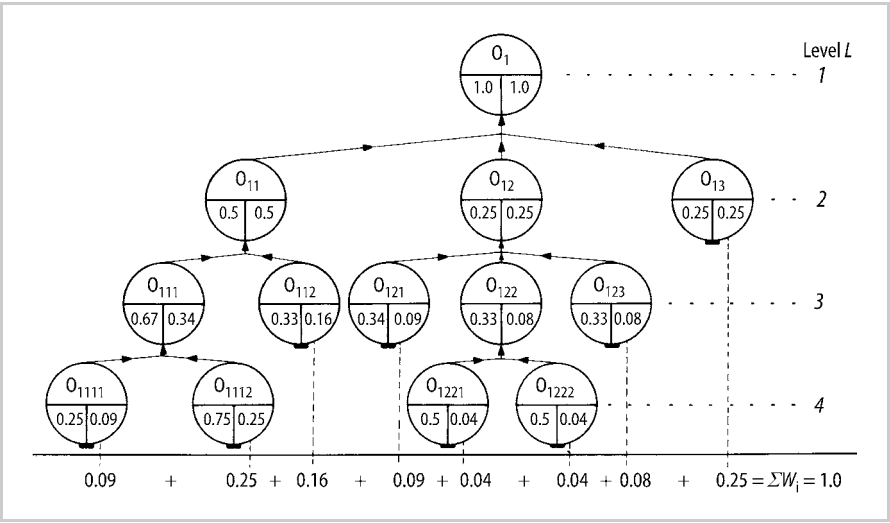


Figure 3.29. Objectives tree with weighting factors. After [3.76]

weightings are 0.5, 0.25 and 0.25. The sum of the weighting factors for any one level must always be equal to $\sum w_i = 1.0$. Next comes the weighting of the objectives of the third level with respect to the subobjectives of the second level. Thus the relative weights of O_{111} and O_{112} with respect to the higher objectives O_{11} were fixed at 0.67 and 0.33. The remaining objectives are treated in similar fashion. The relative weighting of an objective at a particular level with respect to the objective O_1 is found by multiplication of the weighting factor of the given objective level by the weighting factors of the higher objective levels. Thus the subobjective O_{1111}, which has a weighting of 0.25 with respect to the subobjective O_{111} of the next higher level, has a weighting of $0.25 \times 0.67 \times 0.5 \times 1 = 0.09$ with respect to O_1.

Such step-by-step weighting generally produces a realistic ranking because it is much easier to weight two or three subobjectives with respect to an objective on a higher level than to confine the weighting to one particular level only, especially the lowest. Figure 6.33 gives a concrete example of the recommended procedure.

Guideline VDI 2225 tries to dispense with weightings and relies instead on evaluation criteria of approximately equal importance. Weighting factors (2×, 3×) are, however, used for pronounced differences. Kesselring [3.36], Lowka [3.50] and Stahl [3.68] have examined the influences of such weighting factors on the overall value of the solution. Their conclusion was that they exert a significant influence whenever the variants to be evaluated have very distinct properties, and whenever the corresponding evaluation criteria have great importance.

Compiling Parameters

The setting up of evaluation criteria and the determination of their importance is followed, in the next step, by the assignment to them of known (or analytically determined) parameters. These parameters should either be quantifiable or, if that is impossible, be expressed by statements framed as concretely as possible. It has proved very useful to assign such parameters to the evaluation criteria in an evaluation chart before proceeding to the actual evaluation. Figure 3.30 shows an example of such a chart for an internal combustion engine, with appropriate magnitudes entered in the relevant variant columns. The reader will see that the verbal formulation of the evaluation criteria strongly resembles that of the parameters.

In Cost–Benefit Analysis these parameters are referred to as objective parameters (objective criteria) that are compiled with evaluation criteria in a chart. A concrete example is given in Figure 6.55.

In Guideline VDI 2225, in contrast, evaluation follows immediately upon the setting up of evaluation criteria (see Figure 6.41).

Assessing Values

The next step is the assessment of values and hence the actual evaluation. These "values" derive from a consideration of the relative scale of the previously determined parameters, and are thus more or less subjective in character.

The values are expressed by points. Cost–Benefit Analysis employs a range from 0 to 10; Guideline VDI 2225 a range from 0 to 4 (see Figure 3.31). The advantage of the wider range is that, as experience has shown, classification and evaluation

Evaluation criteria			Objective Parameters		Variant V_1 (e.g. Eng.$_1$)			Variant V_2 (e.g. Eng.$_2$)			…	Variant V_j			…	Variant V_m		
No.		Wt.		Unit	Magn. m_{i1}	Value v_{i1}	Weighted value WV_{i1}	Magn. m_{i2}	Value v_{i2}	Weighted value WV_{i2}		Magn. m_{ij}	Value v_{ij}	Weighted value WV_{ij}		Magn. m_{im}	Value v_{im}	Weighted value WV_{im}
1	Low fuel consumption	0.3	Fuel consumption	$\frac{g}{kWh}$	240			300			…	m_{1j}			…	m_{1m}		
2	Light weight construction	0.15	Mass per unit power	$\frac{kg}{kW}$	1.7			2.7			…	m_{2j}			…	m_{2m}		
3	Simple production	0.1	Simplicity of components	–	low			average			…	m_{3j}			…	m_{3m}		
4	Long service life	0.2	Service life	km	80 000			150 000			…	m_{4j}			…	m_{4m}		
⋮	⋮	⋮	⋮	⋮	⋮			⋮				⋮				⋮		
i		W_i			m_{i1}			m_{i2}			…	m_{ij}			…	m_{im}		
⋮	⋮	⋮	⋮	⋮	⋮			⋮				⋮				⋮		
n		W_n			m_{n1}			m_{n2}			…	m_{nj}			…	m_{nm}		
		$\sum_{i=1}^{n} W_i = 1$																

Figure 3.30. Correlation of evaluation criteria and parameters in an evaluation chart

are greatly facilitated by the use of a decimal system that reflects percentages. The advantage of the smaller range is that, in dealing with what are so often no more than inadequately known characteristics of the variants, rough evaluations are sufficient and, indeed, may be the only meaningful approach. They involve the following assessments:

- far below average
- below average
- average
- above average
- far above average.

It is useful to begin with a search for variants with extremely good and bad qualities for a particular criterion and to assign appropriate points to them. Points 0 and 4 (or 10) should only be awarded if the characteristics are really extreme, that is, unsatisfactory or very good (ideal). Once these extreme points have been assigned, the remaining variants are relatively easy to fit in.

Before points can be assigned to the parameters of the variants, the evaluator must at least be clear about the assessment range and the shape of the so-called "value function" (see Figure 3.32). A value function connects values and parameter

Value scale			
Use-value analysis		Guideline VDI 2225	
Pts.	Meaning	Pts.	Meaning
0	absolutely useless solution	0	unsatisfactory
1	very inadequate solution		
2	weak solution	1	just tolerable
3	tolerable solution		
4	adequate solution	2	adequate
5	satisfactory solution		
6	good solution with few drawbacks	3	good
7	good solution		
8	very good solution	4	very good (ideal)
9	solution exceeding the requirement		
10	ideal solution		

Figure 3.31. Points awarded in use-value analysis and guideline VDI 2225

magnitudes, and its characteristic shape is determined either with the help of the known mathematical relationship between the value and the parameter or, more frequently, by means of estimates [3.28].

It is useful to draw up a chart in which the parameter magnitudes are correlated step-by-step with the value scale. Figure 3.33 shows such a scheme, incorporating the point systems of Cost–Benefit Analysis and VDI 2225.

All in all, therefore, the assignment of a value, the selection of a value function and the setting up of an assessment scheme may involve strong subjective influences. Cases with a clear, or even experimentally verified, correlation between the values and the parameters are few and far between. One such exception is the evaluation of machine noise, where the correlation between the value (that is, the protection of the human ear) and the parameter (noise level in dB) is clearly defined by ergonomics.

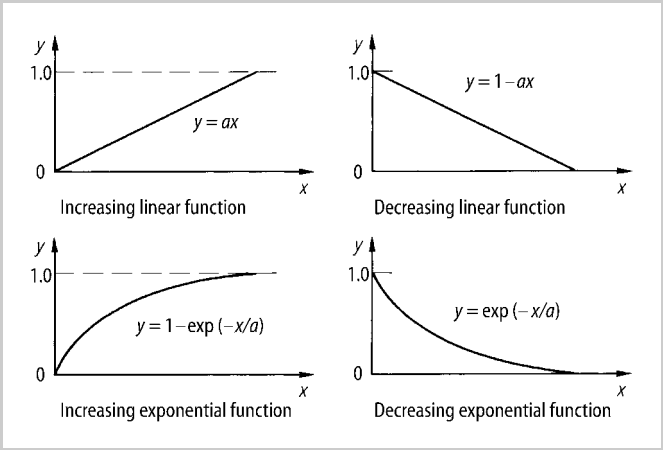


Figure 3.32. Common value functions, after [3.76]; $x \triangleq m_{ij}, y \triangleq v_{ij}$

Value Scale		Parameter magnitudes			
Use-value analysis Pts	VDI 2225 Pts	Fuel consumption g/kWh	Mass per unit power kg/kW	Simplicity of components	Service life
0	0	400	3.5	extremely complicated	$20 \cdot 10^3$
1		380	3.3		30
2	1	360	3.1	complicated	40
3		340	2.9		60
4	2	320	2.7	average	80
5		300	2.5		100
6	3	280	2.3	simple	120
7		260	2.1		140
8	4	240	1.9	extremely simple	200
9		220	1.7		300
10		200	1.5		$500 \cdot 10^3$

Figure 3.33. Chart correlating parameter magnitudes with value scales

Evaluation criteria			Parameters		Variant V_1 (e.g. *Eng.*$_1$)			Variant V_2 (e.g. *Eng.*$_2$)			...	Variant V_j			...	Variant V_m		
No.		Wt.		Unit	Magn. m_{i1}	Value v_{i1}	Weighted value WV_{i1}	Magn. m_{i2}	Value v_{ij}	Weighted value WV_{ij}		Magn. m_{ij}	Value v_{ij}	Weighted value WV_{ij}		Magn. m_{im}	Value V_{im}	Weighted value WV_{im}
1	Low fuel consumption	0.3	Fuel consumption	$\frac{g}{kWh}$	240	8	2.4	300	5	1,5	...	m_{1j}	V_{1j}	WV_{1j}	...	m_{1m}	V_{1m}	WV_{1m}
2	Lightweight construction	0.15	Mass per unit power	$\frac{kg}{kW}$	1.7	9	1.35	2.7	4	0,6	...	m_{2j}	V_{2j}	WV_{2j}	...	m_{2m}	V_{2m}	WV_{2m}
3	Simple production	0.1	Simplicity of components	-	compli-cated	2	0.2	average	5	0,5	...	m_{3j}	V_{3j}	WV_{3j}	...	m_{3m}	V_{3m}	WV_{3m}
4	Long service life	0.2	Service life	km	80 000	4	0.8	150 000	7	1,4	...	m_{4j}	V_{4j}	WV_{4j}	...	$m^4{}_m$	V_{4m}	WV_{4m}
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮		⋮	⋮	⋮		⋮	⋮	⋮
i		W_i			m_{i1}	V_{i1}	WV_{i1}	m_{i2}	V_{i2}	WV_{i2}	...	m_{ij}	V_{ij}	WV_{ij}	...	m_{im}	V_{im}	WV_{im}
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮		⋮	⋮	⋮		⋮	⋮	⋮
n		W_n			m_{n1}	V_{n1}	WV_{n1}	m_{n2}	V_{n2}	WV_{n2}	...	m_{nj}	V_{nj}	WV_{nj}	...	m_{nm}	V_{nm}	WV_{nm}
		$\sum_{i=1}^{n} W_i = 1$				OV_1 R_1	OWV_1 WR_1		OV_2 R_2	OWV_2 WR_2			OV_j R_j	OWV_j WR_j			OV_m R_m	OWV_m WR_m

Figure 3.34. Completed evaluation chart with values (see Figure 3.30)

The values v_{ij} of every solution variant established in respect to every evaluation criterion are added to the list shown in Figure 3.30 in order to produce Figure 3.34.

Whenever the evaluation criteria have a different importance to the overall value of a solution, the weighting factors determined during the second step must also be taken into consideration. To that end, subvalue v_{ij} is multiplied by the weighting factor w_i ($wv_{ij} = w_i \cdot v_{ij}$). Figure 6.55 gives a practical example. The Cost–Benefit Analysis refers to the unweighted values as objective values and the weighted ones as benefit values.

Determining Overall Value

When the subvalues for every variant have been determined, the overall value must now be calculated.

In the evaluation of technical products, the summation of subvalues has become the usual method of calculation but can only be considered accurate if the evaluation criteria are independent. However, even when this condition is only satisfied approximately, the assumption that the overall value has an additive structure seems to be justified.

The overall value of a variant j can then be determined as follows:

Unweighted: $OV_j = \sum_{i=1}^{n} v_{ij}$

Weighted: $OWV_j = \sum_{i=1}^{n} w_i \cdot v_{ij} = \sum_{i=1}^{n} wv_{ij}$

Comparing Concept Variants

On the basis of the summation rule it is possible to assess variants in several ways.

Determining the maximum overall value: In this procedure the variant is judged to be the best if it has the largest overall value:

$$OV_j \rightarrow \max \quad \text{or} \quad OWV_j \rightarrow \max$$

What we have here is a relative comparison of the variants. This fact is made use of in Cost–Benefit Analysis.

Determining the rating: If a relative comparison of the variants is considered to be insufficient and the absolute rating of a variant has to be established, then the overall value must be referred to an imaginary ideal value which results from the maximum possible value as follows:

Unweighted: $R_j = \dfrac{OV_j}{v_{\max} \cdot n} = \dfrac{\sum_{i=1}^{n} v_{ij}}{v_{\max} \cdot n}$

Weighted: $WR_j = \dfrac{OWV_j}{v_{\max} \cdot \sum_{i=1}^{n} w_i} = \dfrac{\sum_{i=1}^{n} w_i \cdot v_{ij}}{v_{\max} \cdot \sum_{i=1}^{n} w_i}$

If the available information about all the concept variants allows cost estimates, then it is advisable to proceed to a separate determination of the *technical rating* R_t and the *economic rating* R_e. The technical rating is calculated in accordance with the rule we have given—that is, by division of the technical overall value of the given variant by the ideal value—and the economic rating is calculated similarly, but by reference to comparative costs. The latter procedure is suggested in VDI 2225, which relates the manufacturing costs determined for a variant to the comparative manufacturing costs C_o. In that case, the economic rating becomes $R_e = (C_o/C_{variant})$. It is possible to put, say, $C_o = 0.7 \times C_{admissible}$ or $C_o = 0.7 \times C_{minimum}$ for the cheapest variant. If the technical and economic ratings have been determined separately, then the determination of the "overall rating" of a particular variant may prove interesting. For that purpose, Guideline VDI 2225 suggests a so-called s-diagram (strength diagram) with the technical rating R_t as the abscissa and the economic rating R_e as the ordinate (see Figure 3.35). Such diagrams are particularly useful in the appraisal of variants during further developments, because they show up the effects of design decisions very clearly.

In some cases it is useful to derive the overall rating from these partial ratings and to express it in numerical form, for instance for computer processing. To that end, Baatz [3.1] has proposed two procedures, namely:

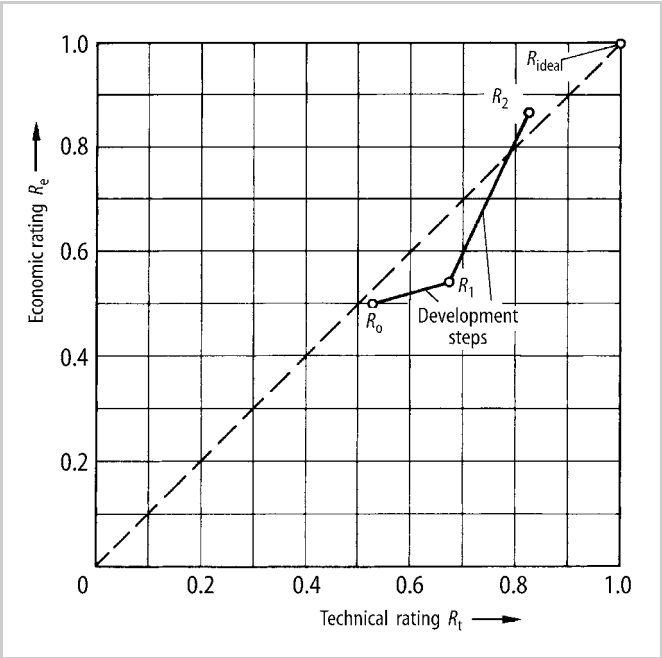


Figure 3.35. Rating diagram. After [3.36, 3.71]

- the straight-line method, based on the arithmetic mean:

$$R = \frac{R_t + R_e}{2} \tag{3.5}$$

and

- the hyperbolic method, which involves multiplying both ratings and then reducing to values between 0 and 1:

$$R = \sqrt{R_t \times R_e} \tag{3.6}$$

The two methods have been combined in Figure 3.36.

Where there are large differences between the technical and economic ratings, the straight-line method might compute a higher overall rating than is the case with lower but balanced partial ratings. Because balanced solutions should be preferred, however, the hyperbolic method is the better of the two; it helps to balance large differences in rating by its progressive reduction effect. The greater the imbalance, the greater the reduction effect on the overall value.

Rough comparison of solution variants: The method we have described relies on differentiated value scales. It is useful whenever the "objective" parameters can be stated with some accuracy and whenever clear values can be assigned to

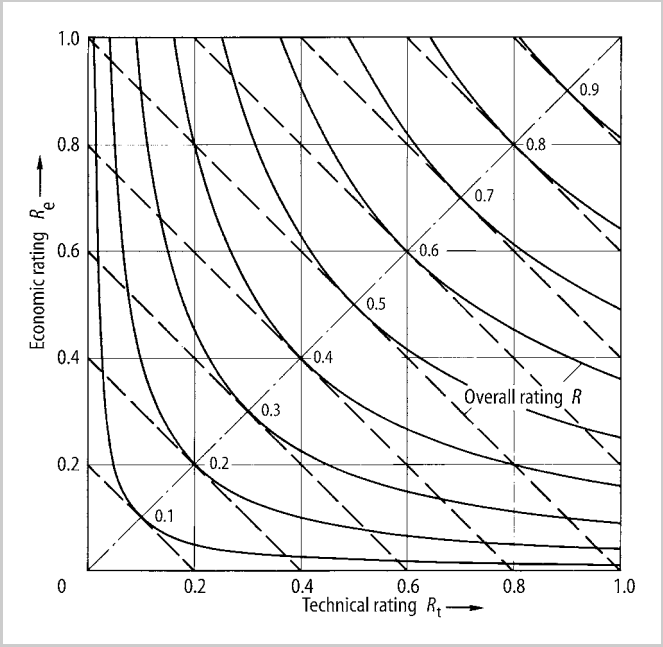


Figure 3.36. Determining the overall rating by the straight-line and hyperbolic methods. After [3.1]

In comparison with Variant	Variant 1	2	3	4	5	6	7
1	-	1	0	1	0	1	0
2	0	-	0	1	0	0	0
3	1	1	-	1	0	1	0
4	0	0	0	-	0	0	0
5	1	1	1	1	-	1	1
6	0	1	0	1	0	-	0
7	1	1	1	1	0	1	-
Sum	3	5	2	6	0	4	1
Rank	4	2	5	1	7	3	6

1 ≙ better 0 ≙ not better

Figure 3.37. Binary evaluation of solution variants. After [3.15]

them. If these conditions cannot be satisfied, relatively fine evaluations based on a differentiated value scale constitute a questionable and expensive method. The alternative here is a rough evaluation involving the application of a particular evaluation criterion to two variants at a time and the selection of the best in each case. The results are entered into a so-called *dominance matrix* [3.15] (see Figure 3.37). From the sum of the columns it is possible to establish a ranking order. If such matrices of individual criteria are combined into an overall matrix, an overall ranking order can be established, either by addition of the preference frequencies or by addition of all the column sums. While this method is comparatively easy and quick, it is not nearly as informative as the other procedures we have discussed.

Estimating Evaluation Uncertainties

The possible errors or uncertainties of the proposed evaluation methods fall into two main groups, namely subjective errors and procedure-inherent shortcomings.

Subjective errors can arise through:

- Abandonment of the neutral position, that is, through bias and partiality. The bias may be hidden from designers, for instance when they compare their own designs with those of others. Hence an evaluation by several persons, if possible from various departments, is always advisable. It is equally important

to refer to the different variants in neutral terms, for instance as A, B, C rather than as "Smith's Proposal", etc., since otherwise unnecessary identifications and emotional overtones may be introduced. Systematisation of the procedure also helps to reduce subjective influences.

- Comparison of variants by application of (the same) evaluation criteria not equally suited to all the variants. Such mistakes arise even during the determination of the parameters and their association with the evaluation criteria. If it is impossible to determine the parameter magnitudes of individual variants for certain evaluation criteria, then these criteria must be reformulated or dropped in case they lead to mistaken evaluations of the individual variants.
- Evaluation of variants in isolation instead of successively by application of the established evaluation criteria. Each criterion must be applied to all the variants in turn (row-by-row in the evaluation chart) to eliminate any bias in favour of a particular variant.
- Pronounced interdependence of the evaluation criteria.
- Choice of unsuitable value functions.
- Incompleteness of evaluation criteria. This defect can be minimised if one of the checklists for design evaluation appropriate to the relevant design phase is followed (see Figures 6.22 and 7.148).

Procedure-inherent shortcomings of the recommended evaluation methods are the result of the almost inevitable "prognostic uncertainty" arising from the fact that the predicted parameter magnitudes and also the values are not precise, but subject to uncertainty and to random variation. These mistakes can be greatly reduced by estimates of the mean error.

With regard to prognostic uncertainty, it is therefore advisable not to express the parameters in figures unless this can be done with some accuracy. It is preferable to use verbal estimates (for instance high, average, low) which do not claim to be precise. Numerical values, by contrast, are dangerous because they introduce a false sense of certainty.

Uncertainties in the evaluation are not only caused by prognostic uncertainty, but also through uncertainties in the formulation of requirements and solution descriptions. To be able to process such vague information in a quantitative way, fuzzy logic, and its extension into fuzzy-MADM (multi-attribute decision making), can be used [3.49]. These procedures use so-called fuzzy sets to describe these imprecise numbers and ranges and calculate their combined averages. The result is a fuzzy overall value for every solution variant.

A more detailed analysis of evaluation procedures for the purpose of judging their reliability and also for comparative purposes has been carried out by Feldmann [3.15] and Stabe [3.67]. The latter also provides an extensive bibliography. If there is an adequate number of evaluation criteria, and if the subvalues of a particular variant are fairly balanced, then the overall value will be subject to a balancing statistical effect, and partly too optimistic and partly too pessimistic individual values will more or less balance out.

Searching for Weak Spots

Weak spots can be identified from below average values for individual evaluation criteria. Careful attention must be paid to them, particularly in the case of promising variants with good overall values, and they ought if possible to be eliminated during further development. The identification of weak spots may be facilitated by graphs of the subvalues—for instance, by the so-called value profiles illustrated in Figure 3.38. In it, the lengths of the bars correspond to the values and the thicknesses to the weightings. The areas of the bars then indicate the weighted subvalues, and the cross-hatched area the overall weighted value of a solution variant. It is clear that, in order to improve a solution, it is essential to improve those subvalues that provide a greater contribution to the overall value than the rest. This is the case with the evaluation criteria that have an above average bar thickness (great importance) but a below average bar length. Apart from a high overall value, it is important to obtain a balanced value profile, with no serious weak spots. Thus, in Figure 3.38, variant 2 is better than variant 1, although both have the same overall weighted value.

There are also cases in which a minimum permissible value is stipulated for all sub-values; that is, any variant that does not fulfil this condition has to be rejected, and all variants that do fulfil it are developed further. In the literature this procedure is described as the "determination of satisfactory solutions" [3.76].

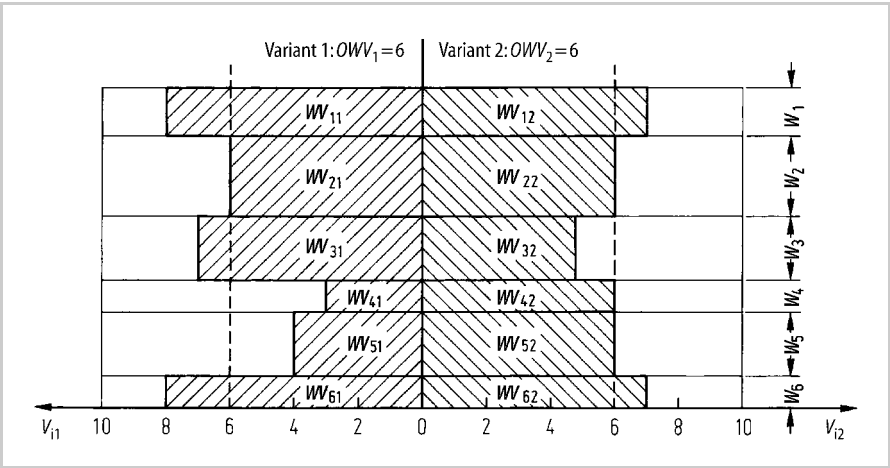


Figure 3.38. Value profiles for the comparison of two variants ($\sum w_i = 1$)

2. Comparison of Evaluation Procedures

Table 3.4 lists the individual steps in the evaluation procedures we have described and also the similarities and differences between Cost–Benefit Analysis and Guideline VDI 2225, which are based on similar principles.

Table 3.4. Individual steps in evaluation, and comparison between use-value analysis and Guideline VDI 2225

Step		Cost–Benefit Analysis	VDI Guideline 2225
1	*Identification of objectives* or *evaluation criteria* for the evaluation of concept variants with the aid of the requirements list and a checklist	Construction of a hierarchically related system of design objectives (objectives tree) based on the requirements list and other general requirements	Compilation of important technical characteristics and also of the minimum demands and wishes of the requirements list
2	*Analysis of the evaluation criteria* for the purpose of determining their *weighting* to the overall value of the solution. If necessary, determination of weighting factors	Step-by-step weighting of the objective criteria (evaluation criteria) and if necessary elimination of unimportant criteria	Determination of weighting factors only if evaluation criteria differ markedly in importance
3	*Compilation of parameters* applicable to the concept variants	Construction of an objective parameter matrix	Not generally included
4	*Assessment of the parameter magnitudes* and assignment of values (0–10 or 0–4 points)	Construction of objective value matrix, with the help of a points system or value functions; 0–10 points	Assessment of characteristics by points (0–4 points)
5	*Determination of the overall value* of the individual concept variants, generally by reference to an ideal solution (rating)	Construction of a use-value matrix with due regard to the weightings; determination of overall values by summation	Determination of a technical rating by summation, with or without weightings based on an ideal solution. If necessary determination of an economic rating based on manufacturing costs
6	*Comparison of concept variants*	Comparison of overall use-values	Comparison of the technical and economic ratings. Construction of an s-(strength) diagram
7	*Estimation of evaluation uncertainties*	Estimation of objective parameter scatter and use–value distribution	Not explicitly included
8	*Search for weak spots* for the purpose of improving selected variants	Construction of use-value profiles	Identification of characteristics with a few points only

The individual steps of Cost–Benefit Analysis are more highly differentiated and more clear-cut but involve more work than those of Guideline VDI 2225. The latter is more suitable when there are relatively few and roughly equivalent evaluation criteria, which is frequently the case during the conceptual phase, and also for the evaluation of certain form design areas during the embodiment phase.

The essence of evaluation procedures has been described on the basis of existing evaluation methods. However, these methods have been consolidated and the terms clarified. Specific suggestions for the use of these methods during the conceptual phase are given in Section 6.5.2, and during the embodiment phase in Section 7.6.

4 Product Development Process

In the previous chapters we examined the fundamentals on which design work should be built to best advantage. They form the basis of a systematic approach which practising designers can follow, regardless of their speciality. The approach is not based on one method but applies known and less well known methods where they are most suitable and useful for specific tasks and working steps.

4.1 General Problem Solving Process

An essential part of our own problem solving method involves step-by-step *analysis* and *synthesis*. In it we proceed from the *qualitative* to the *quantitative*, each new step being more concrete than the last.

In the following sections we propose plans and procedures that should be regarded as mandatory for the general problem solving process of planning and designing technical products, and as guidance for the more concrete phases of the design process. These plans and procedures assist in identifying what, in principle, has to be done, but of course they must be adapted to specific problem situations.

All procedural plans proposed in this book have to be considered as *operational guidelines for action* based on the pattern of technical product development and the logic of stepwise problem solving. According to Müller [4.17], they are process models that are suitable for describing in a rational way the approach necessary to make complex processes comprehensible and transparent.

Thus, these procedural plans are not descriptions of *individual thinking processes* as described in Section 2.2.1, and are not determined by personal characteristics. In a practical application of these procedural plans, the operational guidelines for action blend with individual thinking processes. This results in a set of individual planning, acting and controlling activities based on general procedures, specific problem situations and individual experiences.

As discussed in Section 2.2.1, the suggested procedural plans are meant to be guidelines and not rigid prescriptions. However, they have to be regarded as essentially sequential because, for example, a solution cannot be evaluated before it has been found or elaborated. On the other hand, the procedural plans have to be adapted to specific situations in a flexible manner. It is, for example, possible to leave out certain steps or order them in another sequence. It may be necessary or useful to

repeat certain steps at a higher information level. Furthermore, special procedures (adapted from the more general plans proposed here) may be appropriate in specific product domains.

Given the complexity of the product development process and the many methods that have to be applied, not adopting a procedural plan would leave designers with an unmanageable number of possible approaches. It is therefore necessary for designers to learn about the design process and the application of individual methods, as well as the working and decision making steps proposed in the procedural plans.

The activity of planning and designing was described in Section 2.2.3 as information processing. After each information output, it might become necessary to improve or increase the value of the result of the last working step. That is, to repeat the working step at a higher information level, or to execute other working steps until the necessary improvements have been achieved.

Repeating working steps is the process of *iteration* by which one approaches a solution step-by-step until the result seems satisfactory. The so-called iteration loop can also be observed in the basic thinking processes, for example in the TOTE model (see Section 2.2.1). Such iteration loops are almost always required and occur continuously within and between steps. The reasons for this are that the interrelationships are often so complex that the desired solution cannot be achieved in one step and that information is frequently needed from a subsequent step. The iteration arrows in procedural plans clearly indicate this fact. In subsequent chapters, strategies for reducing, or even avoiding, such iteration loops are presented. It is therefore important that the procedural plans proposed are not considered rigid and purely sequential.

A systematic approach aims to keep the iteration loops as small as possible in order to make design work effective and efficient. It would be a disaster, for example, if the design team had to start again at the beginning having reached the end of a product development. This would correspond to an iteration loop covering the whole of the product development process.

The division in working and decision making steps ensures necessary and permanent links between *objectives*, *planning*, *execution* (organisation) and *control* [4.3, 4.29]. With these links, we can, following Krick [4.15] and Penny [4.21], construct a basic scheme for the general problem solving process (see Figure 4.1).

Every task involves an initial *confrontation* of the problem, which involves elucidating what is known or not (yet) known. The intensity of this confrontation depends on the knowledge, ability and experience of the designers, and on the particular field in which they are engaged. In all cases, however, more detailed *information* about the task itself, about the constraints, about possible solution principles and about known solutions for similar problems is extremely useful since it clarifies the precise nature of the requirements. This information can also reduce confrontation and increase confidence that solutions can be found.

Next comes the *definition* phase, where the essential problems (the crux of the task) are defined on a more abstract plane, in order to set the objectives and main constraints. Such solution-neutral definitions open the way to an unconstrained

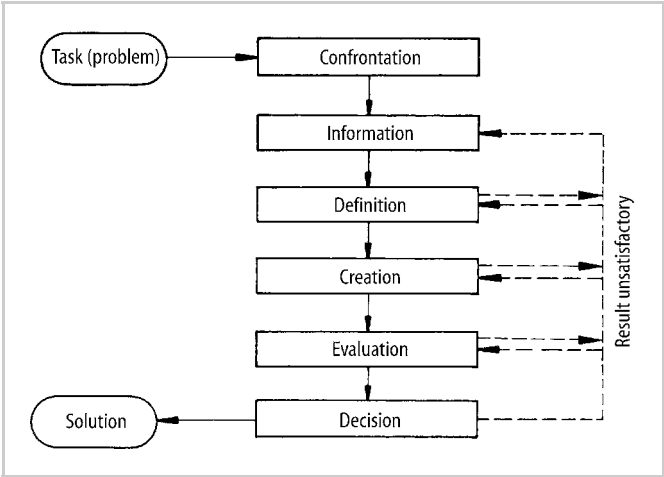


Figure 4.1. General problem solving process

search for solutions because this abstract definition encourages a search for more unconventional solutions.

The next step is *creation*, where solutions are developed by various means and then varied and combined using methodical guidelines. If the number of variants is large, there must also be an *evaluation* which is then used to select the best variant through a *decision*. Because each step of the design process must be evaluated, evaluation serves as a check on progress towards the overall objective.

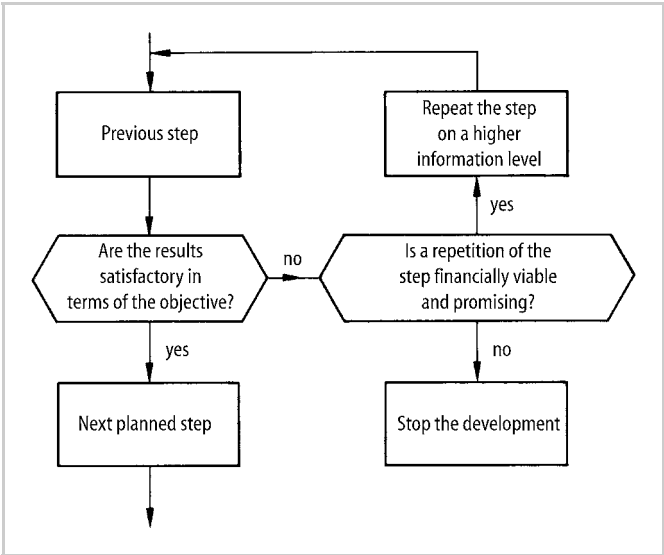


Figure 4.2. General decision making process

Decisions involve the following considerations (see Figure 4.2):

- If the result from the previous step meets the objective, the next step can be started.
- If the results are incompatible with the objective, the next step should not be taken.
- If resources permit repetition of the previous step (or if necessary several preceding steps), and good results can be expected, the step must be repeated on a higher information level.
- If the answer to the previous question is no, the development must be stopped.

Even if the results of a particular step do not meet the objectives, they might nevertheless prove interesting if the objectives are wholly or partly changed. In this case, there should be an investigation to see whether the objectives can be changed or if the results can be used for other applications. This whole process, leading from confrontation through creation to decision, must be repeated in each successive, increasingly concrete, phase of the design process.

4.2 Flow of Work During the Process of Designing

Today's conditions for product design and development demand careful planning of:

- the required activities for the proposed project
- the timing and scheduling of these activities
- the project and product costs.

The activities and their durations strongly depend on the type of task, in particular whether the task is for an original, adaptive or variant design.

4.2.1 Activity Planning

The flow of work during the process of designing has been described in both general terms as well as domain and product-specific terms in VDI Guidelines 2221 and 2222 [4.24, 4.25] (see Figure 1.9). In line with these guidelines, the next sections provide an extensive description of this flow of work, focused on mechanical engineering. The description is essentially based on the fundamentals of technical systems (see Section 2.1), the fundamentals of the systematic approach (see Section 2.2), and the general problem solving process (see Section 4.1). The aim is to adapt the general statements to the requirements of the mechanical engineering design process and to incorporate the specific working and decision making steps for this domain. In principle, the planning and design process proceeds from the planning and clarification of the task, through the identification of the required functions, the elaboration of principle solutions, the construction of modular structures, to the final documentation of the complete product [4.18].

In addition to the planning of the specific tasks described in the guidelines mentioned above, it is useful and common to divide the planning and design process into the following *main phases*:

- Planning and task clarification: specification of information
- Conceptual design: specification of principle solution (concept)
- Embodiment design: specification of layout (construction)
- Detail design: specification of production.

As we will see later on, it is not always possible to draw a clear borderline between these main phases. For example, aspects of the layout might have to be addressed during conceptual design, or it might be necessary to determine some production processes in detail during the embodiment phase. Neither is it possible to avoid backtracking, for example during embodiment design when new auxiliary functions may be discovered for which principle solutions have to be found. Nevertheless, the division of the planning and control of a development process into main phases is always helpful.

The working steps proposed for each of the main phases are termed the *main working steps* (see Figure 4.3). The results of these main working steps provide the basis for the subsequent working steps. Many lower level working steps are required to realise these results, such as collecting information, searching for solutions, calculating, drawing and evaluating. Each of these working steps is accompanied by indirect activities such as discussing, classifying and preparing. The *operational main working steps* listed in the procedural plans proposed in this chapter are considered to be the most useful strategic guidelines for a technical domain. Guidelines that are not listed include, for example, those related to basic problem solving, collecting information and verifying results. This is because they can usually only be recommended in relation to a specific problem and a particular designer. Recommendations for such elementary working steps will, where possible, be given in the sections describing individual methods and those dealing with practical applications.

After the main phases, and some of the more important main working steps, *decision making steps* are required. The decision making steps listed are the main ones—those that end a main phase or working step, which after an appropriate assessment of the results, allow the main flow of work to proceed. It is also possible, because the result of a decision making step was unsatisfactory, that certain steps will have to be repeated. The smallest possible iteration loop is desirable.

Again, the individual test and decision making steps (see for example the TOTE model in Section 2.2.1) that are required for every single action have not been listed separately. This would have been impossible because such decisions are determined by the approach of individual designers and by particular problem situations.

The decision to stop a development that ceases to be viable, as discussed in Section 4.1, is not mentioned explicitly in the individual decision making steps of the procedural plans. One should, however, always explicitly consider this possibility because an early and clear decision to halt a hopeless situation will, in the end, minimise disappointment and cost.

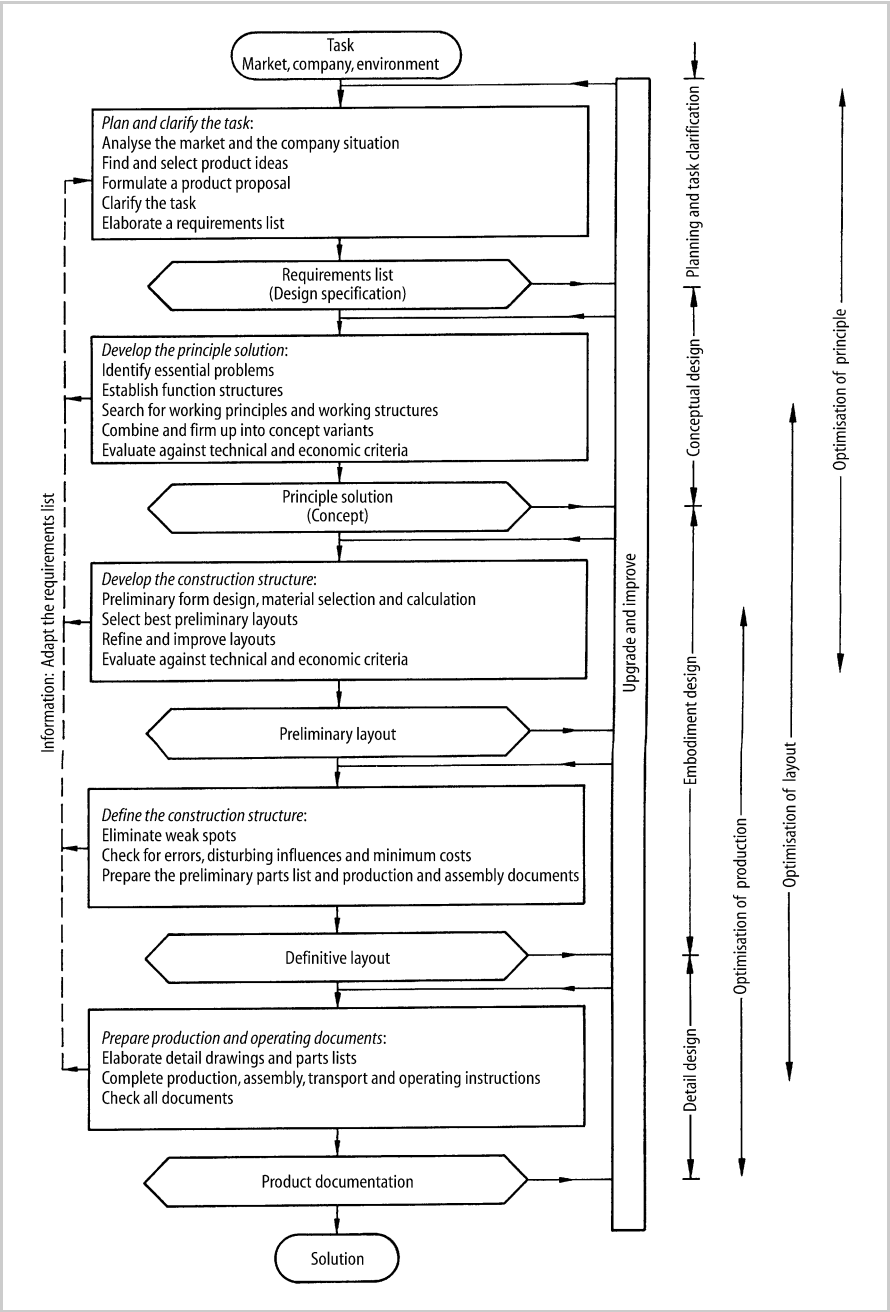


Figure 4.3. Steps in the planning and design process

In all cases procedural plans should be applied in a flexible manner and adapted to the particular problem situation. At the end of each main working and decision step, the overall approach should be assessed and adjusted if necessary.

The four main phases are outlined below.

1. Planning and Task Clarification

The product development task is given to the engineering department by the marketing department, or a special department responsible for product planning, see also Sections 3.1 and 5.1.

Irrespective of whether the task is based on a product proposal stemming from a *product planning* process or on a specific customer order, it is necessary to clarify the given task in more detail before starting product development. The purpose of this *task clarification* is to collect information about the requirements that have to be fulfilled by the product, and also about the existing constraints and their importance.

This activity results in the *specification of information* in the form of a *requirements list* that focuses on, and is tuned to, the interests of the design process and subsequent working steps (see Section 5.2). The conceptual design phase and subsequent phases should be based on this document, which must be updated continuously (this is indicated by the information feedback loop in Figure 4.3).

2. Conceptual Design

After completing the task clarification phase, the conceptual design phase determines the principle solution. This is achieved by abstracting the essential problems, establishing function structures, searching for suitable working principles and then combining those principles into a working structure. Conceptual design results in the specification of a *principle solution* (concept).

Often, however, a working structure cannot be assessed until it is transformed into a more concrete representation. This concretisation involves selecting preliminary materials, producing a rough dimensional layout, and considering technological possibilities. Only then, in general, is it possible to assess the essential aspects of a solution principle and to review the objectives and constraints (see Section 2.1.7). It is possible that there will be several principle solution variants.

The representation of a principle solution can take many forms. For existing building blocks, a schematic representation in the form of a function structure, a circuit diagram or a flow chart may be sufficient. In other cases a line sketch might be more suitable, and sometimes a rough scale drawing is necessary.

The conceptual design phase consists of several steps (see Chapter 6), none of which should be skipped if the most promising principle solution is to be found. In the subsequent embodiment and detail design phases it is extremely difficult or impossible to correct fundamental shortcomings of the solution principle. A lasting and successful solution is more likely to spring from the choice of the most appropriate principles than from exaggerated concentration on technical details.

This claim does not conflict with the fact that problems may emerge during the detail design phase, even in the most promising solution principles or combinations of principles.

The solution variants that have been elaborated must now be evaluated. Variants that do not satisfy the demands of the requirements list have to be eliminated; the rest must be judged by the methodical application of specific criteria. During this phase, the chief criteria are of a technical nature, though rough economic criteria also begin to play a part (see Sections 3.3.2 and 6.5.2). Based on this evaluation, the best concept can now be selected.

It may be that several variants look equally promising, and that a final decision can only be reached on a more concrete level. Moreover, various form designs may satisfy one and the same concept. The design process now continues on a more concrete level referred to as embodiment design.

3. Embodiment Design

During this phase, designers, starting from a concept (working structure, principle solution), determine the construction structure (overall layout) of a technical system in line with technical and economic criteria. Embodiment design results in the specification of a *layout*.

It is often necessary to produce several *preliminary layouts* to scale simultaneously or successively in order to obtain more information about the advantages and disadvantages of the different variants.

After sufficient elaboration of the layouts, this design phase also ends with an evaluation against technical and economic criteria. This results in new knowledge on a higher information level. Frequently, the evaluation of individual variants may lead to the selection of one that looks particularly promising but which may nevertheless benefit from, and be further improved by, incorporating ideas and solutions from the others. By appropriate combination and the elimination of weak spots, the best layout can then be obtained.

This *definitive layout* provides a means to check function, strength, spatial compatibility, etc., and it is also at this stage (at the very latest) that the financial viability of the project must be assessed. Only then should work start on the detail design phase.

4. Detail Design

This is the phase of the design process in which the arrangement, forms, dimensions and surface properties of all of the individual parts are finally laid down, the materials specified, production possibilities assessed, costs estimated, and all the drawings and other production documents produced [4.28] (see also [4.26]). The detail design phase results in the *specification of information* in the form of *production documentation*.

It is important that designers should not relax their vigilance at this stage, otherwise their ideas and plans might change out of all recognition. It is a mistake to think that detail design poses subordinate problems lacking in importance or

interest. As we said earlier, difficulties frequently arise from lack of attention to detail. Quite often corrections must be made during this phase and the preceding steps repeated, not so much with the overall solution in mind, as to improve assemblies and components as well as reduce costs.

5. Overall Design Process

In the flow diagram (see Figure 4.3), the main themes are:

- optimisation of principle
- optimisation of layout
- optimisation of production.

Clearly the description above is a generalisation of actual processes. In practice a clear distinction between the working steps and their results cannot always be made, nor is it necessary to do so. However, it is useful for designers to be aware of the main process flow and tasks described in order to plan their work and to avoid forgetting something.

Figure 4.3 does not include the production of models and prototypes because the information they supply may be needed at any point in the design process and so cannot be fitted into any particular slot. In many cases, it is even necessary to develop models and prototypes during the conceptual phase, particularly when they are intended to clarify fundamental questions in, say, the precision engineering, electronics and mass production industries. Due to the one-off nature of heavy and process engineering, the cost and time required to produce prototypes normally makes them uneconomic or infeasible. However, it is possible to test parts of the proposed plant or equipment by building partial prototypes within existing plant and equipment or by using specific test facilities. In batch production it is common to produce prototypes well before production starts and also to undertake a pre-production run to ensure that production will run smoothly. These pre-production products can still be sold.

Figure 4.3 also does not indicate when work has to be subcontracted, because this depends upon the type of product.

The execution of orders is usually part of product development, although in the case of size ranges and modular products it can take place quite late in the process.

If on receiving an order, only existing documents are used, and only production instructions, subcontractor orders, parts lists, etc., need to be compiled, no product development is required. So apart from tender drawings, layout drawings and assembly plans, no further design work is needed, and in many cases these drawings and plans can be generated automatically using variant design software.

Upon looking at Figure 4.3, and after reading about the methods described in the following chapters, practising designers may well object to the process on the basis that they lack the time to go through every one of the many steps. They should bear in mind that:

- Most of the steps are performed in any case—albeit unconsciously—although they are often carried out too quickly, leading to unforeseen consequences.

- This deliberate step-by-step procedure, on the other hand, ensures that nothing essential has been overlooked or ignored, and is therefore indispensable in the case of original designs.
- In the case of adaptive designs, it is possible to resort to time-tested approaches and to reserve the procedure described for where it offers special benefits; for example, when improving a specific detail, in which case the steps should be undertaken focusing on this detail.
- If designers are expected to produce better results, then they must be given the extra time the systematic approach demands, although experience has shown that only a little extra time is needed for a stepwise procedure.
- Scheduling becomes more accurate if the step-by-step method is followed rigorously.

4.2.2 Timing and Scheduling

Products will only be successful when they:

- satisfy the customer needs (requirements)
- reach the market at the right time
- are sold at the right price.

This section focuses on the second prerequisite, because designers often underestimate the importance of time-to-market and are not familiar with the methods and tools used for timing and scheduling. We only introduce the basic approaches. Details have to be obtained from the literature.

Two constraints determine the planning difficulty:

- the project or design result must be finished at a certain point in time, and intermediate results are required on specific dates
- not every task can be executed by every member of the team, i.e. there is a resource constraint.

Network planning is one of the most important planning tools [4.7, 4.8]. A network plan is used to estimate the overall project length and resource requirements. The graphical representation shows the logical links between the required project tasks and the resources assigned to these tasks.

Creating a network plan involves performing three main steps:

- Structure analysis to identify and describe the links and dependencies between the project tasks.
- Time analysis to identify the necessary duration for each task along with a feasible starting date for each of the main steps.
- Resource analysis to allocate the various tasks to individual team members. In the first instance this should be based on their competences, followed by their

Table 4.1. Procedure for creating a network plan

	Activity	Explanation
1.	Determine product structure	In general the structure of an existing similar product is adapted
2.	Determine the tasks necessary to create the individual product elements	For every product element and for the overall product, the tasks include the following to an appropriate level: • solution finding • investigation • embodiment • calculation
3.	Establish logical and temporal dependencies between individual tasks	Dependencies between tasks have to be identified and documented as unambiguous IF–THEN statements: e.g. IF the shaft diameter is determined, THEN the shaft–hub connection can be fixed
4.	Establish the duration of the tasks	• interview those with relevant experience • compare with similar tasks • document completed tasks • estimate
5.	Fix milestones (these are used to check whether the work and schedule have been achieved; a milestone trend analysis allows the prediction of the success or failure of a project)	**Types of milestones:** **Event-driven:** The content of a milestone has to be defined precisely. A milestone is reached when the available working results meet the defined content of that milestone **Application:** Mostly used as the final milestone for the design of an assembly **Time-driven:** The milestone is reached at a certain point in time or after a certain time interval has elapsed **Application:** For large tasks when it is not possible to define clear intermediate results **Point of no return:** Event or point in time after which the results achieved must not be changed further **Application:** Securing intermediate results, e.g. against customers changing requirements **Review:** Point in time at which clearly defined results have to be explicitly signed off or released **Application:** The embodiment of expensive and complex assemblies or components is signed off by production or, in some cases, by the customer
6.	Determine necessary and possible float times for the tasks	Float times serve to manage risk in order to avoid endangering the project plan when delays occur and are applied, in particular, for novel tasks
7.	Create network plan (usually using special software tools: e.g. Microsoft Project, Super-Project-Expert)	A network plan shows in graphical and tabular form all of the dependencies between the tasks and milestones, and is used to determine the course of a project
8.	Create project calendar	The project calendar shows the exact number of working days available for the duration of the project
9.	Select resources and allocate them to tasks in the network plan	The selection is based on the required competencies and the availability of resources during the planned period of the project
10.	Create a resource calendar and allocate to the network plan	For every employee an individual calendar is created showing his or her available working time during the duration of the project: holidays, training days, etc. must be taken into account

Table 4.1. (continued)

	Activity	Explanation
11.	Run through the plan	After the resources and the individual calendars are allocated to the network plan, the first run-through is undertaken
12.	Evaluate the plan	• Can the project milestone be achieved? • What is the critical path? (i.e. the sequence of tasks with no float times that determines the overall duration of the project)
13.	Optimise the plan	The plan can be optimised and corrected by: • increasing the resources available • moving deadlines • reducing the number of tasks • changing the sequence of tasks • altering the content of the tasks
14.	Sign off the plan	The project plan is released through the signature of the manager responsible, and, where appropriate, by the customer
15.	Monitor the project	All project parameters, such as deadlines, costs and risks are continuously monitored and reported

availability, which can be limited because they may be absent due to training courses, illness, holidays, etc., or because they have been allocated to other projects.

In general the product structure is used as the basis for planning the task structure. The product structure determines the main assembly groups and components that have to be designed and, as a consequence, the majority of the tasks.

Table 4.1 shows the procedure for creating a network plan and the individual steps. Figure 4.4 shows part of a network plan, in this case a Gantt chart. The individual tasks are represented by bars. Their dependencies result from logical or possible working sequences, e.g. input/output requirements, where one task must have been completed before the next can be started.

A network plan not only shows project duration, resource requirements and allocation of team members to tasks, but also float times and the critical path of the project. The float times indicate how much the start or end of a task or series of tasks can be delayed without jeopardising the overall lead time of the project. The critical path contains those tasks that have no float times and therefore determine the overall duration of the project.

4.2.3 Planning Project and Product Costs

The cost price is the basis for determining the selling price and is therefore crucial to the success of the product. The cost price is influenced by the production costs and the associated project costs. Design and development are the costliest items contributing to the project cost, so engineering departments carry a great responsibility.

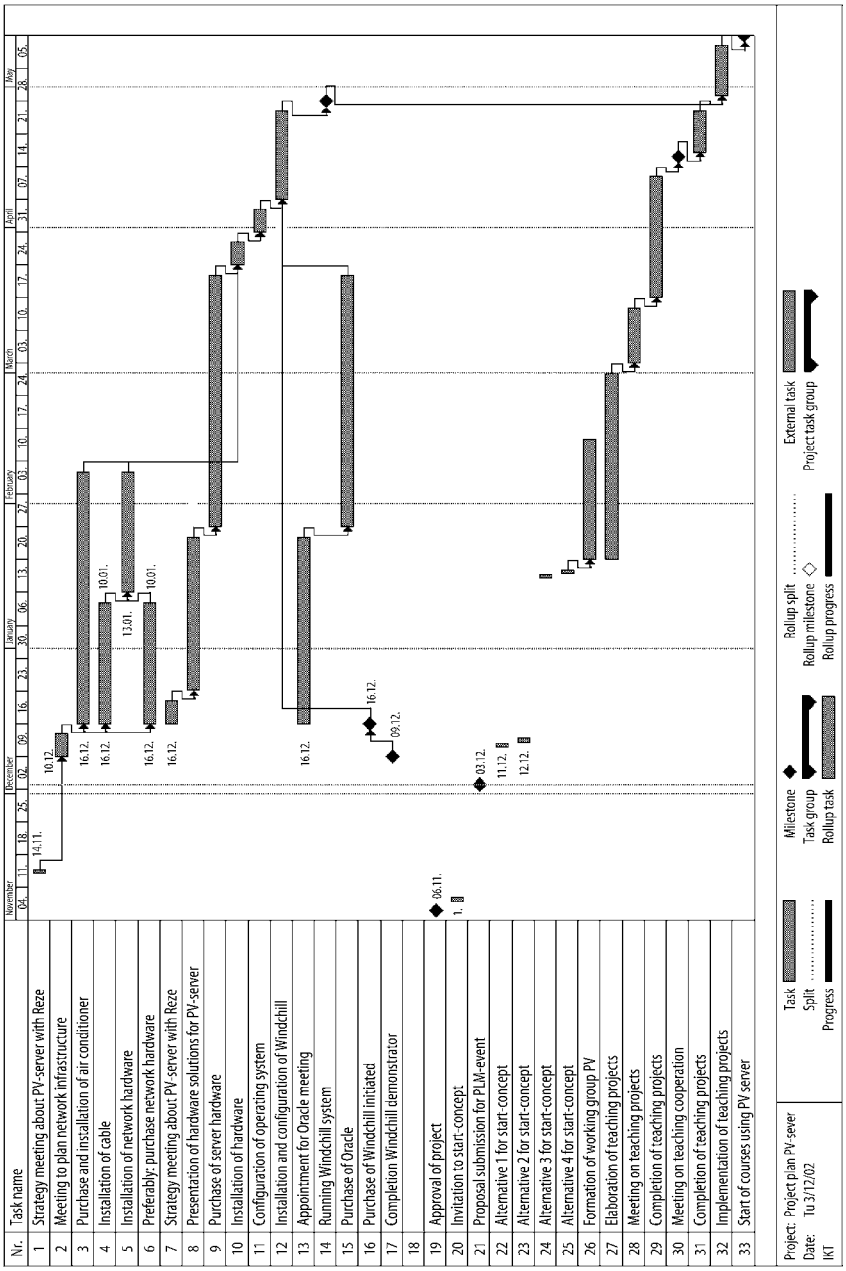


Figure 4.4. Example of a network plan

In order to meet the target cost price, engineering departments not only have to keep production costs to a minimum (see Chapter 11 for more details), but also the design and development costs. Depending on the batch size, the latter costs can represent a large share of the cost price.

To estimate the design and development costs, a network plan can also be used because the main costs incurred by engineering departments are staff costs. Support costs, such as facilities, CAD systems, external consultants, etc., are usually much lower. Using the network plan, costs can be assigned to the allocated resources using the appropriate hourly rate. The distribution of the costs with time can be represented by a cost plan [4.9], which is important when estimating the project budget.

4.3 Effective Organisation Structures

4.3.1 Interdisciplinary Cooperation

Designers cannot work independently of their environment—they depend on the results produced by others and others depend on their results. They are members of their departments, which in turn are parts of the company. Only the coordinated activities of all participants will lead to a satisfactory overall result [4.11, 4.22]. To achieve this, the responsibilities, tasks, etc., for every individual are specified by the organisational and operational structures:

- The *organisational structure* specifies the responsibilities and tasks for individuals, departments and standing committees, and relates these in a hierarchy.
- The *operational structure* specifies the various procedures.

The design and development process is made more efficient through the following actions:

- reducing inner iterations, i.e. repetition of the same activity within a working step
- reducing outer iterations, i.e. jumping back to a working step that has already been completed or even repeating a design phase
- omitting working steps
- executing working steps in parallel.

In particular, the last action has the potential to reduce the overall project time significantly. To achieve these four actions, the following prerequisites must be met:

- The product has to be structured in such a way that the properties of its systems, subsystems and system elements can be modelled precisely and unambiguously during every step of the process. Chapter 9 proposes some possible product structures.
- The interfaces between the process steps have to be defined precisely and unambiguously.
- The process steps have to be independent.

When these prerequisites have been met and interdisciplinary teams formed, then *Simultaneous or Concurrent Engineering* can be introduced. Simultaneous or Concurrent Engineering involves goal-oriented, interdisciplinary (interdepartmental) collaboration and parallel working throughout the development of the product, the production process and the sales strategy. It covers the total product life cycle and requires firm project management [4.1]. Experiences of its application in industry can be found in [4.12, 4.14]. Figure 1.4 highlights the intensive information flows that occur between departments. In a simultaneous engineering process the activities of the various departments run in parallel or at least have significant overlap. Intensive contacts with customers are encouraged, many suppliers are integrated in the process, see Figure 4.5 [4.5, 4.13, 4.23], and the product is monitored until the end of its working life.

For the duration of the project, a team is formed consisting not only of members of the design and development department but also those from other departments involved in the product creation process. This team, which is formed as early as possible, is led by a project manager, works independently, but has to report directly to the Board of Management or Head of Development. Departmental boundaries are thereby transcended. The team can operate as a virtual team; that is, without a visible organisational form. Characteristics of team structures and their importance can be found in [4.6, 4.27]. The objectives of this type of organisation and working procedure are:

- shorter development times
- faster product realisation
- reduction of product and product development costs
- improved quality.

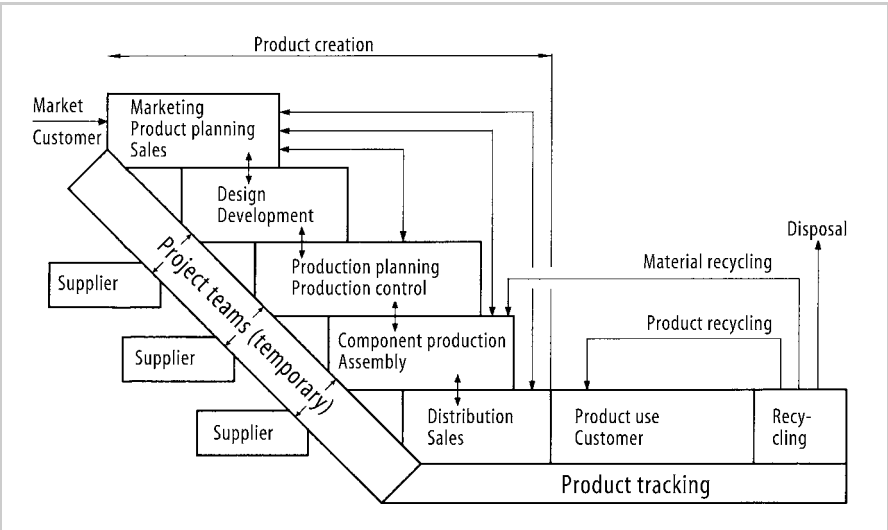


Figure 4.5. Product creation and tracking processes using Simultaneous Engineering, showing the overlapping activities of different disciplines, the formation of a project team and close contact with customers and suppliers

Simultaneous or Concurrent Engineering changes a designer's work as follows [4.20]:

- Working in an interdisciplinary team requires the adaptation of language and terminology.
- A closer, more direct exchange of information takes place through the early involvement of other departments and disciplines.
- More electronic information and communication technologies are used, e.g. data processing systems, CAD, multimedia, etc.
- A project management process with schedules and milestones is imposed so that design work has to be structured more systematically.
- Activities are run in parallel and therefore have to be coordinated accordingly.
- Individual responsibility for the assigned problems and tasks has to be accepted in line with team decisions.
- Contact with suppliers and customers becomes more intense.

It is useful to build a small core team with the experts who are responsible for design, production planning, marketing and sales. The composition of the team depends on the particular problem and type of product. This core team is complemented by experts from quality, assembly, electronics, software, recycling, etc., as and when needed and who may only participate for short periods of time. In such a team the knowledge and experience from neighbouring disciplines (see Figure 4.6) are more or less automatically incorporated into the project. This integration of a wide range of expertise significantly improves the realisation of the project goals and the ability to meet the constraints, as discussed in Section 2.1.7 and in accordance with Figure 2.15.

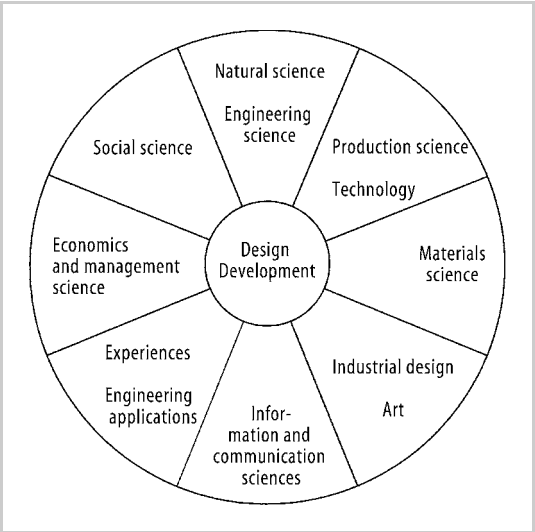


Figure 4.6. Related knowledge domains that support design and development

The advantages of an interdisciplinary team are:

- increased availability of knowledge and mutual stimulation
- better control of the product and the process—achieved by questioning issues and identifying contradictions
- increased motivation through direct participation and information sharing
- immediate responses to situations without the need to seek and wait for approval from higher levels in the hierarchy.

When the focus is on lean production, information and decision chains must become shorter. To facilitate this it is often necessary to build temporary project groups whose members are released from the departmental hierarchy for the duration of the project. The designer who previously worked within the confines of his discipline-based department, where he or she could easily call upon colleagues for advice and support, now has to work much more independently and within less familiar surroundings. To work in such project teams, a number of skills are required that go beyond the usual discipline-based ones [4.19,4.20] (see Figure 4.7). These issues must be taken into account when selecting the project leader.

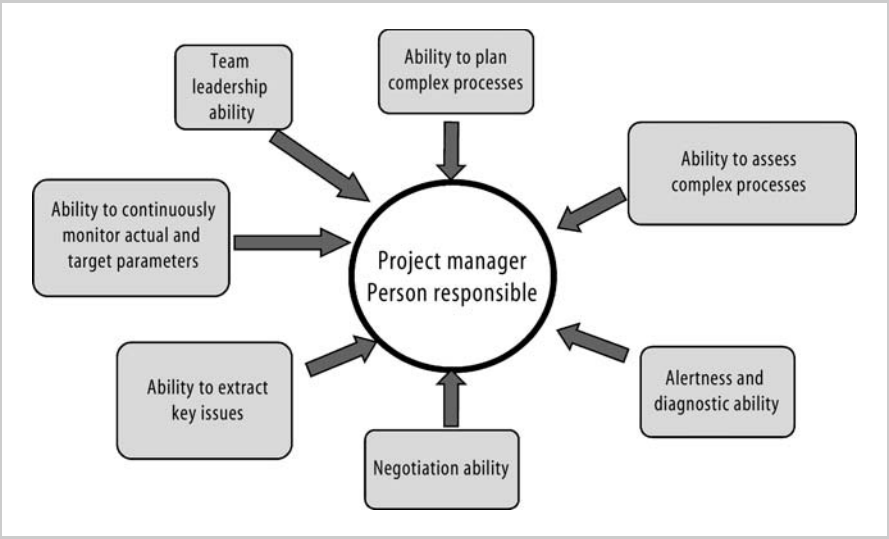


Figure 4.7. Abilities required of project managers

4.3.2 Leadership and Team Behaviour

Strong project leadership is necessary when developing new products in teams that are independent of departmental structures. Project leaders must have a good knowledge of the relevant technology and design methods as well as the characteristics of good problem solvers (see Section 2.2.2). Only then are they able to lead

a team of experts from different fields to achieve the project goals and to cope with the tasks assigned to them [4.20].

Project leaders and their teams can use the systematic approach presented in this book as an effective means of support. They can use it to initiate and check their approach, to select suitable methods, define decision steps (milestones), and apply established design principles. Depending on the problem, project leaders and teams need to be willing to adapt their approaches and methods on the basis of importance and urgency. Project leaders must not be dogmatic in their leadership style, must utilise the manifold skills in the team, must provide every team member with individual freedom of action, and must demonstrate decisiveness when it matters. Leadership involves:
Providing timely information by:

- pointing out deviations from the project plan as early as possible
- managing information in a balanced and uniform manner.

Steering individual activities carefully in line with a systematic approach by:

- planning the main project parameters such as deadlines, costs and resources
- pursuing these project targets
- estimating the effort and consequences of any changes
- updating the project plan when necessary.

Representing the team effectively by:

- managing reporting and documentation
- taking personal responsibility for team presentations, etc.

Fostering team building and mutual trust by:

- making and encouraging decisions in difficult situations.

If project leaders cannot fulfil these requirements, then the simultaneous engineering approach will be difficult to adopt.

Team behaviour also plays an essential role. Teamwork benefits product development and individual team members (see Section 4.3.1), however it can also give rise to the following problems [4.2]:

- groups or teams that work together for a long time tend to oversimplify
- control of team effectiveness can decline
- teams begin to conform, which can lead to the protection of competences and the overestimation of capabilities
- groups who have worked together successfully for a long time develop a self-confidence that is not always justified
- within a team one may find opinionated individuals who dominate others and who need careful management
- some team members may sit back and not pull their weight.

In addition to adopting an understanding leadership style, these problems can be addressed specifically by creating small teams, encouraging an open dialogue, and, if necessary, removing or adding team members. Ideally, teams should be dissolved when their project goals have been achieved.

Dörner and Badke-Schaub [4.2, 4.10] have written about the effectiveness of groups and teams in comparison with individuals. Although general statements are difficult to make, it appears that group opinions settle at a relatively high level. This means that results are never as good as those of the best individuals, but also never as bad as those of the worst individuals. An idea or the work of an individual can stand out from that of the team, but can also be significantly worse.

This implies that surprising proposals from individuals should not be suppressed. On the contrary, these should be developed to a point where a clear comparison with the team result is possible. In a team one cannot rely on, or even expect, valuable individual original contributions to arise, so opportunities should be created to encourage them. Team building does not automatically guarantee good solutions. Company culture and leadership style remain fundamental for effective teamwork and successful individual work.

5 Task Clarification

5.1 Importance of Task Clarification

The design task is generally presented to the design and development department in one of the following forms:

- as a development order (from outside or from the product planning department in the form of a product proposal)
- as a definite order
- as a request based on, for instance, suggestions and criticism by sales, research, test or assembly staff, or originating in the design department itself.

The task description contains not only statements about the product, such as its functionality and performance, but also statements about deadlines and cost targets. The design and development department now faces the problem of identifying the requirements that determine the solution and embodiment and formulating and documenting these quantitatively as far as possible. In order to achieve this, the following questions need to be answered in close cooperation with the client or proposer:

- What are the objectives that the intended solution is expected to satisfy?
- What properties must it have?
- What properties must it not have?

The result or this process is a *requirements list*. This document thus represents the specification against which the success of the design project can be judged.

In so far as this has not already been done in product planning (see Section 3.1), the design and development department should undertake the situation analysis described in 3.1.4 in order to specify the product situation and to identify future developments.

A useful method used to support the preparation of the requirements list is *Quality Function Deployment* (QFD) (see Section 10.5). QFD helps to translate customer wishes into product requirements.

5.2 Setting Up a Requirements List (Design Specification)

The main working steps required to set up a requirements list are shown in Figure 5.1. The procedure involves two stages. In the first stage the obvious requirements are defined and recorded. In the second stage these requirements are refined and extended using special methods.

The following sections describe the contents and format of a requirements list, along with the individual working steps.

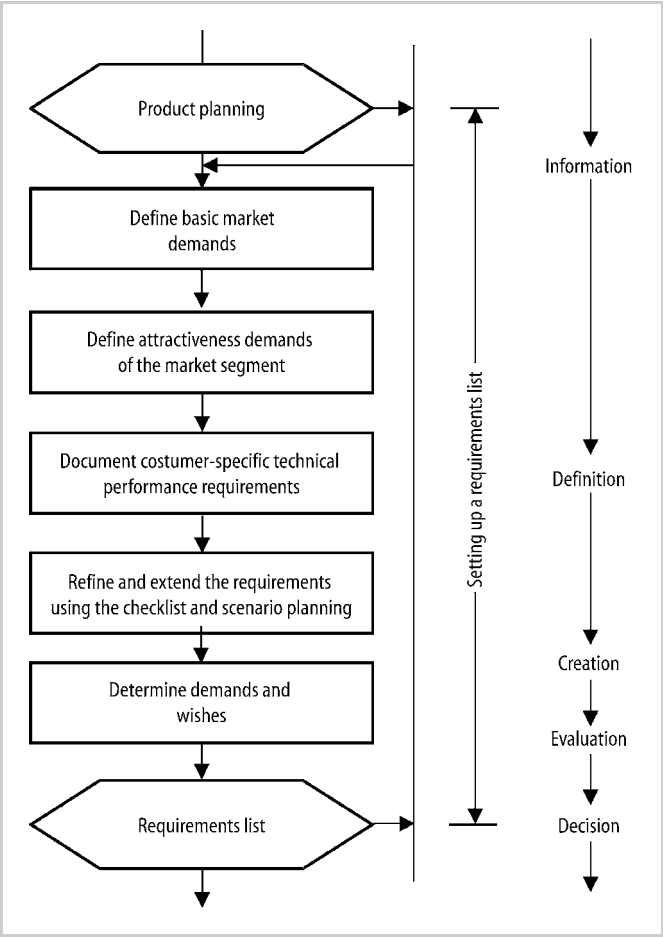


Figure 5.1. Main working steps required to set up a requirements list

5.2.1 Contents

When preparing a detailed requirements list it is essential to clearly elaborate the goals and the circumstances under which they have to be met. The resulting requirements must be identified either as demands or wishes.

Demands are requirements that must be met under all circumstances; in other words, if any of these requirements are not fulfilled the solution is unacceptable (for instance such qualitative demands as "suitable for tropical conditions", "splashproof", etc.). Minimum demands must be formulated as such (for example $P > 20$ kW; $L < 400$ mm).

Wishes are requirements that should be taken into consideration whenever possible, perhaps with the stipulation that they only warrant limited increases in cost, for example, central locking, less maintenance, etc. It is advisable to classify wishes as being of major, medium or minor importance [5.4].

The distinction between demands and wishes is also important at the evaluation stage, since selection (see Section 3.3.1) depends on the fulfilment of demands, while evaluation (see Section 3.3.2) is only performed on variants that already meet the demands.

Even before a certain solution is adopted, a list of demands and wishes should be set up and the quantitative and qualitative aspects tabulated. Only then will the resulting information be adequate:

- Quantity: All data involving numbers and magnitudes, such as number of items required, maximum weight, power output, throughput, volume flow rate, etc.
- Quality: All data involving permissible variations or special requirements, such as waterproof, corrosionproof, shockproof, etc.

Requirements should, if possible, be quantified and, in any case, defined in the clearest possible terms. Special indications of important influences, intentions or procedures may also be included in the requirements list, which is thus an internal digest of all the demands and wishes expressed in the language of the various departments involved in the design process. As a result, the requirements list not only reflects the initial position but, since it is continually reviewed, also serves as an up-to-date working document. In addition, it is a record that can, if necessary, be presented to the Management Board and the sales department so that they may make their objections known before the actual work is started.

5.2.2 Format

The requirements list should at least contain the following information in a structured format (see also Figure 5.2).

- user: company or department
- project or product name
- requirements labelled as demands or wishes
- person responsible for each requirement
- date of issue for the overall requirements list
- date of last change
- version number and/or index number
- page number.

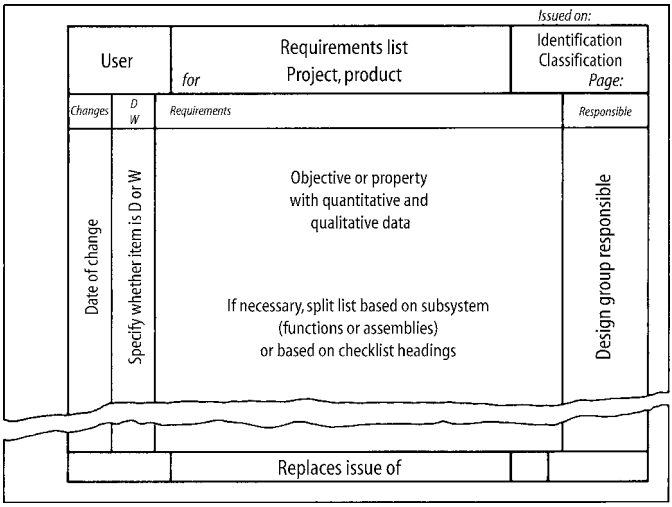


Figure 5.2. Layout of a requirements list

It is helpful if the format of the requirements list becomes a company standard so that it can be used, elaborated and adopted by as many departments as possible. Figure 5.2 is thus no more than a suggestion that can, of course, be modified at will.

It may prove useful to set up the requirements list based on subsystems (functions or assemblies) where these can be identified, or else based on checklist headings (see Figure 5.3). With established solutions, where the assemblies to be developed or improved are already determined, the requirements list must be arranged in accordance with these: special design groups are usually put in charge of the development of each assembly. With motor cars, for instance, the requirements list can be subdivided into engine, transmission and bodywork development.

It is extremely useful to *record the source* of demands and wishes. It is then possible to go back to the proposers of requirements and to enquire into their actual motives. This is particularly important when the question arises of whether or not the demands can be changed in the light of subsequent developments.

Any *changes in*, and *additions to*, the original task that may result from a better understanding of solution possibilities or from possible changes in emphasis must always be entered into the requirements list, which will then reflect the progress of the project at any particular time.

Responsibility for this work is placed on the chief designer. The updated requirements list should be circulated among all departments concerned with the development of the product (management, sales, accounts, research, etc.). The requirements list should only be changed or extended due to a decision of those in charge of the overall project and by following a formal change management procedure.

Main headings	Examples
Geometry	Size, height, breadth, length, diameter, space requirement, number, arrangement, connection, extension
Kinematics	Type of motion, direction of motion, velocity, acceleration
Forces	Direction of force, magnitude of force, frequency, weight, load, deformation, stiffness, elasticity, inertia forces, resonance
Energy	Output, efficiency, loss, friction, ventilation, state, pressure, temperature, heating, cooling, supply, storage, capacity, conversion.
Material	Flow and transport of materials. Physical and chemical properties of the initial and final product, auxiliary materials, prescribed materials (food regulations etc)
Signals	Inputs and outputs, form, display, control equipment.
Safety	Direct safety systems, operational and environmental safety.
Ergonomics	Man-machine relationship, type of operation, operating height, clarity of layout, sitting comfort, lighting, shape compatibility.
Production	Factory limitations, maximum possible dimensions, preferred production methods, means of production, achievable quality and tolerances, wastage.
Quality control	Possibilities of testing and measuring, application of special regulations and standards.
Assembly	Special regulations, installation, siting, foundations.
Transport	Limitations due to lifting gear, clearance, means of transport (height and weight), nature and conditions of despatch.
Operation	Quietness, wear, special uses, marketing area, destination (for example, sulphurous atmosphere, tropical conditions).
Maintenance	Servicing intervals (if any), inspection, exchange and repair, painting, cleaning.
Recycling	Reuse, reprocessing, waste disposal, storage
Costs	Maximum permissible manufacturing costs, cost of tooling, investment and depreciation.
Schedules	End date of development, project planning and control, delivery date

Figure 5.3. Checklist for setting up a requirements list

5.2.3 Identifying the Requirements

As a rule, the first requirements list undertaken is the most difficult to set up. Experience greatly facilitates the compilation of subsequent ones. After a relatively short period, several examples become available that can be used as the starting point for subsequent requirements lists.

The main issue associated with setting up a requirements list is the quantity and quality of the documents and data that are supplied with the design task. Depending on the branch of engineering, not all the expected product properties

will have been defined and documented. The rest of them are expected by the customers but not made explicit, i.e. they are implicit requirements. The following questions therefore need to be answered:

- What is the problem really about?
- Which implicit wishes and expectations are involved?
- Do the specified constraints actually exist?
- What paths are open for development?

It is therefore important for the design and development department to understand the customers or the market segment involved. The basis of the requirements list is often a contract that has been signed with a customer. This contract usually includes the agreed product properties and performance data, product liability regulations, and the guidelines that have to be applied.

In a first exploratory step, the statements and requirements in the contract are translated into product-relevant parameters that designers and engineers can apply. This is relatively straightforward to do because the product specification in a contract involves explicit requirements. A bigger problem is how to deal with the implicit requirements; although they are not expressed they still have a very negative impact if they are not fulfilled. What effects, for example, do statements such as "simple maintenance" have on the embodiment of the product, and how can such statements be formulated as specific requirements? How difficult it is to formulate a requirements list depends on the type of customer and, in principle, two types can be distinguished:

- *anonymous customers:* these include a particular market segment, those identified by the sales department without a customer order, and those identified by the product planning department.
- *specific customers:* these not only include individual customers who place an order, but also market segments that are served by many companies with similar products in which requirements have become standardised, e.g. those for "compact cars" and "family cars". Although the actual customers in such cases are anonymous, they can, in effect, be treated as specific customers.

According to Kramer [5.3], some specific types of requirement can be formulated for each type of customer.

Basic requirements are always implicit requirements, i.e. they are not articulated by the customer. Their fulfilment is self-evident and vital for the customer. Success or failure of a product is determined by these requirements. For example, for a follow-on product the customer generally expects energy consumption and operating costs to be reduced. It is essential for the design and development department to recognise the importance of these implicit requirements. The sales department or product management must supply information about these requirements, along with the thoughts and expectations of the customers.

Technical performance requirements are explicit requirements. They are articulated by the customer and can usually be specified precisely. For example, a new

engine may have to have 15 kW of power and weigh not more than 40 kg. Such concrete values are used by customers when comparing competing products. The importance of the individual parameters is determined by the customers themselves.

Attractiveness requirements are again implicit requirements. Customers are usually not aware of these; however, they are used to differentiate between competing products. In general customers are not willing to pay higher prices for these additional product properties. Consider an example from a motor car where the number of standard colours and the available combinations of external and internal colour schemes are such requirements.

5.2.4 Refining and Extending the Requirements

Two methods have been developed to refine and extend the requirements list defined thus far:

- follow a checklist
- create scenarios.

The checklist shown in Figure 5.3 is a generic one based on ideas described in Section 2.1.7. The items in this list are checked against the existing task and its requirements in order to obtain further requirements. A further checklist can be found in Ehrlenspiel's book [5.1].

When creating scenarios, the product life from production to disposal is considered and sketched out. For every stage, a scenario is developed and the following questions asked:

- What might happen to the product? Examples: What kind of state might it find itself in? How might it be treated and used? Who might use it or come into contact with it? Where might it be used?
- How should the product react? Examples: What level of tolerance to failure should be built in? How should dangerous situations be avoided?

The answers to these questions are used to formulate further product requirements. Most of these requirements will not be very specific, i.e. they cannot be translated into the product parameters that determine solutions or embodiments. For example, the previously mentioned statement "simple maintenance" needs to be specified in more detail. Kramer [5.3] proposes the following three-step procedure to achieve this.

First step (statement)

- Customer's need: simple maintenance.

Second step (development)

- Customer's requirements:
  1. Provide long maintenance interval

2. Enable simple maintenance
3. Make maintenance procedures easy to learn.

Third step (refinement)

- Provide long maintenance interval:
  1. Maintenance interval at least 5 000 operating hours
  2. Grease cams every 10 000 operating hours.
- Enable simple maintenance:
  1. Fit maintenance access covers with manual locks
  2. Fit cams with lubricating points that fit standard grease guns
  3. Leave space for oil drip tray
  4. Provide locating features to assist when refitting access covers.
- Make maintenance procedures easy to learn:
  1. Add a separate section in the operating manual describing maintenance procedures
  2. Provide labels indicating the locks that need to be undone for maintenance
  3. Indicate the directions of the maintenance operations with etched arrows.

The results of the third step are then added to the requirements list.

When clarifying the task, one should start by collecting the essential functions and the existing task-specific constraints with respect to the energy, material and signal transformations. When all of the information is available, it must be grouped, ordered and labelled.

In 5.2.1 we pointed out the essential differences between demands and wishes. In many cases it is clear from the outset whether a requirement is a demand or a wish. However, a definitive assignment is required before the requirements list is released. If necessary, further information should be collected. Wishes should be formulated such that their weighting can be established. Initially it is often useful to express such weightings qualitatively rather than quantitatively, because the estimates often change as the understanding of the task develops.

5.2.5 Compiling the Requirements List

In the light of arguments advanced in this chapter, the following general method of compiling a requirements list can now be recommended:

1. Identify the requirements:
   - Check the customer contract or the sales documents for technical requirements and define and document them.
   - Refer to the items of the checklist (Figure 5.3) and determine the quantitative and qualitative data.
   - Create scenarios that consider all stages in the product's life and thus derive further requirements.

- Refine by asking:
  - What objectives must the solution satisfy?
  - What properties must it have?
  - What properties must it not have?
- Collect further information.
- Specify demands and wishes clearly.
- If possible, rank wishes as being of major, medium or minor importance.

2. Arrange the requirements in a clear order:
   - Define the main objective and the main characteristics.
   - Split into identifiable subsystems, functions, assemblies, etc., or in accordance with the main headings of the checklist.
3. Enter the requirements list on standard forms and circulate among interested departments, licensees, directors, etc.
4. Examine objections and amendments and, if necessary, incorporate them into the requirements list.

Once the task has been adequately clarified, and the relevant departments are satisfied that the listed requirements are technically and economically attainable, the way is clear for the conceptual design phase.

5.2.6 Examples

Figure 5.4 shows a requirements list for a printed circuit board positioning machine, illustrating the main characteristics of the content and the format of requirements lists. It has been structured according to the main characteristics given in Figure 5.3. The requirements have been split into demands and wishes, and, where possible and necessary, quantified. Modifications and amendments with their dates are also shown. The latter were the result of an intensive discussion of a first draft of the requirements list (first version 21st April 1988).

Requirements lists based on the above recommendations are provided in Figures 6.4, 6.27 and 6.43 as further examples.

5.3 Using Requirements Lists

5.3.1 Updating

In principle, requirements lists should be binding and complete. However, initially a requirements list is always provisional because, as the design process proceeds, it grows and changes. Any attempt to formulate all possible requirements at the start of a project will fail and would cause considerable delays. Looking at the inputs

Siemens		Requirements list for a printed circuit board positioning machine	Issued on 27/04/88 Page: 1
Changes	D W	Requirements	Responsible
		1. Geometry: dimensions of the test sample	Langner's group
		Circuit board:	
	D	Length = 80 – 650 mm	
	D	Breadth = 50 – 570 mm	
	W	Height = 0.1 – 10 mm	
	D	Required hight = 1.6 – 2 mm	
	W	Clearance between basic grid boards ≤ 120 mm	
	D	'Clamping area' ≤ 2 mm (edges of the board)	
		2. Kinematics:	
27/04/88	D	Precise positioning of the test sample	
27/04/88	D	Minimum of 2 mm displacement of the test sample normal to the board	
27/04/88	D	Feedback to transfer position	
	W	separate stations for input and output	
	D	Design of clearance zone	
	W	Minimum handling time (as fast as possible)	
		3. Forces:	
	D	Weight of the test sample ≤ 1.7 kg	
27/04/88	W	Maximum weight of the test sample ≤ 2.5 kg	
		4. Energy:	
	D	Electrical and /or pneumatic (6 – 8 bar)	
		5. Material:	
	D	Free from rust	
	D	Isolation between test sample and testing device	
27/04/88	W	Thermal expansion of testing device adjusted to expansion of printed circuit	
27/04/88	D	Consideration of influence of temperature	
27/04/88	D	Temperature range: 15–40 °C	
27/04/88	D	Humidity: 65 %	
27/04/88	W	Circuit boards: epoxy-fiberglass sheet	
27/04/88	D	No condensation	
		6. Safety:	
27/04/88	D	Operator Safety	
		7. Production:	
		Consideration of tolerance build up	
		8. Operation:	
	D	No contamination inside the testing device	
	D	Destination: production line	
		9. Maintenance:	
	W	Maintenance intervall > 10^6 test operations	
		10. Schedule:	
	D	Embodiment finished by July 1988	
		Replaces issue of 21/04/88	

Figure 5.4. Requirements list for a printed circuit board positioning machine (Siemens AG)

and outputs of individual working steps in the design process, the reasons become clear. For example, in the final stages of designing a motor car, the thicknesses of all of the individual layers of body paint need to be known. However, to develop the concept, these data are not relevant. The paintwork requirements therefore do not have to be specified until much later in the process.

Thus, working with binding yet provisional requirements lists takes into account the fact that not all of the data and requirements are known or have to be known at the beginning of a design process. Only those requirements that are absolutely necessary in order to be able to proceed to the next working step need to be documented. At the start of a project, it is important to specify those parameters and properties that:

- define the particular concept
- influence the product structure
- determine the overall embodiment of the product.

The contents of a requirements list therefore depend on the state of the product design and the stage of the design process. The list has to be continuously amended and extended. Managing requirements lists in this way avoids having to deal with questions and requirements before they can be adequately answered and specified.

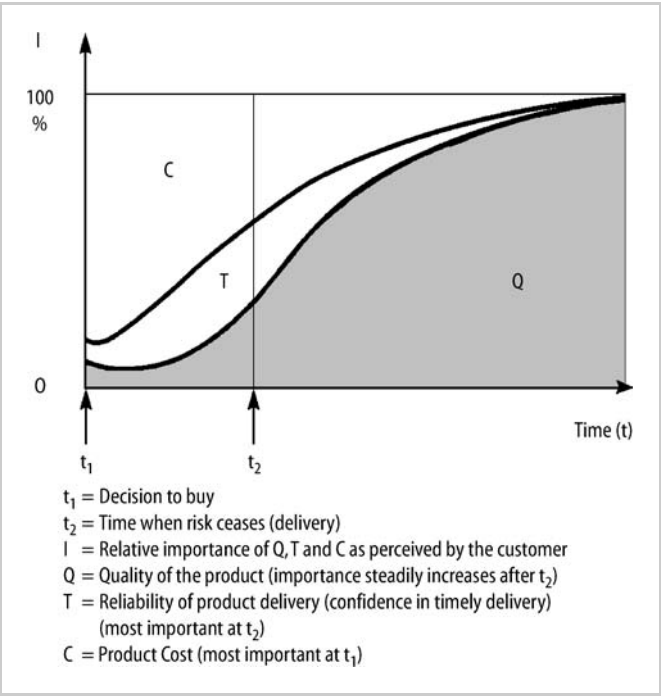


Figure 5.5. Changing appreciation of product quality by the customer

Product requirements frequently change with time, during both product creation and product use.

During Product Creation, customers often change their demands and wishes. This happens when customers gain new knowledge and understanding and when the planned application has been extended. This is typical of capital equipment because of its long development process. For railway rolling stock, for example, it is possible that during the development process the rail network has been extended with the consequence that specified powers and capacities are no longer sufficient.

During product use, customer appreciation of the product can change and, as a consequence, the requirements and their relative importance can also change. For example, the longer a product is in service, the more important quality issues such as long maintenance intervals and reliability become, see Figure 5.5 [5.3].

The fact that requirements can change must be considered when setting up a requirements list. A mutually satisfactory requirements management process is therefore essential in order to ensure good and lasting relations between a company and its customers.

5.3.2 Partial Requirements Lists

It is often beneficial for specialist areas or departments of a company to prepare so-called "partial requirements lists" documenting only their particular requirements. This removes the need for the design and development department to spend time collecting more information and data than are strictly relevant to them, see Figure 5.6 [5.2].

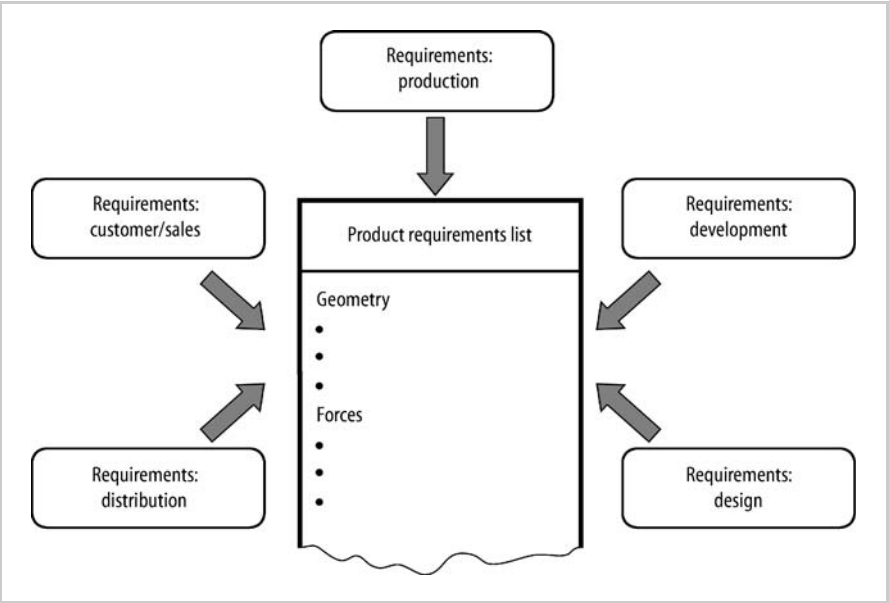


Figure 5.6. Product requirements list compiled from partial requirements lists

The product requirements list is a compilation of all of the partial requirements lists. An important role of project management is to ensure that the partial requirements lists cover all areas and are compatible with each other. Modern engineering data management systems [5.5] support their efficient administration and editing.

5.3.3 Further Uses

Even when the design is not original, and the solution principle and layout are fixed so that nothing more than *adaptations* or *dimensional changes* need to be made in a familiar area, orders should nevertheless be executed on the basis of requirements lists, which can then take the form of templates or questionnaires. These should be constructed in such a way that information for electronic data processing and quality control can be read off directly. As a result, requirements lists become sources of information for direct action.

Beyond that, requirements lists, once compiled, are an invaluable *source of information* about the required or desired properties of the product, and hence extremely helpful for further developments, negotiations with suppliers, etc. Setting up requirements lists for existing products can also provide a very valuable source of information for the subsequent development and rationalisation of those products.

The examination of a requirements list during project meetings or before assessing various designs is an extremely useful procedure. All of those involved are placed in possession of all of the available information and all salient evaluation criteria are brought home to them.

Requirements lists are an important basis for knowledge management systems. Stored in such systems, requirements lists provide a very valuable source of knowledge about previous projects that can often be reused.

5.4 Practical Application of Requirements Lists

In the last few years it has been shown that, at least for original designs, the formulation of a requirements list is a very efficient method for solution development and has been broadly adopted by industry. When used in practice, however, the following issues often arise:

- *Obvious requirements*, such as low-cost production, ease of assembly, are often not included in the requirements list. One should take care that these issues are both addressed and expressed precisely.
- In an early stage of the project it is not always possible to make *precise statements* in the requirements list. The statements have to be amended or corrected during the design and development process.
- A *stepwise development* of the requirements list is very useful when tasks are poorly defined. In these cases the requirements should be formulated more precisely as soon as possible.

- During the formulation of requirements lists or related discussions, *functions* or *solution ideas* are often mentioned. This is not wrong. They can encourage a clearer formulation of the requirements and even lead to the identification of new requirements. The solution ideas or proposals generated should be recorded so that they can be used later in the systematic search for solutions. However, they should not enter—and possibly bias—the requirements list.
- The identification of *deficiencies* and *failures* can initiate requirements that must then be formulated in a solution-neutral way. Failure analysis is often the starting point for a requirements list.
- For *adaptive* or *variant designs*, designers should still make requirements lists for themselves, even when the task is small.
- Setting up requirements lists should not be formalised too strictly. *Guidelines* and *forms* are only a *means* to prevent important issues from being forgotten and to provide a supporting structure. If one deviates from the recommendations in this book, one should a least consider the main characteristics and distinguish between demands and wishes.

6 Conceptual Design

Conceptual design is the part of the design process where—by identifying the essential problems through abstraction, establishing function structures, searching for appropriate working principles and combining these into a working structure—the basic solution path is laid down through the elaboration of a solution principle. Conceptual design *specifies the principle solution*.

From Figure 4.3 we can see that the conceptual phase is preceded by a decision. The purpose of this decision is to answer the following questions based on the requirements list agreed upon during task clarification:

- Has the task been clarified sufficiently to allow the development of a solution in the form of a design?
- Is a conceptual elaboration really needed, or do known solutions permit direct progress to the embodiment and detail design phases?
- If the conceptual stage is indispensable, how and to what extent should it be developed systematically?

6.1 Steps of Conceptual Design

According to the procedural plan outlined in Section 4.2, the conceptual design phase follows the clarification of the task. Figure 6.1 shows the steps involved, correlated in such a way as to satisfy the principles of the general problem solving process set out in Section 4.1.

The reasons for the individual steps have been examined in Section 4.2 and need not be discussed further here. It should, however, be mentioned that refinements of any one of the steps by reiteration on a higher information level should be made whenever necessary. The loops involved have been omitted from Figure 6.1 for the sake of greater clarity.

The individual steps and the appropriate working methods for the conceptual design phase will now be examined in detail.

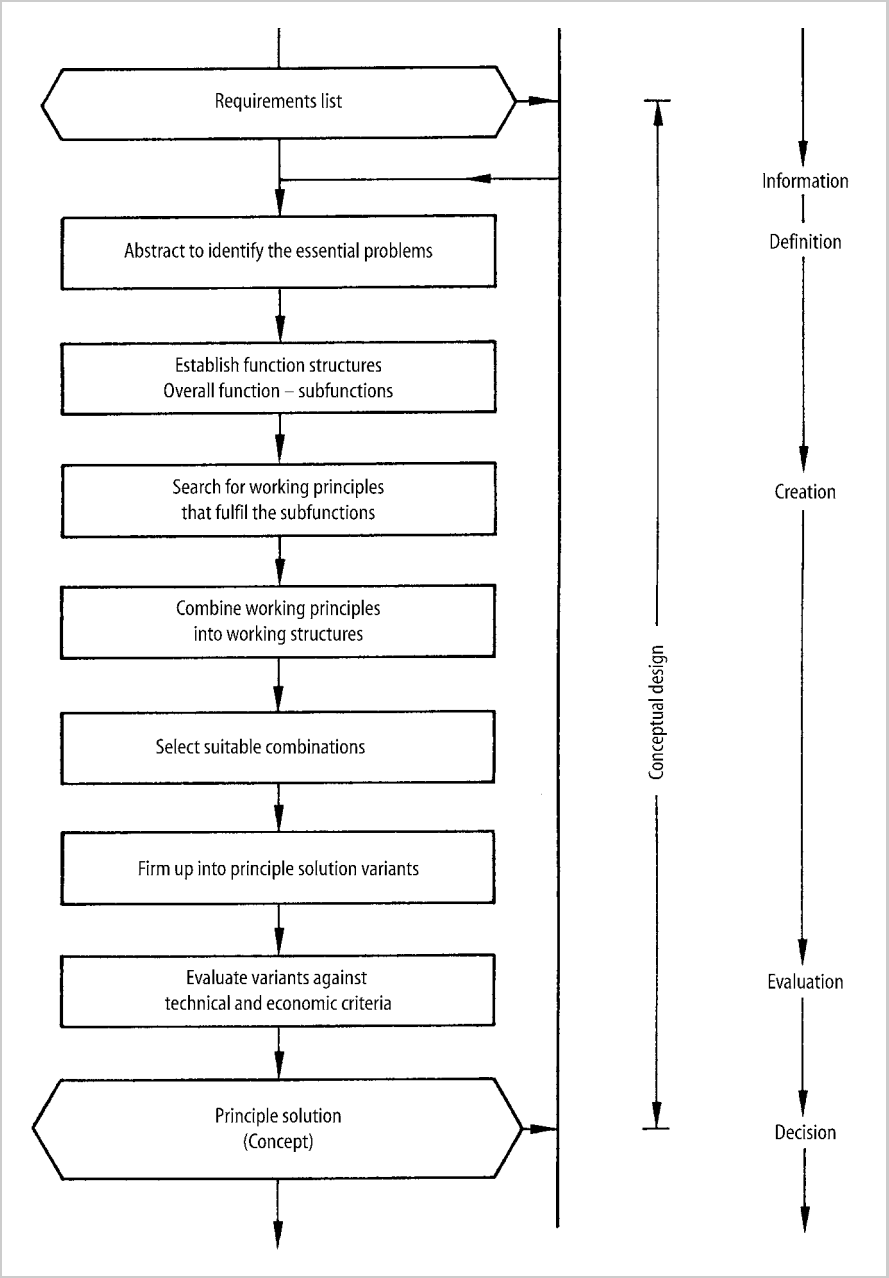


Figure 6.1. Steps of conceptual design

6.2 Abstracting to Identify the Essential Problems

6.2.1 Aim of Abstraction

Solution principles or designs based on traditional methods are unlikely to provide optimum answers when new technologies, procedures, materials, and also new scientific discoveries, possibly in new combinations, hold the key to better solutions.

Every industry and every design office is a store of experiences as well as of prejudices and conventions which, coupled to the wish to minimise risks, stand in the way of better and more economic but unconventional solutions. The client, customer or product planning group might have included specific proposals for a solution in the requirements list. It is also possible that during the discussion of individual requirements, ideas and suggestions for realising a solution will emerge. In the unconscious, at least, certain solutions might exist. Perhaps concrete ideas already exist, however these could be based on fixed ideas and fictitious constraints.

In their search for optimum solutions, designers, far from allowing themselves to be influenced by fixed or conventional ideas, must therefore examine very carefully whether novel and more suitable paths are open to them. In order to solve the problem of fixation and sticking with conventional ideas, *abstraction* is used. This means ignoring what is particular or incidental and emphasising what is general and essential. Such generalisation leads straight to the *crux of the task*. If it is properly formulated, then the overall function and the essential constraints become clear without prejudicing the choice of a particular solution in any way.

As an example, consider the improvement of a labyrinth seal in a high-speed turbine in accordance with a set of requirements. The task is described in detail by means of a requirements list and the formulation of the goal to be achieved. In the abstracting approach, the crux of the task would not so much be the design of a labyrinth seal as that of a shaft seal without physical contact, with due regard being paid to certain operating and spatial constraints, and also to cost limits and delivery times. Specifically, the designer should ask whether the crux is:

- to improve the technical functions, e.g. the sealing quality or safety
- to reduce weight or space
- to significantly lower costs
- to significantly shorten delivery times
- to improve production methods.

All of these questions might have to be satisfied by the overall solution, but their importance may differ from case to case. Nevertheless, due regard must be paid to each of them, since any one of them is likely to provide the impetus for the discovery of a new and better solution principle. New developments involving a proven solution principle, coupled to modifications in production methods, are often imposed by the need to lower costs and shorten delivery times.

Thus, if an improvement in the sealing properties were the crucial requirement in the example we have mentioned, new sealing systems would have to be found.

This would mean studying the flow of fluids in narrow passages and, from the knowledge acquired, providing for better sealing properties, while also satisfying the other subproblems we have mentioned.

If, on the other hand, cost reduction were the crucial point then, after an analysis of the cost structure, one would have to see whether the same physical effects could be produced through the use of cheaper materials, by reducing the number of components or by using a different production process. It is also possible to search for new concepts to achieve a better or at least similar sealing performance for lower cost.

It is the identification of the crux of the task with the functional connections and the task-specific constraints that throws up the essential problems for which solutions have to be found. Once the crux of the task has been clarified, it becomes much easier to formulate the overall task in terms of the essential subproblems as they emerge [6.2, 6.6, 6.13].

6.2.2 Broadening the Problem Formulation

This is the best point in the process to bring in those designers who are actually going to be responsible for the project. Having identified the crux of the task by correct problem formulation, a step-by-step enquiry is now initiated to discover if an extension of, or even a change in, the original task might lead to promising solutions.

An excellent illustration of this procedure has been given by Krick [6.5]. The task he used as an example was an improved method of filling, storing and loading bags of animal feed. An analysis gave the situation shown in Figure 6.2. It would have been a grave mistake to begin immediately by thinking of possible improvements to the existing situation. By proceeding in this way one is likely to ignore other, more useful and more economic solutions. Using abstraction and the systematic extension of what is already known about the task, the following problem formulations are possible, each representing a higher level of abstraction than the last:

1. Filling, weighing, closing and stacking bags of feed.
2. Transferring feed from the mixing silo to stacked bags in the warehouse.
3. Transferring feed from the mixing silo to bags on the delivery truck.
4. Transferring feed from the mixing silo to the delivery truck.
5. Transferring feed from the mixing silo to a delivery system.
6. Transferring feed from the mixing silo to the consumer's storage bins.
7. Transferring feed from ingredient containers to consumer's storage bins.
8. Transferring feed ingredients from their source to the consumer.

Krick has incorporated some of these formulations into a diagram (see Figure 6.3).

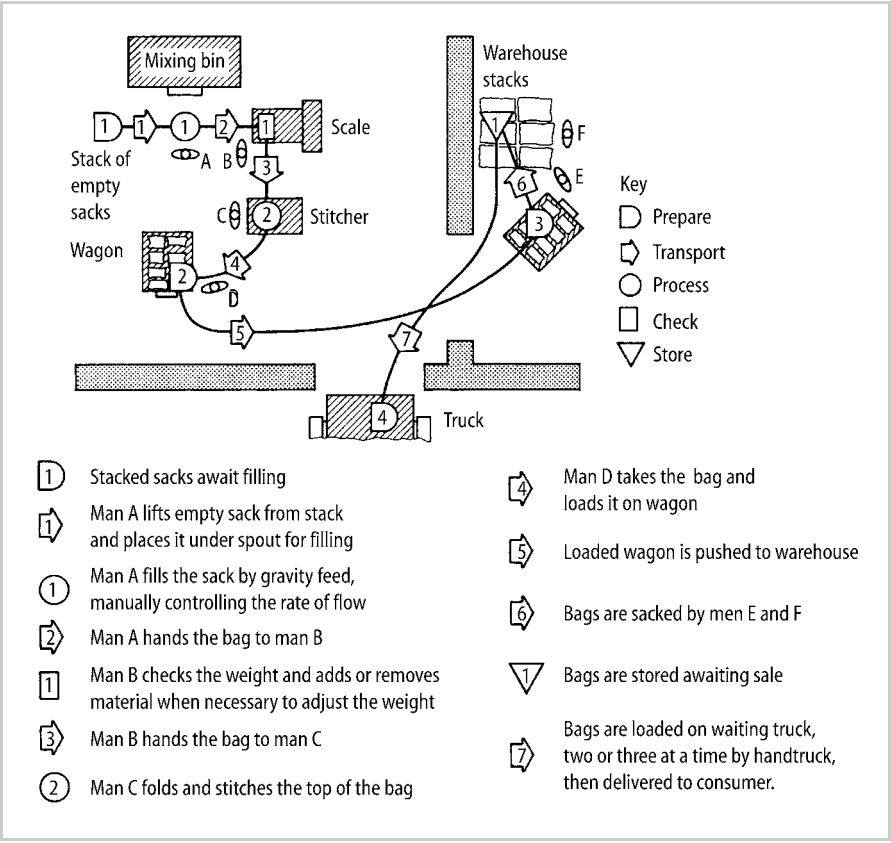


Figure 6.2. The present method of filling, storing and loading bags of feed. After [6.5]

It is characteristic of this approach that the problem formulation is made as broad as possible in successive steps. In other words, the current or obvious formulation is not accepted at face value but *broadened systematically*. Although this may conflict with decisions already taken, it opens up new perspectives. Thus, formulation 8 above is the broadest, the most general and the least circumscribed.

The crux of the task, in fact, is the transport of the correct quantity and quality of feed from the producer to the consumer and not, for instance, the best method of closing or stacking bags, or moving them into the warehouse. With a broader formulation, solutions may appear that render the filling of bags and storing them in the warehouse unnecessary.

How far this process of abstraction is continued depends on the constraints. In the case under consideration, Formulation 8 must be rejected on technical, seasonal and meteorological grounds: the consumption of feed is not confined to harvest time; for various reasons consumers will not want to store feed for a whole year; moreover, they may be reluctant to mix the required ingredients themselves. However, the transfer of feed on demand, for instance, with delivery trucks taking

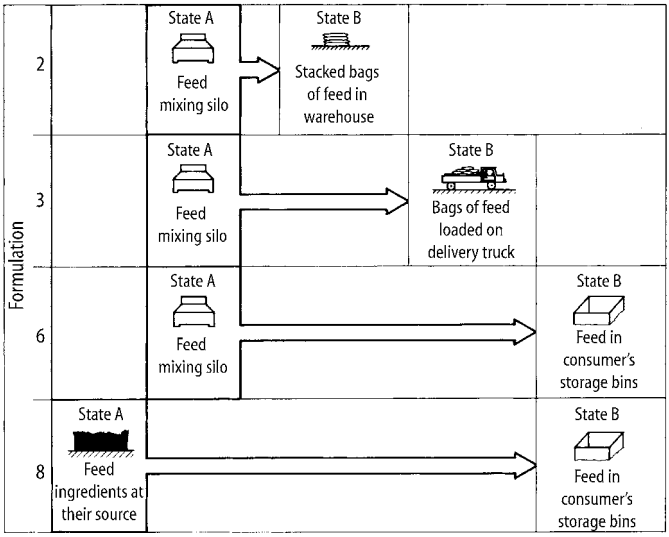


Figure 6.3. Alternative formulations of the feed distribution problem, illustrating progressively broader formulations of a problem. After [6.5]. A = initial state; B = final state

it directly from the mixing silo to consumers' storage bins (Formulation 6), is more economical than intermediate storage in a warehouse and the transport of smaller quantities in bags. In this connection, the reader might recall a development in a different field which culminated in the delivery of ready-mixed concrete direct to building sites in special vehicles.

We have tried to show how comprehensive problem formulation on an abstract plane opens the door to better solutions. This approach, furthermore, helps to raise the influence and responsibility of designers by giving them an overview of the problem and thus involving them in, for instance, environmental protection and recycling. It is useful to analyse the requirements list as set out in the next section.

6.2.3 Identifying the Essential Problems from the Requirements List

The clarification of the task with the help of a requirements list will have helped to focus attention on the problems involved and will have greatly increased the particular level of information (see Chapter 5). Elaborating the requirements list may thus be said to have prepared the way for following steps.

Here the task is to *analyse the requirements list* with respect to the required function and essential constraints in order to confirm and refine the crux of the problem. Roth [6.11] advises that the functional relationships contained in the requirements list should be formulated explicitly and arranged in order of their importance.

That analysis, coupled to the following step-by-step abstraction, will reveal the general aspects and essential problems of the task, as follows:

Step 1. Eliminate personal preferences.

Step 2. Omit requirements that have no direct bearing on the function and the essential constraints.

Step 3. Transform quantitative into qualitative data and reduce them to essential statements.

Step 4. As far as it is purposeful, generalise the results of the previous step.

Step 5. Formulate the problem in solution-neutral terms.

Depending on either the nature of the task or the size of the requirements list (or both), certain steps may be omitted.

Table 6.1 illustrates abstraction based on these steps using the requirements list for a motor vehicle fuel gauge shown in Figure 6.4. The general formulation makes it clear that, with respect to the functional relationships, the problem is the measurement of quantities of liquid, and that this is subject to the essential conditions that the quantity of liquid is changing continuously and that the liquid is in containers of unspecified size and shape.

This analysis thus leads to a definition of the objective on an abstract plane without laying down any particular solution.

In principle, all paths must be left open until such time as it becomes clear which solution principle is the best. Thus designers must question all the constraints they are given and work out with the client or proposer whether or not they should be retained as genuine restrictions. In addition, designers must learn to discard fictitious constraints that they themselves have come to accept, and to that end ask critical questions and test all their presuppositions. Abstraction helps to identify fictitious constraints and to eliminate all but genuine restrictions.

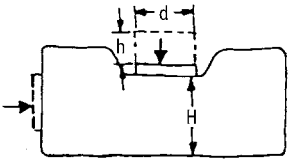
We shall conclude this section with a few examples of purposeful abstraction and problem formulation:

- Do not design a garage door, but look for means of securing a garage in such a way that a car is protected from thieves and the weather.
- Do not design a keyed shaft, but look for the best way of connecting a gear wheel and shaft.
- Do not design a packing machine, but look for the best way of despatching a product safely or, if specific constraints really exist, of packing a product safely, compactly and automatically.
- Do not design a clamping device, but look for a means of keeping a workpiece firmly fixed.

From the above formulations, and this is very helpful for the next step, the final formulation can be derived in a way that does not prejudice the solution, i.e. is *solution-neutral*, and at the same time turns it into a *function*:

- "Seal shaft without contact", not "Design a labyrinth seal".
- "Measure quantity of fluid continuously", not "Gauge height of liquid with a float".
- "Measure out feed", not "Weigh feed in sacks".

TH Darmstadt	Requirements list for fuel gauge	3rd issue 10/07/85 Page 1

Changes	D W	Requirements	Responsible
		• <u>Container</u> 	
		<u>Geometry</u> H = 100 mm – 600 mm Volume: 20 – 160 litres	
	W	2 – 630 litres	
		Shape fixed but unspecified (rigid)	
	W	Container flexible or only partially rigid	
		<u>Material</u>: steel, plastic <u>Connection to container</u> Bayonet socket, clamped connections, top or side:	
	W	d = ∅71 mm, h = 20 mm	
		Tank not pressurised (ventilated) Pressure test for container 0.3 bar	
		<u>Contents, temperature range</u>	

Liquid	Operating Range °C	Storage enviroment °C
Petrol	– 25 to + 65	
Diesel	– 24 to + 65	– 40 to + 100
Engine oil	up to + 140	– 40 to + 100

	W	• <u>Display</u> System with electric input signal – Moving magnet instrument (catalogue) – Bimetallic instrument (catalogue) – Board computer	
		Availabel source of energy: DC at 12 V, 24 V Voltage variation – 10 % to + 25 % of nominal voltage Current consumption max. 300 mA	

	Replaces 2nd issue of 27/06/1973	

Figure 6.4. Requirements list: motor vehicle fuel gauge

TH Darmstadt	Requirements list for fuel gauge	3rd issue 10/07/85 Page 2

Changes	D W	Requirements	Responsible
		• System to be developed	
		Geometry Consider connection constraints to container	
		Kinematics	
	W	No moving parts	
		Energy (see display)	
		Material (see container)	
		Signal	
		∘ Input Minimum measurable content: 3 % of maximum value Reserve tank contents by special signal	
	W	Signal unaffected by angle of liquid surface	
		Possibility of signal calibration	
	W	Possiblity of signal calibration with full container	
		∘ Output	
	W	Output of transmitter: electric signal	
		Output signal accuracy at max. value ±3%	
	W	±2%	
		(together with indicator error ±5%)	
		Under normal conditions, horizontal level, v = const.	
		Able to withstand shocks of normal driving	
		Response sensitivity: 1% of maximum output signal	
	W	0.5% of maximum output signal	
		∘ Connection between input and output	
		Distance container – display: ≠ zero m;	
		3 m–4 m	
	W	1 m–20 m	
		Separate power possible	
		Production large-scale production	

	Replaces 2nd issue of 27/06/1973	

Figure 6.4. (continued)

TH Darmstadt	Requirements list for fuel gauge	3rd issue 10/07/85 Page 3

Changes	D W	Requirements	Responsible
		Test requirements	
		Operating conditions of vehicle	
		Forward acceleration $\pm$ 10 m/s^2	
		Sideways acceleration $\pm$ 10 m/s^2	
		Upward acceleration (vibration), up to 30 m/s^2	
	W	Shocks in forward direction without damage, up to 30 m/s^2	
		Forward tilt up to $\pm 30°$	
		Sideways tilt max 45°	
		Salt spray tests for inside and outside components according to client's requirements (DIN 90905 to be considered)	
		Must conform with heavy vehicle regulations	
		Operation, Maintenance	
	W	Installation by non-specialist	
		Life expectancy 10^4 level changes (full/empty)	
		Minimum of 5 years service life	
		Fuel gauge replaceable	
		Fuel gauge maintenance-free	
	W	Fuel gauge simply modified to suite different container sizes	
		Regulations	
		No regulations relating to explosion safety	
		Quantity	
		10000/day of adjustable type	
		5000/day of the most popular type	
		Costs	
		Manufacturing costs $\leq$ DM 6.00 each (without display)	

	Replaces 2nd issue of 27/06/1973	

Figure 6.4. (continued)

Table 6.1. Procedure during abstraction: motor vehicle fuel gauge based on requirements list given in Figure 6.4

Result of Steps 1 and 2

- Volumes: 20 to 160 litres
- Shape of container: fixed or unspecified (rigid)
- Top or side connection
- Height of container: 100 mm to 600 mm
- Distance between container and indicator: $\neq 0$ m, 3 m to 4 m
- Petrol and diesel, temperature range: $-25\,^{\circ}$C to $65\,^{\circ}$C
- Output of transmitter: unspecified signal
- External energy: DC at 12 V, 24 V. Variation -15% to $+25\%$
- Output signal accuracy at maximum $\pm 3\%$ (together with indicator error $\pm 5\%$)
- Response sensitivity: 1% of maximum signal output
- Possibility of signal calibration
- Minimum measurable content: 3% of maximum value

Result of Step 3

- Various volumes
- Various container shapes
- Various connections
- Various contents (liquid levels)
- Distance between container and indicator: $\neq 0$ m
- Quantity of liquid varies with time
- Unspecified signal
- (with outside energy)

Result of Step 4

- Various volumes
- Various container shapes
- Transmission over various distances
- Measure (continuously changing) quantities of liquid
- (with outside energy)

Result of Step 5 (Problem formulation)

- Measure continuously changing quantities of liquid in containers of unspecified size and shape and indicate the measurements at various distances from the containers.

6.3 Establishing Function Structures

6.3.1 Overall Function

According to Section 2.1.3, the requirements determine the function that represents the intended overall relationship between the inputs and the outputs of a plant, machine or assembly. In Section 6.2 we explained that problem formulation obtained by abstraction does much the same. Hence, once the crux of the overall problem has been formulated, it is possible to indicate an *overall function*

that, based on the *flow of energy, material and signals* can, with the use of a *block diagram*, express the solution-neutral relationship between *inputs* and *outputs*. That relationship must be specified as precisely as possible (see Figure 2.3).

In our example of a fuel gauge (see Figure 6.4), quantities of liquid are introduced into and removed from a container, and the problem is to measure and indicate the quantity of liquid found in the container at any one time. The result, in the liquid system, is a flow of material with the function "store liquid" and, in the measuring system, a flow of signals with the function "measure and indicate quantity of liquid". The second is the overall function of the specific task under consideration, that is, the development of a fuel gauge (see Figure 6.5). That overall function can be broken down into subfunctions in a further step.

6.3.2 Breaking a Function Down into Subfunctions

Depending on the complexity of the problem, the resulting overall function will in turn be more or less complex. By complexity we mean that the transparency of the relationships between inputs and outputs is relatively poor, that the required physical processes are relatively intricate, and that the number of assemblies and components involved is relatively large.

Just as a technical system can be divided into subsystems and elements (see Section 2.1.3), so a complex or *overall function* can be broken down into *subfunctions* of lower complexity. The combination of individual subfunctions results in a *function structure* representing the overall function.

The aims of breaking down complex functions are to:

- determine subfunctions that facilitate the subsequent search for solutions
- combine these subfunctions into a simple and unambiguous function structure.

Let us return to the example of the fuel gauge (see Sections 6.2.3 and 6.3.1). The starting point is the problem formulation for the overall function given in Figure 6.5.

The flow of signals has been treated as the main flow. Associated subfunctions are developed in several steps. As a first step, the contents of the container have to be measured and the resulting signal received. This signal has to be channelled and

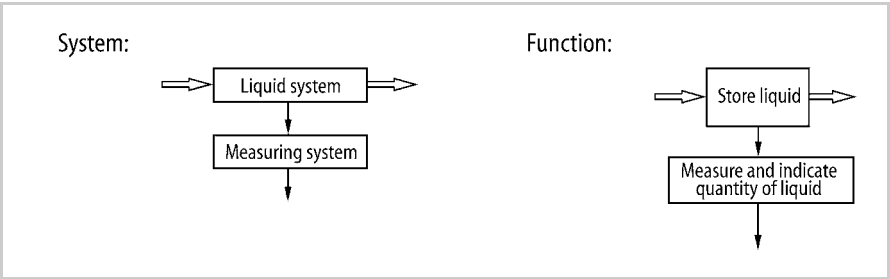


Figure 6.5. Overall functions of the systems involved in measuring the contents of a container. After Figure 6.4 and Table 6.1

finally displayed to the driver to indicate the contents of the container. Thus, three important direct main functions have been identified. Possibly the signal needs to be changed before it can be channelled. Figure 6.6 shows the development and variation of a function structure in accordance with the suggestions set out in this section.

Since the requirements list also provides for measurements in containers of different sizes holding varying initial quantities of liquid, an adjustment of the signal to the respective size of the container is expedient, and is accordingly introduced as an auxiliary function. Measurements in containers of various unspecified shapes will, in certain circumstances, demand the correction of the signal as another auxiliary function. The measuring operation may require a supply of external energy, which must then be introduced as a further flow. Finally, consider the system boundary. If existing indicating instruments are to be used, the device will have to emit an electric output signal. If they are not, then the subfunctions "channel signal" and "indicate signal" must be included in the search for solutions. In this way, a function structure with suitable subfunctions can be developed. The individual subfunctions are of a lower complexity than the overall function and, furthermore, it will become clear which subfunction provides the most useful starting point for the search for solutions.

In our example, this important solution-determining subfunction, that has the working principle upon which the others clearly depend, is "receive signal" (see Figure 6.6). The initial search for solutions should therefore focus on this subfunction. The solution selected for this will largely decide to what extent individual subfunctions can be changed round or omitted. It also allows for better judgement of whether to use existing channelling and display solutions or whether to seek a new solution for these subfunctions, i.e. an extension of the system boundary.

Further recommendations for identifying and formulating subfunctions are now described.

It is useful to start by determining the *main flow* in a technical system, if this is clear. The *auxiliary flows* should only be considered later. When a basic function structure, including the most important links, has been found, it is easier to undertake the next step; that is, to consider the auxiliary flows with their subfunctions and to achieve a further subdivision of complex subfunctions. For this step it is helpful to create a temporary working structure or a solution for the *basic function structure*, without, however, prejudicing the final solution.

The optimum method of breaking down an overall function—that is, the number of subfunction levels and also the number of subfunctions per level—is determined by the relative novelty of the problem and also by the method used to search for solutions. In the case of *original designs*, neither the individual subfunctions nor their relationships are generally known. In that case, the search for and establishment of an optimum function structure constitute some of the most important steps of the conceptual design phase. In the case of *adaptive designs*, on the other hand, the general structure with its assemblies and components is much more well-known, so that a function structure can be obtained by analysing the existing product. Depending on the special demands of the requirements list, that function

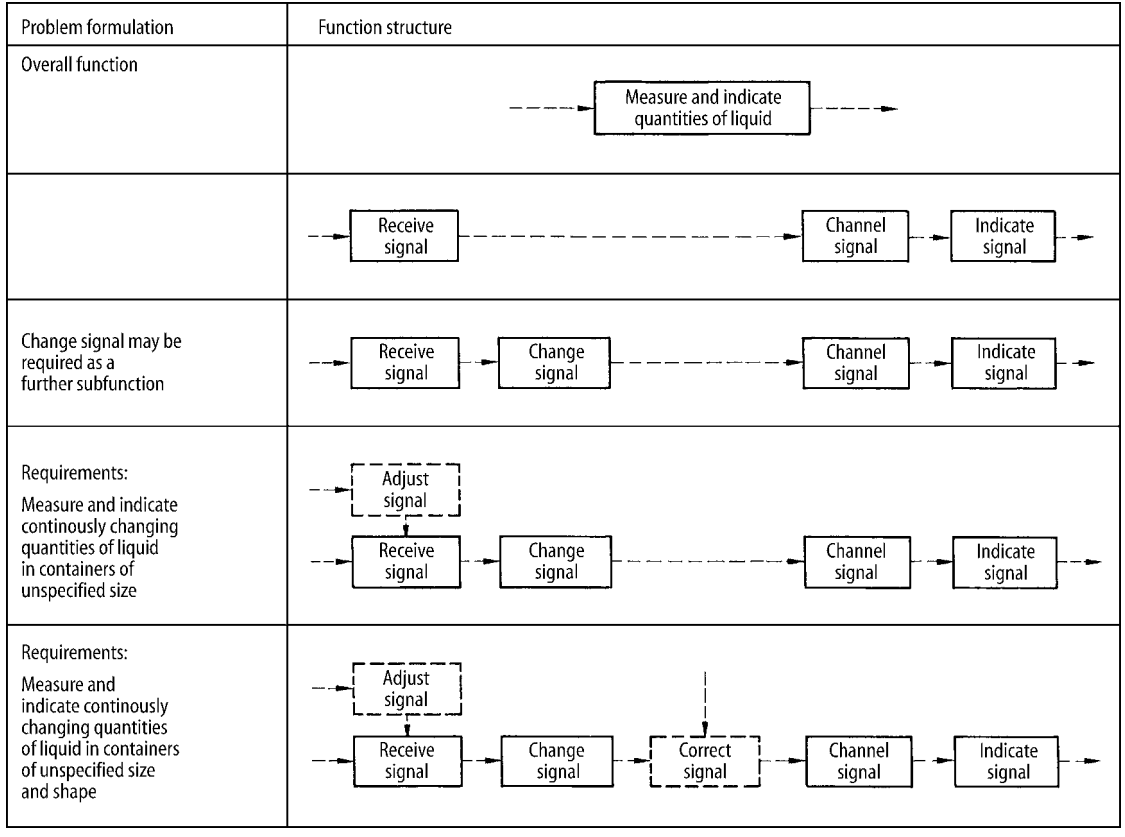


Figure 6.6. Development of a function structure for a fuel gauge

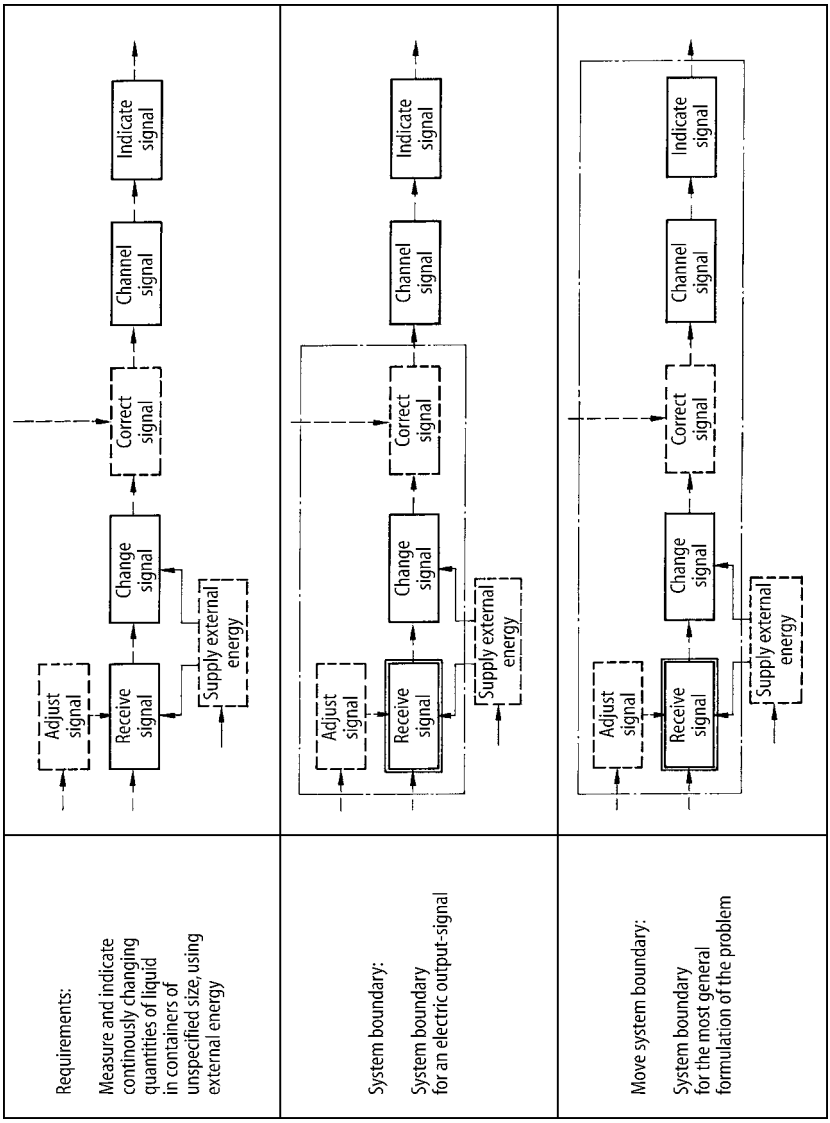


Figure 6.6. (continued)

structure can be modified by the variation, addition or omission of individual subfunctions or by changing the way that they are combined.

Function structures are of great importance in the development of modular systems. For this type of *variant design*, the physical structure—that is, the assemblies and individual components used as building blocks and also their interfaces—must be reflected in the function structure (see also Section 9.2.1).

A further advantage of setting up a function structure is that it permits the *clear definition* of existing subsystems or of those to be newly developed, so that they can be *dealt with separately*. If existing assemblies can be assigned directly as complex subfunctions, the subdivision of the function structure can be discontinued at a fairly high level of complexity. In the case of new assemblies or those requiring further development, however, the division into subfunctions of decreasing complexity must be continued until the search for solutions seems promising. By adapting function structures to the novelty of the task or the subsystem, the use of function structures can save a great deal of time and money.

Apart from aiding in the search for solutions, function structures or their subfunctions can also be used for purposes of classification. Examples are the "classifying criteria" of classification schemes (see Section 3.2.3) and the subdivision of design catalogues.

It may prove expedient not only to set up task-specific functions, but also to elaborate the function structure from *generally valid subfunctions* (see Figure 2.7). The latter recur in technical systems, and may be helpful when searching for a solution since they may lead to the discovery of task-specific subfunctions or because design catalogues may list solutions for them. Defining generally valid functions can also be of use when varying function structures, for example to optimise the energy, material and signal flows. The following list and examples should be helpful in this regard.

Conversion of energy:

- Changing energy (e.g. electrical into mechanical energy)
- Varying energy components (e.g. amplifying torque)
- Connecting energy with a signal (e.g. switching on electrical energy)
- Channelling energy (e.g. transferring power)
- Storing energy (e.g. storing kinetic energy)

Conversion of material:

- Changing matter (e.g. liquefying a gas)
- Varying material dimensions (e.g. rolling sheet metal)
- Connecting matter with energy (e.g. moving parts)
- Connecting matter with signal (e.g. cutting off steam)
- Connecting different types of materials (e.g. mixing or separating materials)
- Channelling material (e.g. mining coal)
- Storing material (e.g. keeping grain in a silo)

Conversion of signals:

- Changing signals (e.g. changing a mechanical into an electrical signal, or a continuous into an intermittent signal)
- Varying signal magnitudes (e.g. increasing a signal's amplitude)
- Connecting signals with energy (e.g. amplifying measurements)
- Connecting signals with matter (e.g. marking materials)
- Connecting signals with signals (e.g. comparing target values with actual values)
- Channelling signals (e.g. transferring data)
- Storing signals (e.g. in databases)

In many cases in industry it may not be expedient to build up a function structure from generally valid subfunctions, because they are, in fact, too general and thus do not provide a sufficiently concrete picture of the relationships to aid the subsequent search for solutions. In general, a clear picture only emerges after adding more task-specific details (see Section 6.3.3).

To illustrate the approach some expamples follow. Figures 6.7 and 6.8 show the function structure of a tensile testing machine with a relatively complex flow of energy, material and signals. In this type of overall function, the function structure is built up step-by-step from subfunctions, with attention initially focused on essential main functions. Thus, on a first functional level, only the subfunctions

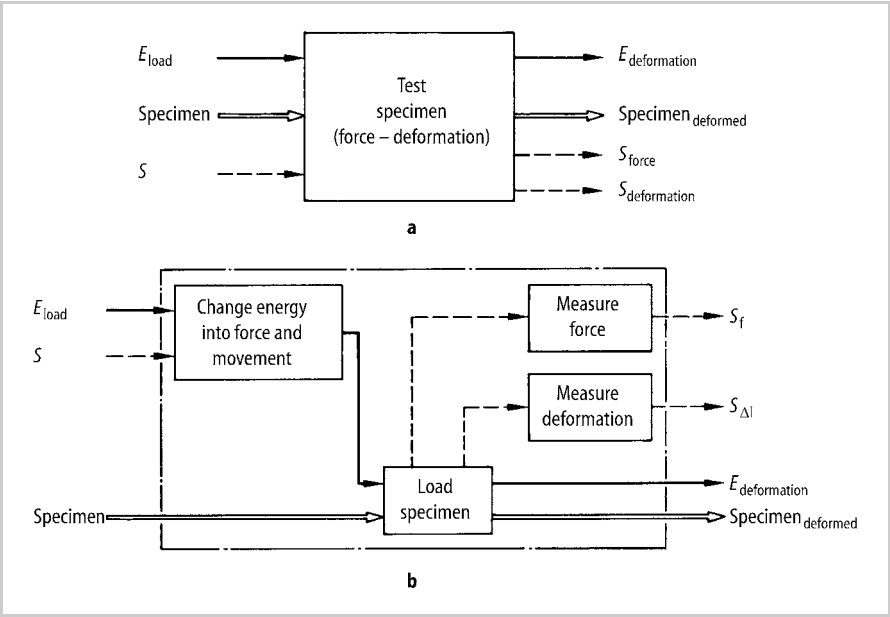


Figure 6.7. Overall function **a** and subfunctions (main functions) **b** of a tensile testing machine

that directly satisfy the required overall function are specified (see Figure 6.7). These are formulated as complex subfunctions, such as "change energy into force and movement" and "load specimen" in our example. Starting with complex subfunctions helps to establish a simple function structure.

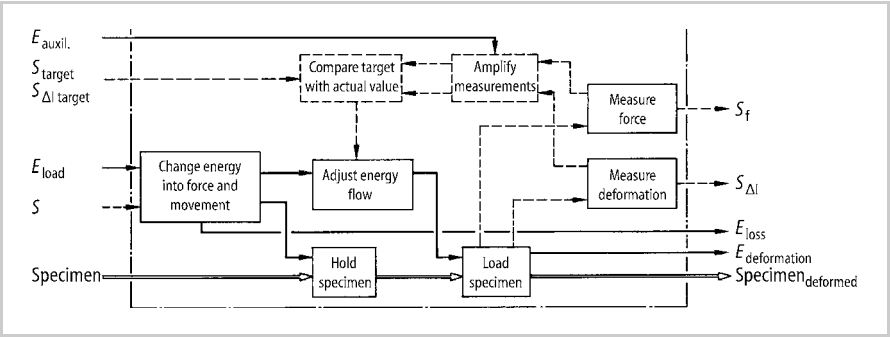


Figure 6.8. Completed function structure for the overall function set out in Figure 6.7

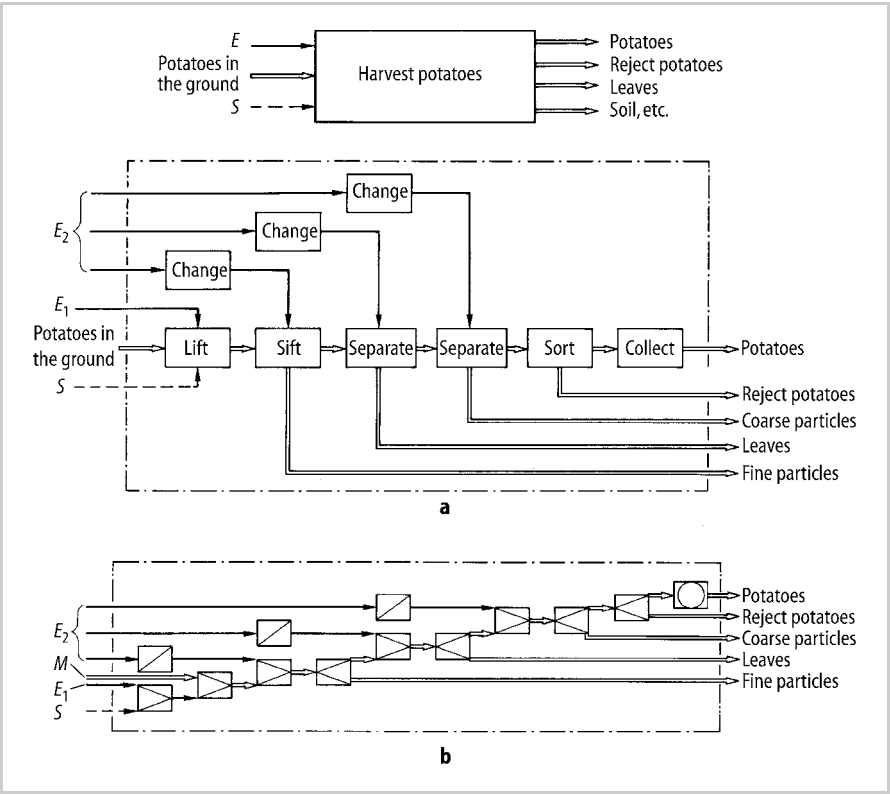


Figure 6.9. a Function structure of a potato harvesting machine **b** For comparison: diagram with generally valid functions based on [6.1], Figure 2.7

In the problem under consideration, the energy and signal flows are of roughly equivalent importance in the search for solutions, while the flow of material—the exchange of specimens—is only essential for the holding function added in Figure 6.8. In this figure, an adjusting function for the load magnitudes and, at the output of the system, the energy lost during the energy flow were also added because both clearly affect the design. The energy required to deform the specimen is lost with the material flow when the specimen is exchanged. Moreover, the auxiliary functions "amplify measurements" and "compare target with actual values" proved indispensable for the adjustment of the energy level.

There are, however, some problems in which variation of the main flow alone cannot lead to a solution, because *auxiliary flows* have a *crucial* bearing on the design and are solution-determining. As an example, let us consider the function structure of a potato harvesting machine. Figure 6.9a shows the overall function and the function structure based on the flow of material (the main flow) and the auxiliary flows of energy and signals. In Figure 6.9b, by comparison, the function structure is represented by means of generally valid functions, in order to emphasise the clear interrelationship of the different flows.

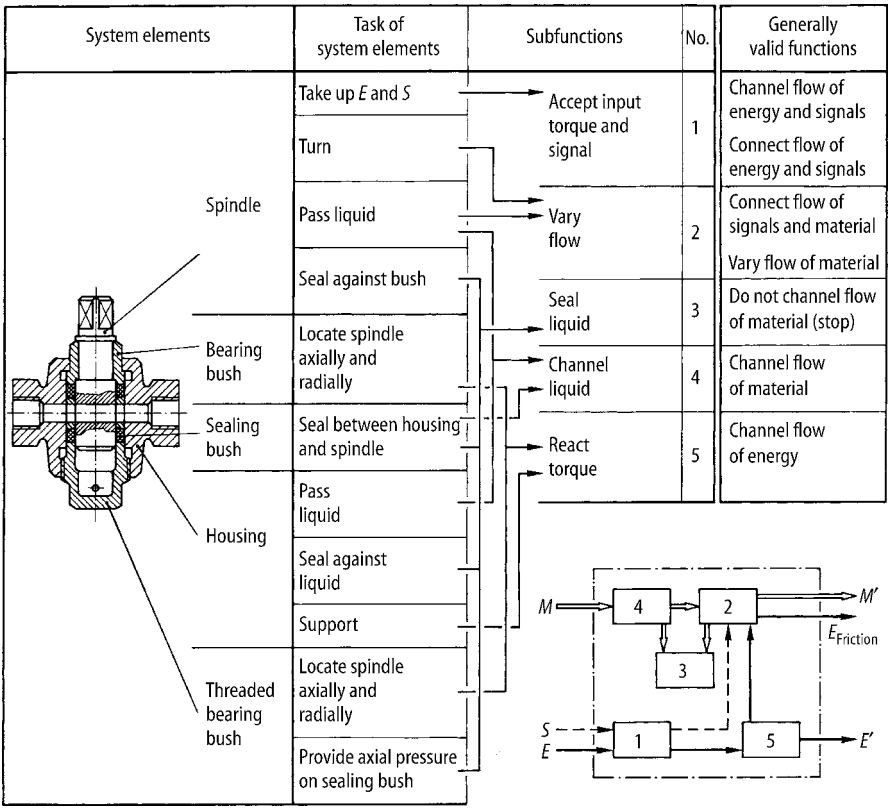


Figure 6.10. Analysis of a flow control valve with respect to its function structure

When generally valid functions are used, the separation into subfunctions is generally more pronounced than it is in the case of task-specific subfunctions. Thus, in the present example, the subfunction "separate" is replaced with the generally valid functions "connect energy with material mix" and "separate material mix" (the reverse of "connect"). The representation, however, is on such an abstract level that it is not easy to understand and requires further interpretation.

Our final example illustrates the derivation of function structures by the *analysis of existing systems*. This method is particularly suitable for developments in which at least one solution with the appropriate function structure is known, and the main problem is the discovery of better solutions. Figure 6.10 shows the steps used in the analysis of a flow control valve (a typical on–off switch), showing the individual tasks of the various elements and the subfunctions satisfied by the system. The function structure can be derived from the subfunctions and then varied in order to improve the product.

The function structure for the one-handed mixing tap examined in Section 6.6 clearly shows that the study of function structures may prove extremely useful, even after the physical effect has been selected, for determining the behaviour of the system at a very early stage of its development, and hence for identifying the structure that best suits the problem under consideration.

6.3.3 Practical Applications of Function Structures

When establishing function structures, we must distinguish between original and adaptive designs. In the case of *original designs*, the basis of a function structure is the *requirements list* and the *abstract formulation of the problem*. Among the demands and wishes, we are able to identify functional relationships, or at least the subfunctions at the inputs and outputs of a function structure. It is helpful to write out the functional relationships arising from the requirements list in the form of sentences and to arrange these in the order of their anticipated importance or in some other logical order [6.11].

In the case of *adaptive designs*, the starting point is the *function structure of the existing solution* obtained by analysing its elements. It helps to develop variants in order to open the path for other solutions, for subsequent optimisation and for the development of modular products. The identification of functional relationships can be facilitated by asking the right questions.

In modular systems, the function structure has a decisive influence on the modules and their arrangement (see Section 9.2). Here, the function structure and that of the assembly is affected not only by functional considerations, but also, and increasingly so, by production needs.

Function structures are intended to facilitate the discovery of solutions: they are not ends in themselves. The degree of detail used depends very much on the novelty of the task and the experience of the designers.

Moreover, it should be remembered that function structures are seldom completely free of physical or formal presuppositions, which means that the number

of possible solutions is inevitably restricted to some extent. Hence, it is perfectly legitimate to conceive a preliminary solution and then abstract this by developing and completing the function structure by a process of iteration.

Anyone setting up a function structure ought to bear the following points in mind:

1. First derive a rough function structure with a few subfunctions from what functional relationships you can identify in the requirements list, and then break this rough structure down, step-by-step, by resolving complex subfunctions. This is much simpler than starting out with more complicated structures. In certain circumstances, it may be helpful to substitute a first solution idea for the rough structure and then, by analysing that first idea, to derive other important subfunctions. It is also possible to begin with subfunctions whose inputs and outputs cross the assumed system boundary. From these, we can then determine the inputs and outputs for the neighbouring functions; in other words, we work from the system boundary inwards.

2. If no clear relationship between the subfunctions can be identified, the search for a first solution principle may, under certain circumstances, be based on the mere *enumeration of identified subfunctions* without logical or physical relationships, but if possible these should be arranged according to the extent to which they have been realised.

3. *Logical relationships* may lead to function structures through which the logical elements of various working principles (mechanical, electrical, etc.) can be anticipated.

4. Function structures are not complete unless the existing or expected flows of energy, material and signals can be specified. Nevertheless, it is useful to begin by focusing attention on the *main flow* because, as a rule, it determines the design and is more easily derived from the requirements. The auxiliary flows then help with the further elaboration of the design, coping with faults, and when dealing with problems of power transmission, control, etc. The complete function structure, comprising all flows and their relationships, can be obtained by iteration; that is, by looking first for the structure of the main flow, completing that structure by taking the auxiliary flows into account, and then establishing the overall structure.

5. When setting up function structures it is useful to know that, in the conversion of energy, material and signals, several *subfunctions recur* in most structures and should therefore be introduced first. Essentially, these are the generally valid functions of Figure 2.7, and they can prove extremely helpful in the search for task-specific functions.

6. For the application of microelectronics, it is useful to consider signal flows as shown in Figure 6.11 [6.6]. This results in a function structure that clearly suggests the modular use of elements to detect (sensors), to activate (actuators), to operate (controllers), to indicate (displays) and, in particular, to process signals using microprocessors.

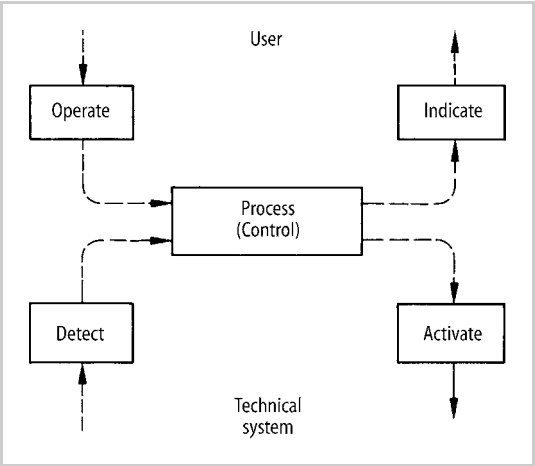


Figure 6.11. Basic signal flow functions for modular use in microelectronics. After [6.6]

7. From a rough structure, or from a function structure obtained by the analysis of known systems, it is possible to derive further *variants* and hence to optimise the solution, by:
   - breaking down or combining individual subfunctions
   - changing the arrangement of individual subfunctions
   - changing the type of switching used (series switching, parallel switching or bridge switching)
   - moving the system boundary.

   Because varying the function structure introduces distinct solutions, the setting up of function structures constitutes a first step in the search for solutions.

8. Function structures should be kept as *simple* as possible, in order to encourage simple and economical solutions. To this end, it is also advisable to aim at the combination of functions for the purpose of obtaining integrated function carriers. There are, however, some problems in which discrete functions must be assigned to discrete function carriers; for instance, when the requirements demand clarity in the solution, or when there is a need for extreme loading and quality. In this connection, the reader is referred to our discussion on the division of tasks (see Section 7.4.2).

9. In the search for solutions, only *promising function structures* should be introduced, which implies that a *selection procedure* (see Section 3.3.1) should be employed, even at this early stage.

10. For the *representation* of function structures it is best to use the *simple and informative symbols* shown in Figure 2.4, supplemented with task-specific verbal clarifications.

11. An *analysis* of the function structure leads to the identification of those subfunctions for which new working principles must be found, and of those for which known solutions can be used. This encourages an efficient approach. The search for solutions (see Section 3.2) then focuses on the subfunctions that are essential for the solution and on which the solutions of other subfunctions depend (see the example in Figure 6.6).

It is sometimes assumed wrongly that auxiliary functions are unimportant. Technical systems do not have functions that are "more important" or "less important". All functions are important because they are needed. Any functions that are not necessary or superfluous functions should be eliminated. It is only in order to reduce effort that designers start their search for solutions with the function that seems most important, i.e. solution-determining. All of the other functions are still necessary and must be fulfilled.

6.4 Developing Working Structures

6.4.1 Searching for Working Principles

Working principles need to be found for the various subfunctions, and these principles must eventually be combined into a working structure. The concretisation of the working structure will lead to the principle solution. A working principle must reflect the physical effect needed for the fulfilment of a given function and also its geometric and material characteristics (see Section 2.1.4). In many cases, however, it is not necessary to look for new physical effects, the form design (geometry and materials) being the sole problem. Moreover, in the search for a solution it is often difficult to make a clear mental distinction between the physical effect and the form design features. Designers therefore usually search for working principles that include the physical process along with the necessary geometric and material characteristics, and combine these into a working structure. Theoretical ideas about the nature and form of function carriers are usually presented by way of diagrams or freehand sketches.

It should be emphasised that the step we are now discussing is intended to lead to several solution variants, that is, a solution field. A solution field can be constructed by varying the physical effects and the form design features. Moreover, in order to satisfy a particular subfunction, several physical effects may be involved in one or several function carriers.

In Section 3.2 we discussed methods and tools for finding solutions. The same methods can be used in the search for working principles. Of particular importance, however, are literature searches, methods for analysing natural and known technical systems, and intuition-based methods (see Section 3.3.2). If preliminary solution ideas are available from product planning or through intuition, systematic analyses of physical processes and the utilization of classification schemes are also helpful (see Section 3.2.3). The last two methods usually provide several solutions.

Other important tools are design catalogues, in particular those proposed by Roth and Koller for physical effects and working principles (see Section 3.2.3)